ON DERIVED GALOIS DEFORMATION RINGS IN CHARACTERISTIC 0 AND A CONJECTURE OF VENKATESH

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ABSTRACT. Let Π be a regular algebraic cuspidal automorphic representation of GL_n over a CM field F, and ρ its associated Galois representation in characteristic 0. With mild assumptions, we use the Taylor–Wiles method to define a free action of the potentially semistable derived deformation ring of ρ on the Π-part of the p-adic cohomology of GL_n/F, confirming a prediction of Venkatesh. As a key step, we establish an equivalence between the derived deformation ring of ρ with the completion at the point corresponding to ρ of the derived deformation ring of the mod p reduction $\bar{\rho}$, generalising well-known results by Kisin.

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1. Introduction

The Langlands programme predicts correspondences between automorphic forms and Galois representations. At a basic level, this can be taken to mean there are bijections between sets of isomorphism classes, but in its modern formulation, the programme posits that these bijections carry geometric structure. The following schematic makes the relationships precise while keeping the exact meaning of terms vague.

Schematic. *Let $\mathcal{A}$ be a space of automorphic forms with an action of a Hecke algebra $\mathbb{T}$. Then there exists a moduli stack $\mathcal{X}$ of Galois representations with a map $\mathcal{X} \to \mathrm{Spec}(\mathbb{T})$ such that*

$$\mathcal{A} \cong \Gamma(\mathcal{X}, \omega_{\mathcal{X}}),$$

the global sections of the dualising complex $\omega_{\mathcal{X}}$.

At a basic level, the schematic says that there exists a morphism which takes a Galois representation to a system of Hecke eigenvalues. In the other direction, thinking of $\mathcal{A}$ as a sheaf on $\mathrm{Spec}(\mathbb{T})$, it says that $\mathcal{A}$ can be considered as a sheaf on a stack of Galois representations, and that this sheaf is the dualising complex. Recalling from algebraic geometry that the dualising complex is generically a trivial line bundle, we may extract the following slogan.

Schematic. *Locally near a generic point, a space of automorphic forms is naturally a free module of rank 1 over a local ring of functions of a moduli stack of Galois representations.*

This slogan has appeared in different guises in a series of works by Venkatesh and collaborators on which the present work builds ([GV18, PV21, Ven19, HV19]). To explain this in detail, let us now fix the meaning of the objects appearing in the slogan.

Fix a prime p, a CM field F with maximal totally real subfield F^+, an isomorphism $\iota\colon \mathbb{C} \cong \overline{\mathbb{Q}}_p$ and a finite extension $E/\mathbb{Q}_p$ assumed large enough. Let $\mathbf{G} = \mathrm{GL}_n/F$ and suppose Π is a regular algebraic cuspidal automorphic representation of $\mathbf{G}(\mathbb{A}_F)$ of weight $\lambda \in (\mathbb{Z}^n_+)^{\mathrm{Hom}(F,E)}$. Fix an open compact subgroup $K \subset \mathbf{G}(\mathbb{A}_F^\infty)$ and consider the locally symmetric space X_K of level K, which is a smooth manifold of dimension $\dim X_K = [F^+\colon \mathbb{Q}]n^2 - 1$ equipped with a locally constant sheaf of E-vector spaces $\mathcal{V}_\lambda$ determined by the weight λ. The singular homology $H_*(X_K, \mathcal{V}_\lambda)$ is a finite-dimensional E-vector space equipped with an action of Hecke operators at the unramified places, and the system of eigenvalues associated to Π defines an eigenspace

$$H_*(X_K, \mathcal{V}_\lambda)_\Pi \subset H_*(X_K, \mathcal{V}_\lambda),$$

which is non-zero in degrees $[q_0, q_0 + l_0]$, where $l_0 = [F^+\colon \mathbb{Q}]n - 1 \in \mathbb{N}$ is the defect of $\mathbf{G}$ and $2q_0 + l_0 = \dim X_K$. The dimension of the Π-eigenspace in degree $q_0 + i$ is $\binom{l_0}{i}$ times a fixed factor depending on the prime-to-p level K^p and not on i. This suggestive numerology underlies the following conjecture.

Conjecture. [PV21, Conj. 1.2.1] *$H_*(X_K, \mathcal{V}_\lambda)_\Pi$ is a free graded module over $\wedge^* V_\Pi$, where V_Π is the dual adjoint Bloch–Kato Selmer group attached to Π (defined below), a vector space of dimension l_0.*

Properly speaking, this is the p-adic realisation of a conjecture for homology with $\mathbb{Q}$-coefficients where the role of V_Π is played by a conjectural motivic cohomology group whose p-adic realisation is V_Π. In order for this statement to fit with the slogan above, we ought to explain how the dual adjoint Bloch–Kato Selmer group can be thought of as a 'local ring of functions' on a stack of Galois representations.

By work of Harris–Lan–Taylor–Thorne [HLTT16] and Scholze [Sch15], one can attach to Π a continuous Galois representation

$$\rho\colon \Gamma_F = \mathrm{Gal}(\overline{F}/F) \to \mathrm{GL}_n(E),$$

uniquely characterised up to isomorphism by its values at Frobenius elements matching the Hecke eigenvalues of Π in a precise way. This representation is unramified outside a finite set of places S; to simplify the exposition in the introduction, we ignore the places in S not above p.

The representation ρ and its Hodge–Tate weights $\mathrm{HT}(\rho)$ define a (classical) deformation functor $D_\rho^{\mathrm{ss}}\colon \mathrm{Art}_E \to \mathrm{Set}$ where Art_E is the category of local artinian E-algebras, given by

$$B \mapsto \{r_B\colon \Gamma_{F,S} \to \mathrm{GL}_n(B) \mid r_B \equiv \rho \bmod \mathfrak{m}_B, r_B \text{ potentially semistable, } \mathrm{HT}(r_B) = \mathrm{HT}(\rho)\}/\sim,$$

where $\sim$ denotes strict equivalence. When ρ is absolutely irreducible, D_ρ^{ss} is pro-represented by a complete noetherian local E-algebra R_ρ^{ss}, whose spectrum $\mathrm{Spec}(R_\rho^{\mathrm{ss}})$ we view as an intersection

$$(\text{deformations of } \rho) \cap (\text{deformations of } \prod_{v\in S} \rho_v \text{ w. local conditions}) \subset (\text{ deformations of } \prod_{v\in S} \rho_v).$$

The construction of ρ from Π furnishes an action

$$R_\rho^{\mathrm{ss}} \circlearrowright H_*(X_K, \mathcal{V}_\lambda)_\Pi,$$

and thus, our space of automorphic forms is lifted to a module over a ring describing infinitesimal deformations of ρ. As we shall see, R_ρ^{ss} can indeed be thought of as a local ring of functions on a stack of Galois representations.

In terms of rings, the intersection above is represented by a tensor product

$$R_\rho^{\mathrm{ss}} \cong R_\rho \otimes_{R_\rho^{\mathrm{loc}}} R_\rho^{\mathrm{loc,ss}}.$$

By [All16, Prop. 1.3.12], the tangent space of R_ρ^{ss} is given by the adjoint Bloch–Kato Selmer group $H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho)$, which is known to vanish – confirming the Bloch–Kato conjecture's prediction – in many cases by the main result of [A'C24]. In other words, $\mathrm{Spec}\, R_\rho^{\mathrm{ss}}$ is a point. However, tangent-obstruction theory predicts the dimension of R_ρ^{ss} to be given by an Euler characteristic which comes out to $-l_0 < 0$. This suggests that the intersection above is non-transverse and that R_ρ^{ss} underlies a 'derived object' which encodes this statement.

A natural candidate is given by replacing $\otimes$ with $\otimes^{\mathbf{L}}$, but the resulting object only has good formal properties when the rings involved are complete intersections of the expected dimension, which is expected to hold but hard to verify for R_ρ. Supposing for the moment that this was known, we would have an isomorphism

$$\mathrm{Tor}_*^{R_\rho^{\mathrm{loc}}}(R_\rho, R_\rho^{\mathrm{loc,ss}}) \cong \bigwedge H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho),$$

and one might hope to define an action of this Tor-algebra on $H_*(X_K, \mathcal{V}_\lambda)_\Pi$. This is the approach taken in [HT17] (using eigenvarieties) and [Mol24] (using completed homology), but one can circumvent the issue entirely by replacing the Tor-algebra with a more natural object.

This is the motivation behind the theory of derived Galois deformation rings developed by Galatius–Venkatesh [GV18], which yields derived enhancements of classical deformation rings for mod p representations. Later, Zhu [Zhu25] gave a more general construction which also works in characteristic 0, In fact, Zhu constructs a derived moduli stack of Galois representation as featured in the beginning of this introduction, whose local rings of functions are derived deformation rings. In both setups, a

derived deformation ring is an animated (or simplicial) ring whose underlying static ring is canonically isomorphic to a classical deformation ring; for example, there exists a derived deformation ring denoted $\mathcal{R}_\rho^{\mathrm{ss}}$ which satisfies

$$\pi_0(\mathcal{R}_\rho^{\mathrm{ss}}) \cong R_\rho^{\mathrm{ss}}$$

The benefit of the derived framework can be explained in terms of tangent complexes. To $\mathcal{R}_\rho^{\mathrm{ss}}$ and R_ρ^{ss} we can associate a cochain complexes of E-vector spaces denoted $\mathrm{T}_E^\bullet(\mathcal{R}_\rho^{\mathrm{ss}})$ and $\mathrm{T}_E^\bullet(R_\rho^{\mathrm{ss}})$, respectively. In both cases, the 0th cohomology is the adjoint Bloch–Kato Selmer group:

$$\mathrm{T}_E^0(\mathcal{R}_\rho^{\mathrm{ss}}) \cong \mathrm{T}_E^0(R_\rho^{\mathrm{ss}}) \cong H_g^1(\Gamma_{F,S}, \operatorname{ad}\rho).$$

For the derived deformation ring, we have

$$\mathrm{T}_E^1(\mathcal{R}_\rho^{\mathrm{ss}}) \cong H_g^2(\Gamma_{F,S}, \operatorname{ad}\rho) \cong H_g^1(\Gamma_{F,S}, \operatorname{ad}\rho(1))^\vee,$$

and $\mathrm{T}_E^i(\mathcal{R}_\rho^{\mathrm{ss}}) = 0$ for $i \geq 2$. In contrast, supposing that $R_\rho^{\mathrm{ss}} \cong E$ we have $\mathrm{T}_E^i(R_\rho^{\mathrm{ss}}) = 0$ for all i. As another example, the unrestricted derived deformation ring of ρ satisfies

$$\mathrm{T}_E^i(\mathcal{R}_\rho) \cong H^{i+1}(\Gamma_{F,S}, \operatorname{ad}\rho),\ i \geq 0,$$

whereas there is no clear-cut description of $\mathrm{T}_E^i(R_\rho)$ in terms of Galois cohomology for $i \geq 1$. Derived deformation rings are convenient precisely because of this close relation to Galois cohomology. One could certainly argue that it means $\mathcal{R}_\rho$ is a more natural object to consider; more importantly, the derived framework is more amenable to computations, since many questions about derived deformations reduce to linear algebra of tangent complexes.

Without further ado, let us state a simplified version of our main result (cf. Conjecture 1 above).

Theorem 1.0.1 (Theorem 4.0.1)**.** *With notation as above and under technical assumptions on F and ρ, there exists a free action of $\pi_*(\mathcal{R}_\rho^{\mathrm{ss}})$ on $H_*(X_K, \mathcal{V}_\lambda)_\Pi$ and a canonical isomorphism $\pi_*(\mathcal{R}_\rho^{\mathrm{ss}}) \cong \wedge V_\Pi$.*

In [GV18], a free action of the crystalline version of $\pi_*(\mathcal{R}_{\mathbf{S}})$ on the *integral* cohomology is defined, in the Fontaine–Laffaille setting and assuming the Calegari–Geraghty vanishing conjecture for mod p cohomology [CG18, Conj. B], together with some other simplifying assumptions (some of which have since been relaxed, see [Cai21, CT22]). Inverting p then yields the desired action. In a general setting, this strategy is not applicable, and the naive generalisation of the integral result is not expected to always hold. Our proof takes a modified approach with the aim of avoiding restrictive local assumptions, at the cost of obtaining only a result in characteristic 0.

The proof of Theorem 1.0.1 builds on the Taylor–Wiles method as applied to derived deformation rings by Galatius–Venkatesh. Since this method concerns mod p deformations and we are interested in a result in characteristic 0, we relate the derived deformation ring of ρ to that of $\bar\rho$ by proving derived analogues of Kisin's widely referenced results [Kis09, Lem. 2.3.3, Prop. 2.3.5], which relate the two notions in the classical setting. For example, we obtain the following result as a consequence of Proposition 3.7.2 and Theorem 3.7.3, which are applicable in broad generality.

Corollary 1.0.2. *Let $\mathcal{R}_{\bar\rho} \in \mathcal{CNL}_k$ and $\mathcal{R}_\rho \in \mathcal{CNL}_E$ denote the derived deformation rings of $\bar\rho$ and ρ, respectively, and let $\xi\colon \mathcal{R}_{\bar\rho} \to \mathcal{O}$ be a point corresponding to an $\mathcal{O}$-lattice $\xi \subset \rho$. Then the derived completion*

$$(\mathcal{R}_{\bar\rho} \otimes_{\mathcal{O}} E)^\wedge_\xi := \lim_r (\mathcal{R}_{\bar\rho} \otimes_{\mathcal{O}} E)/^{\mathbf{L}} I_\xi^r$$

of $\mathcal{R}_{\bar\rho} \otimes_{\mathcal{O}} E$ at $I_\xi = \ker(\pi_0(\mathcal{R}_{\bar\rho}) \otimes_{\mathcal{O}} E \to E)$ is equivalent to $\mathcal{R}_\rho$.

Using the same results, and under our technical assumptions, we prove an equivalence

$$\mathcal{R}_\rho^{\mathrm{ss}} \simeq (\mathcal{R}_{\mathbf{S}} \otimes_{\mathcal{O}} E)^\wedge_\xi,$$

where $\mathcal{R}_{\mathbf{S}}$ represents a mod p (derived) deformation problem $\mathbf{S}$ defined using Kisin's potentially semistable deformation rings (see Corollary 3.7.9). This equivalence allows us to carry out patching arguments in the mod p setting and then pass to characteristic 0.

Let us describe in simplified terms the Taylor–Wiles argument. The goal, following [GV18], is to approximate the mod p deformation ring $\mathcal{R} \coloneqq \mathcal{R}_{\mathbf{S}}$ by a derived tensor product of static rings. For every $N \geq 1$, one defines an augmented deformation problem $\mathbf{S}_N$ of Taylor–Wiles level N and a corresponding derived deformation ring $\mathcal{R}_N \coloneqq \mathcal{R}_{\mathbf{S}_N}$ with underlying static ring $R_N \coloneqq \pi_0(\mathcal{R}_N)$. Descending from Taylor–Wiles level yields an equivalence

$$\mathcal{R} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{S_N} \mathcal{O},$$

where S_N is the $\mathcal{O}$-group ring of a finite p-group. Via this equivalence, one defines a morphism

$$\mathcal{R} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{S_N} \mathcal{O} \to R_N \otimes^{\mathbf{L}}_{S_N} \mathcal{O} \to \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}_N} \mathcal{O}_N =: \mathcal{C}_N$$

where $R_N \twoheadrightarrow \overline{R}_N$ and $S_N \twoheadrightarrow \overline{S}_N$ are artinian quotients and $\mathcal{O}_N \coloneqq \mathcal{O}/\varpi^N$. We think of $\mathcal{C}_N$ as a 'mod ϖ^N approximation' of $\mathcal{R}$. A compactness argument implies the existence of a compatible system of approximations, yielding a map

$$g\colon \mathcal{R} \to \mathcal{C}_\infty = \lim_N \mathcal{C}_N$$

for some sequence of N. This induces a morphism on localised and completed rings

$$(\mathcal{R}_{\mathbf{S}} \otimes_{\mathcal{O}} E)^\wedge_\xi \simeq \mathcal{R}_\rho^{\mathrm{ss}} \to (\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi.$$

We prove that this morphism is an equivalence by comparing tangent complexes. Carrying out this non-constructive argument while retaining sufficient control over the induced morphism of tangent complexes is the main technical difficulty in the proof. Applying an '$R = T$' theorem from [A'C24], we identify $(\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi$ with a derived tensor product of power series rings:

$$(\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi \simeq E[[X_1, \dots, X_s]] \otimes^{\mathbf{L}}_{E[[X_1,\dots,X_{s+l_0}]]} E$$

A'Campo proves that $E[[X_1, \dots, X_s]]$ acts freely on a patched homology module $(M_\infty)^\wedge_\xi$, and descending from Taylor–Wiles level, we obtain a free action

$$\pi_*(\mathcal{R}_\rho^{\mathrm{ss}}) \circlearrowleft (M_\infty)^\wedge_\xi \otimes^{\mathbf{L}}_{E[[X_1,\dots,X_{s+l_0}]]} E \cong H_*(X_K, \mathcal{V}_\lambda)_\Pi.$$

A priori, this action depends on the non-canonical choice made in defining the morphism g above. In [GV18], this possibility is ruled out by establishing compatibility with the action of the derived Hecke algebra introduced in [Ven19]. Thus, so far the naturality part of Conjecture 1 is only known in the setting of [GV18], though we expect it to hold in general.

Outline. Let us outline the content of the sections below. In Section 2, we recall generalities concerning the cohomology of locally symmetric spaces and animated rings. Section 3 discusses the derived deformation theory of Galois representations: we define various derived deformation functors and prove all their necessary properties. In particular, Section 3.7 concerns the link between deformations of a characteristic 0 representation and its mod p reduction. In Section 4, we use the patching and the results of the preceding sections to prove the main result, Theorem 4.0.1.

Notation and conventions. The following notation will be used throughout the article. F is a CM field with ring of integers $\mathcal{O}_F$. When S is a set of places of F, we use the notation S_p for $\{v \in S \colon v \mid p\}$, and the completion of $\mathcal{O}_F$ at a place v is $\mathcal{O}_{F_v}$. Our field of coefficients is denoted E and is a finite extension of $\mathbb{Q}_p$ with ring of integers $\mathcal{O}$, uniformiser ϖ and residue field $k \coloneqq \mathcal{O}/\varpi$. The extension E is tacitly assumed to be large enough, so that all of our Galois representations are defined over E and E contains all embeddings $F \hookrightarrow \overline{\mathbb{Q}}_p$. For any $r \geq 1$, we let $\mathcal{O}_r \coloneqq \mathcal{O}/\varpi^r$. Galois groups are denoted by Γ, e.g. $\Gamma_{F,S} = \mathrm{Gal}(\overline{F}_S/F)$ and $\Gamma_{F_v} = \mathrm{Gal}(\overline{F}_v/F_v)$. When $\bar{\rho}$ or ρ is a representation of a global Galois group such as $\Gamma_{F,S}$, we denote restrictions by subscripts, e.g. $\bar{\rho}_v = \bar{\rho}|_{\Gamma_{F_v}}$.

This article uses the language of quasicategories (from now on, ∞-categories). Generally, we use calligraphic letters to denote ∞-categories, so that e.g. $\mathcal{A}lg_{\mathcal{O}/}$ is the animation of the category $\mathrm{Alg}_{\mathcal{O}}$ of $\mathcal{O}$-algebras (see Section 2.2). When we are interested in both an animated object and its underlying static object, we sometimes distinguish them by fonts; this is always explicitly declared at some point in the text, and if so then e.g. $R = \pi_0(\mathcal{R})$. Unless otherwise specified, limits and colimits are taken in the ∞-categorical sense. We use the symbol $\varprojlim$ to denote the 1-categorical inverse limit whereas the undecorated $\lim$ always denotes a limit in the sense of ∞-categories. We make no typographic distinction between $\varinjlim$ and colim; their conflation is harmless as they yield the same object. Finally, we recall a couple of constructions. If Y is an object of an ∞-category $\mathcal{C}$, we denote by $\mathcal{C}_{/Y}$ the over-∞-category with objects $(X \to Y)$ [Lur09, §1.2.9]. Given an ∞-category $\mathcal{C}$ and a full subcategory $h'\mathcal{C} \hookrightarrow h\mathcal{C}$ of its homotopy category, the pullback of $N(h'\mathcal{C}) \to N(h\mathcal{C}) \leftarrow \mathcal{C}$, where $N(\cdot)$ denotes the nerve, is the ∞-subcategory of $\mathcal{C}$ spanned by the vertices lying in $h'\mathcal{C}$ [Cis19, Ex. 3.9.5].

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2. Preliminaries

The goal of this section is to describe our general setup and collect some useful results from the literature for easy reference.

2.1. **Locally symmetric spaces.** Let $\mathbf{G} = \mathrm{GL}_n/F$. We begin by recalling how automorphic representations of $\mathbf{G}(\mathbb{A}_F)$ contribute to the cohomology of locally symmetric spaces, following the exposition in [Hev24, §2.1]. We let $X^{\mathbf{G}}$ be the symmetric space of $\mathrm{Res}_{F/\mathbb{Q}}\mathbf{G}$ as in [BS73, §2], which is a homogeneous $\mathbf{G}(F \otimes_{\mathbb{Q}} \mathbb{R})$-space.

An open compact subgroup $K \subset \mathbf{G}(\mathbb{A}_F^\infty)$ is *good* if $K = \prod_v K_v$ with $K_v \subseteq \mathbf{G}(\mathcal{O}_{F_v})$ and K is neat. If S is a finite set of places of F, we write $K = K^S \times K_S \coloneqq \prod_{v \in S} K_v \times \prod_{v \notin S} K_v$. Given such a K, we define the locally symmetric space of level K as

$$X_K = \mathbf{G}(F)\backslash(X^{\mathbf{G}} \times \mathbf{G}(\mathbb{A}_F^\infty)/K),$$

a smooth manifold of dimension $[F^+ \colon \mathbb{Q}](n^2-1)$. The defect of $\mathbf{G}$ is the positive integer

$$l_0 \coloneqq [F^+ \colon \mathbb{Q}]n - 1.$$

Let $\lambda = (\lambda_\iota)_{\iota \in \mathrm{Hom}(F,E)} \in (\mathbb{Z}^n_+)^{\mathrm{Hom}(F,E)}$ be a dominant weight and suppose we are given representations $\tau_v \colon I_{F_v} \to \mathrm{GL}_n(E)$ with open kernel for $v \mid p$. Then [SZ99] associate to each τ_v an irreducible smooth $\mathcal{O}$-representation $\sigma^\circ(\tau_v)$ of $\mathrm{GL}_n(\mathcal{O}_{F_v})$, and we let $\sigma^\circ(\tau) = \bigotimes_{v|p} \sigma^\circ(\tau_v)$. From the pair (λ, τ) we then obtain a locally constant sheaf of $\mathcal{O}$-modules $\mathcal{V}^\circ_{\lambda,\tau} = \mathcal{V}^\circ_\lambda \otimes \sigma^\circ(\tau) \in \mathrm{Sh}(X_K)$

where $\mathcal{V}_\lambda^\circ$ is an $\mathcal{O}$-lattice in the highest weight representation of $(\mathrm{GL}_n)_E^{\mathrm{Hom}(F,E)}$ of highest weight λ. We define $\sigma(\tau) \coloneqq \sigma^\circ(\tau) \otimes_{\mathcal{O}} E$ and $\mathcal{V}_{\lambda,\tau} \coloneqq \mathcal{V}^\circ_{\lambda,\tau} \otimes_{\mathcal{O}} E$. The complex

$$R\Gamma(X_K, \mathcal{V}^\circ_{\lambda,\tau}) \in \mathcal{D}(\mathcal{O})$$

carries an action of the *abstract Hecke algebra* $\mathbb{T}^S \coloneqq \mathcal{H}(\mathbf{G}(\mathbf{A}_F^{\infty,S}), K^S)$ where S is a finite set of places such that $K_v = \mathrm{GL}_n(\mathcal{O}_{F_v})$ for $v \notin S$. If Π is a regular algebraic cuspidal representation of $\mathbf{G}$, we let $\mathfrak{p}_{\Pi,\iota} = \ker(\mathbb{T}^S \to \mathrm{End}_{\mathbb{C}}(\otimes'_{v\notin S}\Pi_v^{K_v}))$ and define the Π-part as the localisation

$$H_*(X_K, \mathcal{V}_{\lambda,\tau})_\Pi \coloneqq H_*(X_K, \mathcal{V}_{\lambda,\tau})_{\mathfrak{p}_{\Pi,\iota}},$$

i.e. the eigenspace (over E) corresponding to the system of Hecke eigenvalues of Π. We write

$$\mathbb{T}^S\big(R\Gamma(X_K, \mathcal{V}^\circ_{\lambda,\tau})\big) \subset \mathrm{End}_{\mathcal{D}(\mathcal{O})}\big(R\Gamma(X_K, \mathcal{V}^\circ_{\lambda,\tau})\big)$$

for the corresponding faithful Hecke algebra. There exists an open compact subgroup $K_\tau \subset K$ such that

$$R\Gamma(X_K, \mathcal{V}_{\lambda,\tau})_\Pi = R\Gamma\big(K/K_\tau, R\Gamma(X_{K_\tau}, \mathcal{V}_\lambda) \otimes_E \sigma(\tau)\big),$$

and since taking invariants is an exact functor in characteristic 0, we have

$$H_*(X_K, \mathcal{V}_{\lambda,\tau})_\Pi \cong \big(H_*(X_{K_\tau}, \mathcal{V}_\lambda)_\Pi\big)_{K/K_\tau}.$$

Proposition 2.1.1. [ACC+23, Thm. 2.4.10] *Let $2q_0 = \dim X_K - l_0$, fix a regular algebraic cuspidal representation Π of weight λ and inertial type $\tau = (\tau_v)_{v|p} = (\mathrm{LL}(\Pi_v)|_{I_{F_v}})_{v|p}$. Let m_Π denote $\frac{1}{\dim \sigma(\tau)} \dim_{\mathbb{C}}(\Pi^\infty)^{K^p}$. Then $H_i(X_K, \mathcal{V}_{\lambda,\tau})_\Pi = 0$ for $i \notin [q_0, q_0 + l_0]$ and for $i = 0, \dots, l_0$ we have*

$$\dim H_{q_0+i}(X_K, \mathcal{V}_{\lambda,\tau})_\Pi = m_\Pi \binom{l_0}{i}.$$

2.2. **Animated algebras.** We begin by recalling the definition of animated algebras and its relation to simplicial rings, following the presentation in [ČS24, §5.1]. Let $\mathrm{Alg}_{\mathcal{O}}$ denote the ordinary category of commutative $\mathcal{O}$-algebras and let

$$\mathcal{A}lg_{\mathcal{O}/} \coloneqq \mathcal{A}ni(\mathrm{Alg}_{\mathcal{O}})$$

denote the ∞-category of *animated $\mathcal{O}$-algebras*, i.e. the ∞-category generated under sifted colimits by the subcategory $\mathrm{Alg}_{\mathcal{O}}^{\mathrm{ffg}} \subset \mathrm{Alg}_{\mathcal{O}}$ of finite free $\mathcal{O}$-algebras; it is the ∞-category underlying the simplicial model category $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Alg}_{\mathcal{O}})$ of simplicial $\mathcal{O}$-algebras [Lur09, Cor. 5.5.9.3]. If A is an animated $\mathcal{O}$-algebra, an *A-module* is a module over the underlying $\mathbb{E}_1$-ring of A (see [Lur04, §2.4]), whereas an *animated A-module* is a connective such module. When $A = \pi_0(A)$ is static, the category

$$\mathcal{A}ni(\mathrm{Mod}_A)$$

of animated A-modules is equivalent to the derived category $D^{\geq 0}(A)$ of connective complexes of A-modules, which is equivalent to the localisation of the simplicial model category $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Mod}_A)$ of simplicial A-modules at weak equivalences.

Definition 2.2.1. [Lur04, Def. 2.5.9] An animated $\mathcal{O}$-algebra $A \in \mathcal{A}lg_{\mathcal{O}/}$ is:

(1) *Noetherian* if $\pi_0(A)$ is noetherian and $\pi_i(A)$ is a finitely generated $\pi_0(A)$-module for every $i \geq 0$.
(2) *Local* if $\pi_0(A)$ is local.

The content of the following proposition is found immediately below [Lur04, Rem. 3.1.4].

Proposition 2.2.2. [Lur04] *Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ and suppose B lies in the smallest subcategory of $\mathcal{A}lg_{A/}$ containing $A[X]$ which is closed under finite colimits. Then $\mathrm{Map}_A(B, -)$ commutes with filtered colimits.*

If the assumption of Proposition 2.2.2 holds, we say that B is a *finitely presented A-algebra.*

Proposition 2.2.3. [Lur04, Prop. 3.1.5] *Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ be noetherian, $j \geq 0$ and suppose B is an A-algebra. The following are equivalent:*

1. *The functor $\mathrm{Map}_A(B, -)$ commutes with filtered colimits when restricted to the subcategory $\tau_{\leq j}(\mathcal{A}lg_A)$ of j-truncated A-algebras.*
2. *The A-algebra $\tau_{\leq j}B$ is noetherian and $\pi_0(B)$ is a finitely presented $\pi_0(A)$-algebra.*

If the equivalent conditions in Proposition 2.2.3 hold, we say that B is *of finite presentation to order j as an A-algebra*. If the conditions hold for every $j \geq 0$, we say that B is *almost of finite presentation over A*. Note that if A is a static noetherian ring and $A \twoheadrightarrow B$, then B is almost of finite presentation over A by (2) above.

Given an animated $\mathcal{O}$-algebra $A \in \mathcal{A}lg_{\mathcal{O}/}$, an animated A-module $M \in \mathcal{M}od_{\mathcal{O}}$ and an element $a\colon \{*\} \to A$, we obtain a map $\mathcal{O}[t] \to A$ and define

$$M/^{\mathbf{L}}a \coloneqq M \otimes^{\mathbf{L}}_{\mathcal{O}[t]} \mathcal{O},$$

where the map $\mathcal{O}[t] \to \mathcal{O}$ corresponds to $0 \in \mathcal{O}$. More generally, given $a_1, \ldots, a_m \in A$, we define

$$M/^{\mathbf{L}}(a_1, \ldots, a_m) \coloneqq M \otimes^{\mathbf{L}}_{\mathcal{O}[t_1,\ldots,t_m]} \mathcal{O} \simeq M/^{\mathbf{L}}a_1/^{\mathbf{L}} \ldots /^{\mathbf{L}}a_m.$$

There are natural maps

$$M/^{\mathbf{L}}(a_1^{r+1}, \ldots, a_m^{r+1}) \to M/^{\mathbf{L}}(a_1^r, \ldots, a_m^r)$$

forming an inverse system of animated $\mathcal{O}$-modules. If $M \in \mathcal{A}lg_{\mathcal{O}/}$, this is an inverse system of animated $\mathcal{O}$-algebras and the limit depends only on the closed subspace of $\mathrm{Spec}\,\pi_0(A)$ defined by $(a_1, \ldots, a_m)$, as the following result shows.

Proposition 2.2.4. [Lur04, Prop. 6.1.1] *Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ and let $I = (a_1, \ldots, a_m) \subset \pi_0(A)$ be a finitely generated ideal. Then, for any $B \in \mathcal{A}lg_{\mathcal{O}/}$, the natural map*

$$\operatorname{colim}_r \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A/^{\mathbf{L}}(a_1^r, \ldots a_m^r), B) \to \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B)$$

identifies the left-hand side with the subspace spanned by morphisms $f\colon A \to B$ such that the induced morphism $\pi_0(A) \to \pi_0(B)$ factors through $\pi_0(A)/I^N$ for some $N \geq 1$.

Definition 2.2.5. [ČS24, 5.6.1] Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ be noetherian and let $I = (a_1, \ldots, a_m) \subset \pi_0(A)$. Then A is *derived I-adically complete* if the natural map $A \to \lim_r A/^{\mathbf{L}}(a_1^r, \ldots, a_m^r)$ is a weak equivalence.

Definition 2.2.6. A local noetherian animated $\mathcal{O}$-algebra A is *complete* if A is derived $\mathfrak{m}_{\pi_0(A)}$-adically complete. We let $\mathcal{CNL}_k \subset (\mathcal{A}lg_{\mathcal{O}/})_{/k}$ and $\mathcal{CNL}_E \subset (\mathcal{A}lg_{\mathcal{O}/})_{/E}$ denote the full subcategories of complete noetherian local animated $\mathcal{O}$-algebras with residue field $\pi_0(A)/\mathfrak{m}_{\pi_0(A)}$ equal to k and E, respectively.

Lemma 2.2.7. [BS15, Lem. 3.4.13] *Let $A \in \mathcal{M}od_{\mathcal{O}}$ and suppose $\pi_i(A)$ is ϖ-adically complete in the usual sense for every $i \geq 0$. Then A is derived ϖ-adically complete.*

2.3. **Bloch–Kato Selmer groups.** In this section, we recall the definition of the Bloch–Kato Selmer group mentioned in the introduction, and prove a dimension formula. We also define a complex whose cohomology is given by Bloch–Kato Selmer groups, which will be related to derived deformation rings in Corollary 3.7.9. See [Mil86] for a complete reference on Galois cohomology.

Let S be a finite set of places of F containing S_p and suppose $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ is a continuous representation which is de Rham at all places above p. Let $v \in S$. If $v \mid p$, then

$$H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho) \coloneqq \ker\left(H^1(\Gamma_{F_v}, \operatorname{ad}\rho) \to H^1(\Gamma_{F_v}, \operatorname{ad}\rho \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}})\right),$$

where the map on the right is the natural one induced by the tensor product and B_{dR} is Fontaine's ring of periods. If $v \nmid p$, let $I_{F_v} \subset \Gamma_{F_v}$ denote the inertia subgroup and set

$$H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho) \coloneqq \ker\left(H^1(\Gamma_{F_v}, \operatorname{ad}\rho) \to H^1(I_{F_v}, \operatorname{ad}\rho)\right).$$

where the map on the right is the one induced by the inclusion $I_{F_v} \subset \Gamma_{F_v}$. The (geometric adjoint) *Bloch–Kato Selmer group* of ρ is the space of global classes with restrictions lying in the groups defined above, i.e.

$$H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho) \coloneqq \ker\left(H^1(\Gamma_{F,S}, \operatorname{ad}\rho) \to \prod_{v\in S} \frac{H^1(\Gamma_{F_v}, \operatorname{ad}\rho)}{H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho)}\right).$$

We use lowercase letters to denote the dimension of cohomology groups or subspaces thereof, e.g.

$$h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho) \coloneqq \dim_E H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho).$$

Proposition 2.3.1. *Let $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ be a continuous representation, such that:*

1. *ρ is absolutely irreducible.*
2. *For every $v \in S$, the Weil-Deligne representation $\mathrm{WD}(\rho_v)$ is generic.*
3. *For every $v \in S_p$, ρ_v is potentially semistable with distinct Hodge-Tate weights.*

Then, with $l_0 = [F^+\colon \mathbb{Q}]n - 1$,

$$h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho) = h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1)) - l_0.$$

Proof. By the Greenberg–Wiles formula,

$$h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho) - h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1)) =$$
$$\sum_{v\in S}\left(h^1_g(\Gamma_{F_v}, \operatorname{ad}\rho) - h^0(\Gamma_{F_v}, \operatorname{ad}\rho)\right) + h^0(\Gamma_{F,S}, \operatorname{ad}\rho) - h^0(\Gamma_{F,S}, \operatorname{ad}\rho(1)) - \sum_{v\mid\infty} h^0(\Gamma_{F_v}, \operatorname{ad}\rho).$$

Assumption (1) implies $h^0(\Gamma_{F,S}, \operatorname{ad}\rho) = 1$ and $h^0(\Gamma_{F,S}, \operatorname{ad}\rho(1)) = 0$, and assumption (2) implies that the local terms coming from $v \in S \setminus S_p$ vanish. Thus, we have

$$h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho) - h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1)) = \sum_{v\in S_p}\left(h^1_g(\Gamma_{F_v}, \operatorname{ad}\rho) - h^0(\Gamma_{F_v}, \operatorname{ad}\rho)\right) + 1 - [F^+\colon \mathbb{Q}]n^2.$$

Now, when $\mathrm{WD}(\rho_v)$ is generic, the sum over $v \in S_p$ above equals the sum of total multiplicities of the negative Hodge-Tate weights of $\operatorname{ad}\rho$ [BK90, Cor. 3.8.4], and by assumption (3) this number is $[F^+\colon \mathbb{Q}]n(n-1)/2$. Since $l_0 = [F^+\colon \mathbb{Q}]n - 1$, the result follows. □

2.4. **Poitou–Tate duality and local conditions.** In this section, we construct complexes with local conditions whose cohomology are given by Bloch–Kato Selmer groups. These will be used later on to relate Bloch–Kato Selmer groups to derived deformation rings. In the setting of mod p derived deformations, such complexes are constructed in [GV18, §B] (the idea behind the construction was known before, see e.g. [Nek06, §6]). In fact, the authors give a more general construction than we shall describe here; we restrict to the case where the local conditions come from a Selmer system in degree 1. The arguments are essentially the same as those of *loc.cit.*; we include details for the reader's convenience.

Throughout this section, we assume that $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ is potentially semistable at all places above p. We let $C^\bullet(\Gamma_{F,S}, \mathrm{ad}\,\rho)$ denote the usual complex of continuous cochains, whose cohomology is $H^*(\Gamma_{F,S}, \mathrm{ad}\,\rho)$, and similarly for Γ_{F_v} and $\mathrm{ad}\,\rho(1)$. Given a Selmer system $\mathcal{L} = (\mathcal{L}_v)_{v\in S}$, i.e. a collection of subspaces

$$\mathcal{L}_v \subseteq H^1(\Gamma_{F_v}, \mathrm{ad}\,\rho),$$

we let $\mathcal{L}_v^\perp \subseteq H^1(\Gamma_{F_v}, \mathrm{ad}\,\rho(1))$ be the orthogonal complement of $\mathcal{L}_v$ with respect to the cup product, and $\mathcal{L}^\perp = (\mathcal{L}_v^\perp)_{v\in S}$. As in [GV18, §B.3], the Selmer system $\mathcal{L}$ defines subcomplexes

$$C^\bullet_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) \subseteq C^\bullet(\Gamma_{F_v}, \mathrm{ad}\,\rho)$$

for $v \in S$, where

- $C^0_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) = C^0(\Gamma_{F_v}, \mathrm{ad}\,\rho)$.
- $C^1_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) = \ker\big(Z^1(\Gamma_{F_v}, \mathrm{ad}\,\rho) \to H^1(\Gamma_{F_v}, \mathrm{ad}\,\rho)/\mathcal{L}_v\big)$.
- $C^j_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) = 0$, for $j \geq 2$.

Here, $Z^1(\Gamma_{F_v}, \mathrm{ad}\,\rho) \subset C^1(\Gamma_{F_v}, \mathrm{ad}\,\rho)$ denotes the cocycles. We define

$$C^\bullet_{\mathcal{L}}(\Gamma_{F,S}, \mathrm{ad}\,\rho) = \mathrm{fib}\left(C^\bullet(\Gamma_{F,S}, \mathrm{ad}\,\rho) \to \bigoplus_{v\in S} \frac{C^\bullet(\Gamma_{F_v}, \mathrm{ad}\,\rho)}{C^\bullet_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho)}\right),$$

i.e. the cone given in degree i by

$$C^i_{\mathcal{L}}(\Gamma_{F,S}, \mathrm{ad}\,\rho) = C^i(\Gamma_{F,S}, \mathrm{ad}\,\rho) \oplus \bigoplus_{v\in S} \frac{C^{i-1}(\Gamma_{F_v}, \mathrm{ad}\,\rho)}{C^{i-1}_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho)}$$

whose cohomology we denote by

$$H^*_{\mathcal{L}}(\Gamma_{F,S}, \mathrm{ad}\,\rho) = H^*\big(C^\bullet_{\mathcal{L}}(\Gamma_{F,S}, \mathrm{ad}\,\rho)\big).$$

Note that the degree-wise quotient $C^\bullet(\Gamma_{F_v}, \mathrm{ad}\,\rho)/C^\bullet_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho)$ is the complex

$$0 \to C^1(\Gamma_{F_v}, \mathrm{ad}\,\rho)/C^1_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) \to C^2(\Gamma_{F_v}, \mathrm{ad}\,\rho) \to \dots,$$

whose cohomology groups are given by

$$0, \quad H^1(\Gamma_{F_v}, \mathrm{ad}\,\rho)/\mathcal{L}_v, \quad H^2(\Gamma_{F_v}, \mathrm{ad}\,\rho).$$

It follows that we have a long exact sequence

$$\begin{aligned}
0 &\longrightarrow H^0_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow H^0(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow 0 \\
&\longrightarrow H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow H^1(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow \textstyle\bigoplus_{v\in S} H^1(\Gamma_{F_v}, \operatorname{ad}\rho)/\mathcal{L}_v \\
&\longrightarrow H^2_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow H^2(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow \textstyle\bigoplus_{v\in S} H^2(\Gamma_{F_v}, \operatorname{ad}\rho) \\
&\longrightarrow H^3_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho).
\end{aligned}$$

Similarly, we define $H^*_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))$ as the cohomology of the complex

$$C^\bullet_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) = \operatorname{fib}\left(C^\bullet(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to \bigoplus_{v\in S} \frac{C^\bullet(\Gamma_{F_v}, \operatorname{ad}\rho(1))}{C^\bullet_{\mathcal{L}_v^\perp}(\Gamma_{F_v}, \operatorname{ad}\rho(1))}\right),$$

where $C^\bullet_{\mathcal{L}_v^\perp}(\Gamma_{F_v}, \operatorname{ad}\rho(1))$ denotes the complex

$$0 \to C^0(\Gamma_{F_v}, \operatorname{ad}\rho(1)) \to \ker\left(Z^1(\Gamma_{F_v}, \operatorname{ad}\rho) \to H^1(\Gamma_{F_v}, \operatorname{ad}\rho)/\mathcal{L}_v\right) \to 0 \to \dots.$$

We now prove a duality statement for these cohomology groups. The following theorem is a characteristic 0 version of [GV18, Thm. B.1] (in the special case where the local conditions come from a Selmer system in degree 1).

Theorem 2.4.1. *With notation as above, there is a natural perfect pairing*

$$H^i_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \times H^{3-i}_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to E.$$

Proof. The case $i = 0$ follows from Poitou–Tate duality: we have $H^0_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong H^0(\Gamma_{F,S}, \operatorname{ad}\rho)$, and, by Poitou–Tate duality,

$$H^3_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \cong \operatorname{coker}\left(H^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to \bigoplus_{v\in S} H^2(\Gamma_{F_v}, \operatorname{ad}\rho(1))\right) \cong H^0(\Gamma_{F,S}, \operatorname{ad}\rho)^\vee,$$

see e.g. [NSW00, §6, (8.6.10)] for a statement with torsion coefficients, from which the characteristic 0 statement follows from passage to the limit. It remains to verify the statement for $i = 1$, i.e. to construct a perfect pairing

$$H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \times H^2_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to E.$$

We follow the argument of [GV18, §B.5]. By the long exact sequence above,

$$H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong \ker\left(Z^1(\Gamma_{F,S}, \operatorname{ad}\rho) \to \prod_{v\in S} H^1(\Gamma_{F_v}, \operatorname{ad}\rho)/\mathcal{L}_v\right),$$

whereas an element of $H^2_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))$ is represented by a tuple

$$(x', (y'_v)_{v\in S}) \in Z^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \times \prod_{v\in S} C^1(\Gamma_{F_v}, \operatorname{ad}\rho(1))$$

such that $x'_v + \mathrm{d}y'_v \in \mathcal{L}_v^\perp$ for every $v \in S$, where $(\cdot)_v$ denotes the restriction map at v, and d is the differential. Given x and $(x', (y'_v)_{v\in S})$ as above, we choose an arbitrary $z \in C^2(\Gamma_{F,S}, E(1))$ such that $\mathrm{d}z = x \smile x'$ and obtain elements

$$-x_v \smile y'_v + z_v \in C^2(\Gamma_{F_v}, E(1)).$$

One verifies that this expression is closed, and hence we obtain elements of $H^2(\Gamma_{F_v}, E(1)) \cong E$. Taking the sum over v produces an element of E, which we define to be $\langle x, (x', (y'_v)_{v\in S})\rangle$.

This pairing is independent of choice of z since $\sum[z_v] = 0$. Furthermore, replacing y'_v by $y'_v + w'_v$ for some $w'_v \in C^1_{\mathcal{L}^\perp}(\Gamma_{F_v}, \operatorname{ad}\rho(1))$ changes the local contribution at v by $x_v \smile w'_v$, which is cohomologically trivial since x_v and w'_v define orthogonal classes. Lastly, if we replace x by $x - \mathrm{d}a$, one may choose $z - a \smile x'$ in place of z, and similarly for x'. Thus, this is a well-defined pairing and it remains to prove that it is non-degenerate.

To see this, we consider the following two-part filtrations:

$$Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) \coloneqq \{[x] \in H^1(\Gamma_{F,S}, \operatorname{ad}\rho) \mid \forall v \in S\colon [x_v] = 0\,\} \subseteq H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho),$$
$$Q^1_{\mathcal{L}^\perp} \coloneqq \operatorname{coker}\left(H^1(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to \prod_{v\in S} H^1(\Gamma_{F,S}, \operatorname{ad}\rho(1))/\mathcal{L}_v^\perp\right) \subseteq H^1_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)).$$

The subspace $Q^1_{\mathcal{L}^\perp}$ is identified with the classes represented by tuples $(x, (y'_v)_{v\in S})$ such that

$$x'_v = x'_v + \mathrm{d}y'_v = 0$$

for all $v \in S$. It follows that the pairing respects this filtration; i.e. the restriction

$$Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) \times Q^1 \to E$$

is identically 0. Moreover, we have an isomorphism

$$H^1_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))/Q^1 \cong Ш^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)).$$

The problem thus reduces to verifying that the induced pairings

$$Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) \times Ш^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to E$$
$$H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho)/Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) \times Q^1_{\mathcal{L}^\perp} \to E$$

In the first case, the obtained pairing is the Tate pairing ([NSW00, Chap. VIII, Thm. 8.6.7]). In the second case, we choose a representative $(x', (y'_v)_{v\in S}) = (0, (y'_v)_{v\in S})$ for the right-hand class such that $\mathrm{d}y'_v = 0$, and $z = 0$. The local contribution at v of the pairing reduces to $-x_v \smile y'_v$, which is the usual local pairing between in degree 1 (up to a minus sign). □

We record the following special case which will be referenced later in Corollary 3.7.9.

Corollary 2.4.2. *Let* $\mathcal{L} = \left(H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho)\right)_{v\in S_p} \cup \left(H^1(\Gamma_{F_v}, \operatorname{ad}\rho)\right)_{v\in S\setminus S_p}$*, and suppose* $\mathrm{WD}(\rho_v)$ *is generic for* $v \in S_p$*. Then, with notation as above, there is a natural isomorphism*

$$H^i_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong H^{3-i}_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))^\vee.$$

In particular, $H^2_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1))^\vee$.

Proof. The first statement follows from Theorem 2.4.1. Now, if $v \in S_p$ and $\mathrm{WD}(\rho_v)$ is generic, the inclusion

$$H^1_f(\Gamma_{F,S}, \operatorname{ad}\rho) \subseteq H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho)$$

is an equality by [All16, Lem. 1.1.7], where H^1_f denotes the Bloch–Kato Selmer group defined using the crystalline period ring B_{cris} in place of B_{dR}. Since the orthogonal of $H^1_f(\Gamma_{F,S}, \operatorname{ad}\rho)$ is $H^1_f(\Gamma_{F,S}, \operatorname{ad}\rho(1))$ (see [BK90, Prop. 3.8]), the second statement follows. □

3. Derived deformation theory

3.1. **Artinian algebras.** In this section, we recall the notion of artinian animated algebras from [Lur04].

Definition 3.1.1. The category $\mathcal{A}rt_k$ (resp. $\mathcal{A}rt_E$) of *artinian animated $\mathcal{O}$-algebras over k* (resp. over E) is defined as the sub-∞-category of $(\mathcal{A}lg_{\mathcal{O}/})_{/k}$ (resp. $(\mathcal{A}lg_{\mathcal{O}/})_{/E}$) spanned by $(B \to k)$ (resp. $(B \to E)$) such that

(1) $\pi_0(B)$ is an artinian ring with residue field k (resp. E).
(2) $\pi_*(B)$ is a finitely generated $\pi_0(B)$-module.

Any artinian animated $\mathcal{O}$-algebra is truncated. Note that if $B \in \mathcal{A}rt_E$ then $\pi_0(B)$ is an E-algebra and $\pi_*(B)$ is a finite-dimensional E-vector space.

Proposition 3.1.2. *Let $A \in \mathcal{CNL}_k$ and $B \in \mathcal{A}rt_k$. Then, for any ideal $I \subset \pi_0(A)$, the natural morphism*

$$\operatorname*{colim}_s \operatorname{Map}_{/k}(A/^{\mathbf{L}}I^s, B) \to \operatorname{Map}_{/k}(A, B)$$

is an equivalence.

Proof. The maximal ideal of $\pi_0(B)$ is nilpotent, hence given any map $A \to B$, the induced morphism $\pi_0(A) \to \pi_0(B)$ factors through a power of $I \subset \pi_0(A)$. We conclude using Proposition 2.2.4. □

A useful description of artinian algebras allowing for proofs by induction is given in terms of small extensions.

Definition 3.1.3. [Lur04, Def. 3.3.1] Let $B \in \mathcal{A}lg_{\mathcal{O}/}$ and $M \in \mathcal{M}od_B$ a connective B-module. A *small extension* of B by M is a pullback

$$\begin{array}{ccc} B' & \longrightarrow & B \\ \downarrow & & \downarrow{\scriptstyle s} \\ B & \xrightarrow[(\mathrm{id},0)]{} & B \oplus M[1] \end{array}$$

where $B \oplus M[1]$ is the trivial square-zero extension (see below [Lur04, Prop. 3.1.5]) and s is a section of the projection $B \oplus M[1] \to B$.

Lemma 3.1.4. [Lur04, Lem. 6.2.6] *Let $(B \to E) \in \mathcal{A}rt_E$. Then there exists a factorisation*

$$B \simeq B_m \to B_{m-1} \to \cdots \to B_0 \simeq E$$

in $\mathcal{A}rt_E$ such that B_j is a small extension of B_{j-1} by $E[n_j]$ for an increasing sequence $n_j \geq 0$ and $\dim_E(\pi_*(B_j)) = j + 1$. The analogous statement is true for $(B \to k) \in \mathcal{A}rt_k$.

Lemma 3.1.5. *If $\mathcal{R} \in \mathcal{CNL}_k$ and $B \in \mathcal{A}rt_k$, then $\operatorname{Map}_{/k}(\mathcal{R}, B)$ has finitely many connected components and finite homotopy groups.*

Proof. We argue by induction on m, the length of a series of small extensions for B as in Lemma 3.1.4. For $m = 0$, we have $B \simeq k$, and $\operatorname{Map}_{/k}(\mathcal{R}, k)$ is contractible. For the induction step, suppose the claim holds for every $B' \in \mathcal{A}rt_k$ given by series of length $m' \leq m$. By Lemma 3.1.4, there is an algebra $B' \in \mathcal{A}rt_k$ and an equivalence

$$B \simeq B' \times_{B' \oplus k[n+1]} B'$$

for some $n \geq 0$, where B' is given by a series of length $m' < m$. Applying $\mathrm{Map}_{/k}(\mathcal{R}, -)$ gives a pullback

$$\mathrm{Map}_{/k}(\mathcal{R}, B) \simeq \mathrm{Map}_{/k}(\mathcal{R}, B') \times_{\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1])} \mathrm{Map}_{/k}(\mathcal{R}, B')$$

and the associated long exact sequence of homotopy groups

$$\cdots \to \pi_{i+1}\big(\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1])\big) \to \pi_i\big(\mathrm{Map}_{/k}(\mathcal{R}, B)\big) \to \big(\pi_i\big(\mathrm{Map}_{/k}(\mathcal{R}, B')\big)\big)^2 \to \cdots.$$

By the induction hypothesis, $\pi_i\big(\mathrm{Map}_{/k}(\mathcal{R}, B')\big)$ is finite. To complete the proof, it therefore suffices to prove the same statement for $\pi_{i+1}\big(\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1])\big)$. Now, we have a map

$$\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1]) \longrightarrow \mathrm{Map}_{/k}(\mathcal{R}, B')$$

induced by the projection $B' \oplus k[n+1] \to B'$, whose fibre over a map f equals

$$\mathrm{Map}_{/B'}(\mathcal{R}, B' \oplus k[n+1]) \simeq \mathrm{Map}_{/k}(\mathcal{R}, k \oplus k[n+1]).$$

The homotopy groups of the fibre are

$$\pi_i \mathrm{Map}_{/k}(\mathcal{R}, k \oplus k[n+1]) \cong \mathrm{T}_k^{n+1-i}(\mathcal{R}),$$

which is finite by Proposition 3.6.4 below and the noetherianity assumption on $\mathcal{R}$. The corresponding fibre sequence therefore implies the sought statement, which completes the induction step. □

3.2. **Integral models of artinian E-algebras.** In this section, we introduce a notion of integral models of artinian E-algebras. The definition is made in analogy with [Kis09, 2.3.5] and will be used in Section 3.7. Note that we have an equivalence of ∞-categories

$$((\mathcal{A}lg_{\mathcal{O}/})_{/E})_{/(B \to E)} \simeq (\mathcal{A}lg_{\mathcal{O}/})_{/B}.$$

Definition 3.2.1. Let $(B \to E) \in \mathcal{A}rt_E$. The ∞-category $\mathcal{I}ntB$ of *integral models of* B is defined as the sub-∞-category of $(\mathcal{A}lg_{\mathcal{O}/})_{/B}$ spanned by all $(A \to B)$ such that:

(1) $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module.
(2) The induced map $\pi_*(A) \to \pi_*(B)$ is injective.
(3) The induced map $A \otimes^{\mathbf{L}}_{\mathcal{O}} E \to B \otimes^{\mathbf{L}}_{\mathcal{O}} E$ is a weak equivalence.

By Proposition 2.2.3(2) and Definition 3.2.1(1), any $A \in \mathcal{I}ntB$ is derived ϖ-adically complete and almost of finite presentation over $\mathcal{O}$. When B is clear from context, we sometimes denote objects of $\mathcal{I}ntB$ simply by A. The key property of integral models we shall need is the following.

Proposition 3.2.2. *For any $B \in \mathcal{A}rt_E$, the ∞-category $\mathcal{I}ntB$ is filtered.*

Before embarking on the proof of Proposition 3.2.2, we establish some preliminary results about integral models.

Lemma 3.2.3. *Let $(B \to E) \in \mathcal{A}rt_E$ and $(A \to B) \in \mathcal{I}ntB$. The natural map*

$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B \times_E \mathcal{O}) \to \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B)$$

is a weak equivalence.

Proof. Since $\mathcal{O}$ and E are static and $\pi_0(A)$ is a finitely generated $\mathcal{O}$-module, we have equivalences of mapping spaces

$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, E) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(\pi_0(A), E) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(\pi_0(A), \mathcal{O}) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, \mathcal{O}).$$

Consequently,

$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B \times_E \mathcal{O}) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B) \times_{\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A,E)} \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, \mathcal{O}) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B).$$

□

To prove Proposition 3.2.2, we will show that integral models can be constructed via small extensions, in the following sense.

Lemma 3.2.4. *Let $B \in \mathcal{A}rt_E$ be given by a sequence of small extensions*

$$B \simeq B_m \to B_{m-1} \to \cdots \to B_0 \simeq E$$

as in Lemma 3.1.4. Then, for any $A \in \mathcal{I}ntB$, the morphism $A \to \mathcal{O}$ factors as

$$A \simeq A_m \to A_{m-1} \to \cdots \to A_0 \simeq \mathcal{O},$$

where, for every $j \geq 1$, $A_j \in \mathcal{I}ntB_j$ is a small extension of A_{j-1} by $\varpi^{-r_j}\mathcal{O}[n_j]$ for a non-decreasing sequence $0 \leq n_1 \leq \cdots \leq n_m$ and some $r_j \geq 0$.

Before embarking on the proof of the Lemma, let us describe the homotopy groups of a small extension. Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ be n-truncated and suppose $M \in \mathcal{M}od_A$ has homotopy concentrated in degree $n-1$, so that $M[1]$ has homotopy concentrated in degree n. Given a small extension $A' = A \times_{A \oplus M[1]} A$, we obtain a long exact sequence

$$\cdots \to \pi_{i+1}(A) \to \pi_{i-1}(M) \to \pi_i(A') \to \pi_i(A) \to \pi_{i-2}(M) \to \dots$$

which implies that $\pi_i(A') \cong \pi_i(A)$ for every $i \neq n$ and that

$$\pi_{n-1}(M) \cong \ker(\pi_n(A') \to \pi_n(A)).$$

Thus, a single small extension of the form considered in Lemma 3.2.4 only changes the homotopy groups in the top degree by adding a module which is free of rank 1 over $\mathcal{O}$.

Proof of Lemma 3.2.4. Suppose we have a factorisation $B \simeq B_m \to \cdots \to B_0$. We prove the lemma by induction on m. The case $m = 0$ is trivial since $\mathcal{I}ntE \simeq \{\mathcal{O}\}$. For the induction step, suppose the statement holds for any length $m-1$ sequence $B'_{m-1} \to \cdots \to B'_0$. Let $n = n_m$ denote the largest integer such that $\pi_n(B) \neq 0$, and let $A \in \mathcal{I}ntB_m$. The kernel

$$I = \ker\left(\pi_*(A) \hookrightarrow \pi_*(B_m) \to \pi_*(B_{m-1})\right) \subset \pi_n(A)$$

is a $\pi_0(A)$-submodule of $\pi_*(A)$, which is free of rank 1 as an $\mathcal{O}$-module. Using [Lur04, Prop. 3.3.3, Prop. 3.3.5], we construct a morphism $A \to A_{m-1}$ in $\mathcal{A}lg_{\mathcal{O}/}$, where A_{m-1} ('A/I' in the notation of *loc. cit.*) satisfies the following properties:

(1) For any $Y \in \mathcal{A}lg_{\mathcal{O}/}$, the map $A \to A_{m-1}$ induces an identification

$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A_{m-1}, Y) \hookrightarrow \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, Y)$$

with the subspace spanned by maps $A \to Y$ for which $I \subset \ker(\pi_*(A) \to \pi_*(Y))$.

(2) The induced map $\pi_*(A) \to \pi_*(A_{m-1})$ is an isomorphism in degree $i \neq n$ and for $i = n$ is the map $\pi_n(A) \to \pi_n(A)/I$.

(3) A is a small extension of A_{m-1} by $I[n]$, i.e. there is an equivalence

$$A \simeq A_m = A_{m-1} \times_{A_{m-1} \oplus I[n+1]} A_{m-1}.$$

From (1)-(3), we deduce that the map $A_{m-1} \to B_{m-1}$ defines an object of $\mathcal{I}ntB_{m-1}$ and that the square

$$\begin{array}{ccc} A & \longrightarrow & A_{m-1} \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_{m-1} \end{array}$$

commutes up to homotopy. By the induction hypothesis, $A_{m-1} \to \mathcal{O}$ factors as a series of small extensions of the required form, and we are done. □

Lemma 3.2.5. *Let $B \in \mathcal{A}rt_E$. Then $B \times_E \mathcal{O}$ is equivalent in $(\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$ to a filtered colimit of algebras of the form of A_m in Lemma 3.2.4.*

Proof. To prove the second claim, we use induction on m again. The case $m = 0$ is trivial since $E \times_E \mathcal{O} \simeq E0$. Suppose the statement is true for any B' given by a length $m-1$ sequence of small extensions. Let $n = n_m + 1$ and note that

$$E[n] = \operatorname*{colim}_{r \geq 0} \varpi^{-r}\mathcal{O}[n].$$

Since filtered colimits commute with finite limits, under the induction hypothesis we have

$$\begin{aligned} B_m \times_E \mathcal{O} &\simeq (B_{m-1} \times_{B_{m-1} \oplus E[n]} B_{m-1}) \times_E \mathcal{O} \\ &\simeq (B_{m-1} \times_E \mathcal{O}) \times_{(B_{m-1} \times_E \mathcal{O}) \oplus E[n]} (B_{m-1} \times_E \mathcal{O}) \\ &\simeq \operatorname*{colim}_{A_{m-1}} (A_{m-1} \times_{A_{m-1} \oplus E[n]} A_{m-1}) \end{aligned}$$

where the colimit runs over a filtered diagram of $A_{m-1} \in \mathcal{I}ntB_{m-1}$ of the sought form. We have equivalences

$$\begin{aligned} A_{m-1} \oplus E[n] &\simeq A_{m-1} \oplus \operatorname*{colim}_{r \geq 0} \varpi^{-r}\mathcal{O}[n] \\ &\simeq \operatorname*{colim}_{r \geq 0} A_{m-1} \oplus \varpi^{-r}\mathcal{O}[n]. \end{aligned}$$

Indeed, A_{m-1} preserves $\mathcal{O}[n]$ inside $E[n]$, since if $\mathcal{E}nd$ denotes the derived endomorphism ring, we have

$$A_{m-1} \to \mathcal{E}nd_{\mathcal{O}}(E[n]) \simeq \pi_0\big(\mathcal{E}nd_{\mathcal{O}}(E)\big) \cong E,$$

and hence the action map $A_{m-1} \to E$ factors through $\pi_0(A_{m-1})$ and lands in $\mathcal{O}$. It follows that

$$\operatorname*{colim}_{A_{m-1}} (A_{m-1} \times_{A_{m-1} \oplus E[n]} A_{m-1}) \cong \operatorname*{colim}_{A_{m-1}} \operatorname*{colim}_{r \geq 0} (A_{m-1} \times_{A_{m-1} \oplus \varpi^{-r}\mathcal{O}[n]} A_{m-1}),$$

and thus B_m is a filtered colimit of the sought form. □

We are now in a position to prove the main result of this section, Proposition 3.2.2.

Proof of Proposition 3.2.2. By the local characterisation of filtered ∞-categories [Lur18, Tag 02PS], it suffices to prove the following statement: Given $A, A' \in \mathcal{I}ntB$ and a morphism of anima

$$\gamma \colon \partial\Delta^n \to \mathrm{Map}_{/B}(A, A')$$

for some $n \geq 0$, there exists an $A'' \in \mathcal{I}ntB$ and a morphism $g \colon A' \to A''$ such that the composition

$$g \circ \gamma \colon \partial\Delta^n \to \mathrm{Map}_{/B}(A, A') \to \mathrm{Map}_{/B}(A, A'')$$

is null-homotopic. In other words, we would like to define the dotted arrow in the diagram

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{\gamma} & \mathrm{Map}_{/B}(A, A') \\ \downarrow & & \downarrow{\scriptstyle g_*} \\ \Delta^n & \dashrightarrow & \mathrm{Map}_{/B}(A, A'') \end{array}$$

Since B is final in $(\mathcal{A}lg_{\mathcal{O}/})_{/B}$ and by Lemma 3.2.5, we have

$$\{*\} \simeq \mathrm{Map}_{/B}(A, B) \simeq \mathrm{Map}_{/B}(A, B \times_E \mathcal{O}) \simeq \mathrm{Map}_{/B}(A, \operatorname*{colim}_{A''} A''),$$

where A'' runs over a filtered subcategory of $\mathcal{I}ntB$. Thus, the composition

$$\partial\Delta^n \to \mathrm{Map}_{/B}(A, A') \to \mathrm{Map}_{/B}(A, \operatorname*{colim}_{A''} A'')$$

is null-homotopic. Now, since $A' \in \mathcal{I}ntB$ is a finitely presented $\mathcal{O}$-algebra by Proposition 2.2.3, the map $A' \to \operatorname{colim} A''$ factors through some A''. It follows that there exists a $g\colon A' \to A''$ such that the composition $g \circ \gamma$ is null-homotopic. □

Lemma 3.2.6. *Let $(B \to E) \in \mathcal{A}rt_E$. We have the following:*

(1) *There is an equivalence $\operatorname*{colim}_{A \in \mathcal{I}ntB} A \to B \times_E \mathcal{O}$.*
(2) *There is a canonical isomorphism of graded rings $\pi_*(B \times_E \mathcal{O}) \cong \pi_*(B) \times_E \mathcal{O}$.*
(3) *Any $A \in \mathcal{I}ntB$ is derived ϖ-adically complete.*
(4) *For any $A \in \mathcal{I}ntB$ and $r \geq 1$, $(A/^{\mathbf{L}}\varpi^r \to k) \in \mathcal{A}rt_k$.*

Proof. (1) By Lemma 3.2.3, any $(A \to B) \in \mathcal{I}ntB$ defines a map $A \to B \times_E \mathcal{O}$. Taking the colimit over $\mathcal{I}ntB$, we produce

$$\operatorname*{colim}_{A \in \mathcal{I}ntB} A \to B \times_E \mathcal{O},$$

and we claim that this map is a weak equivalence. Since $\mathcal{I}ntB$ is filtered (Proposition 3.2.2) and filtered colimits commute with π_*, it is enough to prove that the map

$$\operatorname*{colim}_{A \in \mathcal{I}ntB} \pi_*(A) \to \pi_*(B \times_E \mathcal{O})$$

is an isomorphism. Since $\pi_*(A) \to \pi_*(B)$ is injective for every $A \in \mathcal{I}ntB$ (Definition 3.2.1(3)), injectivity follows. To see that the map is surjective, we use the isomorphism provided by Lemma 3.2.5,

$$\pi_*(B \times_E \mathcal{O}) \cong \operatorname*{colim}_{A''} \pi_*(A''),$$

a filtered colimit over objects $A'' \in \mathcal{I}ntB$. Surjectivity follows.

(2) There is a Mayer-Vietoris sequence

$$\cdots \to \pi_n(B \times_E \mathcal{O}) \to \pi_n(B) \times \pi_n(\mathcal{O}) \to \pi_n(E) \to \ldots,$$

and hence for every $n \geq 1$, we have $\pi_n(B \times_E \mathcal{O}) \cong \pi_n(B)$ since $\pi_n(\mathcal{O}) = \pi_n(E) = 0$. For $n = 0$, the sequence implies $\pi_0(B \times_E \mathcal{O}) \cong \ker(\pi_0(B) \times \mathcal{O} \to E) \cong \pi_0(B) \times_E \mathcal{O}$.

(3) Since $A \in \mathcal{I}ntB$, $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module and hence ϖ-adically complete. Thus, A is derived ϖ-adically complete by Lemma 2.2.7.

(4) Let $A \in \mathcal{I}ntB$ and $r \geq 1$. From Quillen's spectral sequence [Qui67, Thm. 2.6]

$$E^2_{i,j}\colon \operatorname{Tor}_i^{\mathcal{O}[t]}\left(\pi_j(A), \mathcal{O}\right) \implies \pi_{i+j}(A/^{\mathbf{L}}\varpi^r),$$

we deduce that $\pi_0(A/^{\mathbf{L}}\varpi^r) \cong \pi_0(A)/\varpi^r$, which is artinian since $\pi_0(A)$ is finitely generated and free as an $\mathcal{O}$-module. Moreover, for any $n \geq 1$, the spectral sequence induces a short exact sequence

$$0 \to \pi_n(A)/\varpi^r \to \pi_n(A/^{\mathbf{L}}\varpi^r) \to \pi_{n-1}(A)[\varpi^r] \to 0.$$

Since A is truncated, $\pi_*(A/^{\mathbf{L}}\varpi^r)$ is finitely generated over $\pi_0(A)$, hence over $\pi_0(A/^{\mathbf{L}}\varpi^r)$. □

Lemma 3.2.7. *Let $(A \to B) \in \mathcal{I}ntB$. Then the map $A \to E$ factors through $\mathcal{O} \to E$, and we have a colimit-reflecting lifting*

$$\begin{array}{ccc} \mathcal{I}ntB & \dashrightarrow & (\mathcal{A}lg_{\mathcal{O}/})_{/O} \\ \downarrow & & \downarrow \\ (\mathcal{A}lg_{\mathcal{O}/})_{/B} & \longrightarrow & (\mathcal{A}lg_{\mathcal{O}/})_{/E}. \end{array}$$

Proof. The functor $\mathcal{I}ntB \to (\mathcal{A}lg_{\mathcal{O}/})_{/B}$ factors through

$$(\mathcal{A}lg_{\mathcal{O}/})_{/\operatorname{colim}_{A \in \mathcal{I}ntB} A} \simeq (\mathcal{A}lg_{\mathcal{O}/})_{/B \times_E \mathcal{O}} \simeq (\mathcal{A}lg_{\mathcal{O}/})_{/B} \times_{(\mathcal{A}lg_{\mathcal{O}/})_{/E}} (\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}},$$

where we have used Lemma 3.2.6(1) in the first equivalence. Thus, we obtain the sought lifting. Reflection of colimits follows from the fact that colimits in an over-∞-category $\mathcal{C}_{/X}$ are computed in $\mathcal{C}$ [Lur09, Prop. 1.2.13.8]. □

3.3. **Topologies on animated algebras.** In this section, we discuss the *ind-ϖ-adic topology* in the derived setting which we, following [Zhu25, §2.4], will use to define derived moduli stacks of continuous representations. Let $\mathrm{Set}^{\mathrm{cp}}$ denote the category of finite sets. The starting point is the observation that any static $\mathcal{O}$-module M defines an ind-profinite set, also denoted M:

$$M = \{\{M'/\varpi^r\}_{r \geq 1}\}_{M' \subset M} \in \mathrm{Ind}(\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}})),$$

where M' runs over the filtered category of finitely generated submodules of M. In particular, if M is finitely generated then its associated ind-profinite set is isomorphic to a profinite set in the subcategory $\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}}) \subset \mathrm{Ind}(\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}}))$. The following result of Zhu allows us to generalise these observations from static $\mathcal{O}$-modules to animated $\mathcal{O}$-modules, thus encoding a notion of ind-ϖ-adic topology in the derived setting.

Lemma 3.3.1. [Zhu25, Lem. 2.53] *Let Γ be a profinite set and define*

$$\begin{aligned} C_{\mathrm{sc}}(\Gamma, -)\colon \mathrm{Mod}_{\mathcal{O}} &\to \mathrm{Mod}_{\mathcal{O}} \\ M &\mapsto \mathrm{Map}_{\mathrm{Ind}(\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}}))}(\Gamma, M). \end{aligned}$$

Then $C_{\mathrm{sc}}(\Gamma, -)$ is exact and lax symmetric monoidal and therefore extends to a t-exact functor $\mathcal{M}od_{\mathcal{O}} \to \mathcal{M}od_{\mathcal{O}}$ which preserves colimits. Moreover, $C_{\mathrm{sc}}(\Gamma, -)$ lifts to a functor $\mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}lg_{\mathcal{O}/}$ which preserves finite limits and sifted colimits.

Lemma 3.3.2. *Let $A \in \mathcal{M}od_{\mathcal{O}}$ and suppose $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module. Then there is a weak equivalence*

$$C_{\mathrm{sc}}(\Gamma, A) \simeq \lim_{r \geq 1} C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r)$$

which respects the monoidal structure. Hence, if $A \in \mathcal{A}lg_{\mathcal{O}/}$ then the equivalence holds in $\mathcal{A}lg_{\mathcal{O}/}$.

Proof. For any $A \in \mathcal{M}od_{\mathcal{O}} \simeq D^{\geq 0}(\mathcal{O})$, we have natural morphisms (of $\mathcal{O}$-algebras, if $A \in \mathcal{A}lg_{\mathcal{O}/}$)

$$C_{\mathrm{sc}}(\Gamma, A) \to C_{\mathrm{sc}}(\Gamma, \lim A/^{\mathbf{L}}\varpi^r) \to \lim C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r).$$

If A is derived ϖ-adically complete – in particular, if $\pi_*(A)$ is finitely generated over $\mathcal{O}$ (Lemma 2.2.7) – the first map is an equivalence. Let $P(A)$ be the property that the second map is a weak equivalence. Then $P(\mathcal{O}[0])$ holds, $P(A)$ implies $P(A[1])$ and for any fibre sequence $A \to B \to C \to A[1]$, if P holds for two of A, B, C then it holds for the third by the five lemma. Moreover, for any A and B, $P(A \oplus B)$ is implied by $P(A)$ and $P(B)$ since $C_{\mathrm{sc}}(\Gamma, -)$ commutes with colimits (Lemma 3.3.1). Finally, any retract $A \to B \to A$ induces a commutative diagram

$$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & A \\ \downarrow & & \downarrow & & \downarrow \\ \lim_r C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r) & \longrightarrow & \lim_r C_{\mathrm{sc}}(\Gamma, B/^{\mathbf{L}}\varpi^r) & \longrightarrow & \lim_r C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r) \end{array}$$

where the composition of horizontal arrows is the identity, so that if $P(B)$ holds then the induced map $\pi_*(A) \to \pi_*(\lim_r C_{\mathrm{sc}}(\Gamma, B/^{\mathbf{L}}\varpi^r))$ is both injective and surjective, i.e. $P(A)$ holds. It follows that $P(A)$ holds for any perfect animated $\mathcal{O}$-module, since the category $\mathcal{D}_{\mathrm{perf}}(\mathcal{O})$ of perfect complexes (i.e. complexes C with finitely generated homology $H_*(C)$) is the smallest strictly full, saturated, triangulated subcategory of $\mathcal{D}(\mathcal{O})$ containing $\mathcal{O}[0]$ ([Sta23, Tag 0ATI]). □

3.4. **Moduli stacks of Galois representations.** In this section, we discuss moduli stacks of representations of a profinite group Γ valued in a smooth affine group scheme $\mathbf{G}/\mathcal{O}$ of finite type, following [Zhu25, §2.4]. We only consider what is referred to as *strongly continuous representations* in *loc. cit.*

For $m \in \mathbb{N}$, we let $\mathcal{O}[\mathbf{G}^m]$ denote the coordinate ring of $\mathbf{G}^m$. With boundary maps defined using the group law on $\mathbf{G}$ and degeneracies defined by inclusions $\mathbf{G}^{m_1} \times \{\mathrm{id}\} \times \mathbf{G}^{m_2} \to \mathbf{G}^{m_1+m_2+1}$, we obtain a cosimplicial animated $\mathcal{O}$-algebra $\mathcal{O}[\mathbf{G}^\bullet] \in \mathcal{F}un(\Delta, \mathcal{A}lg_{\mathcal{O}/})$. Given $A \in \mathcal{A}lg_{\mathcal{O}/}$, there is an equivalence (see [Zhu25, Rem. 2.9]):

$$\mathrm{Map}_{\mathcal{A}ni(\mathrm{Grp})}\big(\Gamma, \mathbf{G}(A)\big) \simeq \mathrm{Map}_{\mathcal{F}un(\Delta, \mathcal{A}lg_{\mathcal{O}/})}\big(\mathcal{O}[\mathbf{G}^\bullet], C_{\mathrm{disc}}(\Gamma^\bullet, A)\big),$$

where for any $[m] \in \Delta$, $C_{\mathrm{disc}}(\Gamma^m, -)$ is the animation of the functor which maps a static $\mathcal{O}$-module M to the set of all functions $\Gamma^m \to M$, cf. the definition of $C_{\mathrm{sc}}(\Gamma^\bullet, -)$ in Lemma 3.3.1. Replacing $C_{\mathrm{disc}}(\Gamma^\bullet, -)$ by $C_{\mathrm{sc}}(\Gamma^\bullet, -)$, we obtain the following definition.

Definition 3.4.1. [Zhu25, Def. 2.56] Let Γ be a profinite group and $\mathbf{G}/\mathcal{O}$ an affine group scheme of finite type. The *derived moduli stack of framed* $\mathbf{G}$*-valued* Γ*-representations* is the functor

$$\begin{aligned} \mathcal{X}^{\square}_{\Gamma,\mathbf{G}} \colon \mathcal{A}lg_{\mathcal{O}/} &\to \mathcal{A}ni \\ A &\mapsto \mathrm{Map}_{\mathcal{F}un(\Delta, \mathcal{A}lg_{\mathcal{O}/})}\big(\mathcal{O}[\mathbf{G}^\bullet], C_{\mathrm{sc}}(\Gamma^\bullet, A)\big). \end{aligned}$$

Lemma 3.4.2. [Zhu25, Lem. 2.57] *If $A \in \mathcal{A}lg_{\mathcal{O}/}$ is static and $\mathbf{G} = \mathrm{GL}_n$, then $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(A)$ is the set of homomorphisms $\rho_A \colon \Gamma \to \mathrm{GL}_n(A)$ such that A^n is a filtered colimit of Γ-stable finitely generated $\mathcal{O}$-modules with continuous Γ-action.*

Lemma 3.4.3. *Let $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}} \colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ be the functor of Definition 3.4.1. Then $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}$ commutes with finite limits, and for every $j \geq 0$, the restriction to j-truncated algebras commutes with filtered colimits. Moreover, if $A \in \mathcal{A}lg_{\mathcal{O}/}$ is an algebra such that $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module, we have an equivalence*

$$\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(\lim_r A/^{\mathbf{L}}\varpi^r) \simeq \lim_r \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(A/^{\mathbf{L}}\varpi^r).$$

Proof. The statement for filtered colimits is contained in [Zhu25, Prop. 2.59]. By definition, $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}$ is given by the composition

$$\mathrm{Hom}_{\mathcal{F}un(\Delta,\mathcal{A}lg_{\mathcal{O}/})}(\mathcal{O}[\mathbf{G}^{\bullet}],-)\circ C_{\mathrm{sc}}(\Gamma,-)\circ\iota,$$

where $\iota\colon \mathcal{A}lg_{\mathcal{O}/}\to\mathcal{F}un(\Delta,\mathcal{A}lg_{\mathcal{O}/})$ is the inclusion of constant cosimplicial objects. Being a right adjoint, ι preserves all limits. The functor $C_{\mathrm{sc}}(\Gamma^{\bullet},-)$ preserves finite limits and limits of the form $\lim A/^{\mathbf{L}}\varpi^{r}$ for A as in the statement by Lemma 3.3.2. Finally, $\mathrm{Hom}_{\mathcal{F}un(\Delta,\mathcal{A}lg_{\mathcal{O}/})}(\mathcal{O}[\mathbf{G}^{\bullet}],-)$ preserves all limits. This completes the proof. □

3.5. **Derived deformation functors.** In this section, we define various derived deformation functors of Galois representations in terms of the stack $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}$ of Definition 3.4.1. From now on, we will only consider the case $\mathbf{G}=\mathrm{GL}_n/\mathcal{O}$.

Definition 3.5.1. Let $\mathcal{X}\colon \mathcal{A}lg_{\mathcal{O}/}\to\mathcal{A}ni$ be a functor and fix $\bar\rho\in\mathcal{X}(k)$. We define

$$\begin{aligned}\mathcal{X}_{\bar\rho}\colon \mathcal{A}rt_k &\to \mathcal{A}ni\\ (B\to k)&\mapsto \mathcal{X}(B)\times_{\mathcal{X}(k)}\{\bar\rho\},\end{aligned}$$

which we extend to $\mathcal{P}ro(\mathcal{A}rt_k)$ by setting $\mathcal{X}_{\bar\rho}(\{A_i\}):=\lim_i\mathcal{X}_{\bar\rho}(A_i)$. Similarly, if $\rho\in\mathcal{X}(E)$, we obtain a functor $\mathcal{X}_{\rho}\colon\mathcal{A}rt_E\to\mathcal{A}ni$ given by $\mathcal{X}(B\to E)=\mathcal{X}(B)\times_{\mathcal{X}(E)}\{\rho\}$.

Definition 3.5.2. Let Γ be a profinite group and suppose $\bar\rho\in\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(k)$. The *unrestricted framed derived deformation functor* of $\bar\rho$ is the functor

$$\mathcal{D}^{\square}_{\bar\rho}:=(\mathcal{X}^{\square}_{\Gamma,\mathbf{G}})_{\bar\rho}\colon\mathcal{A}rt_k\to\mathcal{A}ni.$$

Similarly, if $\rho\in\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(E)$, we let

$$\mathcal{D}^{\square}_{\rho}:=(\mathcal{X}^{\square}_{\Gamma,\mathbf{G}})_{\rho}\colon\mathcal{A}rt_E\to\mathcal{A}ni.$$

Remark 3.5.3. [Zhu25, Eq. (2.6)] The derived deformation functors introduced above for a mod p representation coincide with the ones introduced by Galatius–Venkatesh in [GV18, Def. 5.4].

The functors in Definition 3.5.2 correspond to framed deformations, and give rise to unframed deformation functors by passing to quotients with respect to conjugation. Let $\mathbf{Z}\subset\mathbf{G}$ denote the centre and define $\mathbf{PG}:=\mathbf{G}/\mathbf{Z}\colon\mathcal{A}lg_{\mathcal{O}/}\to\mathcal{A}ni$. We define functors $\widehat{\mathbf{PG}}_k,\widehat{\mathbf{G}}_k\colon\mathcal{A}rt_k\to\mathcal{A}ni$ by

$$\widehat{\mathbf{PG}}_k(B\to k):=\mathbf{PG}(B)\times_{\mathbf{PG}(k)}\{\mathrm{id}\},$$

and similarly for $\widehat{\mathbf{G}}_k$. Similarly, we define $\widehat{\mathbf{G}}_E,\widehat{\mathbf{PG}}_E\colon\mathcal{A}rt_E\to\mathcal{A}ni$. Defined as above, the groups $\widehat{\mathbf{G}}_k$ and $\widehat{\mathbf{PG}}_k$ act on $\mathcal{D}^{\square}_{\bar\rho}$ by conjugation, i.e. there is a simplicial diagram

$$(\widehat{\mathbf{PG}}_k^{\bullet}\times\mathcal{D}^{\square}_{\bar\rho})=\Big(\dots\ \widehat{\mathbf{PG}}_k^{2}\times\mathcal{D}^{\square}_{\bar\rho}\ \substack{\rightrightarrows\\ \rightrightarrows}\ \widehat{\mathbf{PG}}_k\times\mathcal{D}^{\square}_{\bar\rho}\ \rightrightarrows\ \mathcal{D}^{\square}_{\bar\rho}\Big)\in\mathcal{F}un\big(\Delta^{\mathrm{op}},\mathcal{F}un(\mathcal{A}rt_k,\mathcal{A}ni)\big).$$

By taking a colimit, we obtain the functor of unframed deformations of $\bar\rho$.

Definition 3.5.4. With notation as above, the *functor of derived (unframed) deformations of* ρ is defined as the geometric realisation (i.e. colimit)

$$|\widehat{\mathbf{PG}}_k^{\bullet}\times\mathcal{D}^{\square}_{\bar\rho}|:=\operatorname*{colim}_{\Delta^{\mathrm{op}}}\big(\widehat{\mathbf{PG}}_k^{\bullet}\times\mathcal{D}^{\square}_{\bar\rho}\big)\in\mathcal{F}un(\mathcal{A}rt_k,\mathcal{A}ni).$$

The functor $\mathcal{D}_{\rho}$ of derived deformations of a characteristic 0 representation ρ is defined analogously.

Remark 3.5.5. Definition 3.5.4 is modelled after classical Galois deformation theory. An alternative way of defining unframed deformation functors is passing to the quotient at the level of stacks by forming $\mathcal{X}_{\Gamma,\mathbf{G}} \coloneqq |\mathcal{X}^{\square}_{\Gamma,\mathbf{G}} \times \mathbf{PG}^{\bullet}|$ as in [Zhu25, Def. 2.20], then defining $\mathcal{D}_{\bar\rho} \coloneqq (\mathcal{X}_{\Gamma,\mathbf{G}})_{\bar\rho}$ for a point $\bar\rho \in \mathcal{X}_{\Gamma,\mathbf{G}}(k)$. This yields the same functor up to equivalence, which can be proved by considering the natural map $(\mathcal{X}^{\square}_{\Gamma,\mathbf{G}})_{\bar\rho} \to (\mathcal{X}_{\Gamma,\mathbf{G}})_{\bar\rho}$ and comparing tangent complexes (see Section 3.6).

Definition 3.5.6. A (mod p) *derived deformation problem* is a tuple

$$\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$$

where S is a finite set of finite places of F, $\bar\rho \in \mathcal{X}^{\square}_{\Gamma_{F,S},\mathbf{G}}(k)$, and $\{\mathcal{D}_v\}_{v\in S}$ is a set of functors $\mathcal{D}_v \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni)$ equipped with actions by $\widehat{\mathbf{PG}}_k$ and equivariant maps $\mathcal{D}_v \to \mathcal{D}^{\square}_{\bar\rho_v}$. Similarly, a *characteristic 0 derived deformation problem* is a tuple $(\rho, S, \{\mathcal{D}_v\}_{v\in S})$ where $\rho \in \mathcal{X}^{\square}_{\Gamma_{F,S},\mathbf{G}}(E)$ and $\mathcal{D}_v \to \mathcal{D}^{\square}_{\rho_v}$.

Definition 3.5.7. Let $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ be a deformation problem. We define a functor

$$\mathcal{D}^{\square}_{\mathbf{S}} \coloneqq \mathcal{D}^{\square}_{\bar\rho} \times_{\prod_{v\in S}\mathcal{D}^{\square}_{\bar\rho_v}} \prod_{v\in S} \mathcal{D}_v \colon \mathcal{A}rt_k \to \mathcal{A}ni$$

which we call the *functor of framed derived deformations of type* $\mathbf{S}$. Since the morphisms $\mathcal{D}_v \to \mathcal{D}^{\square}_{\bar\rho_v}$ are equivariant under conjugation, we may form the geometric realisation

$$\mathcal{D}_{\mathbf{S}} = |\widehat{\mathbf{PG}}^{\bullet}_k \times \mathcal{D}^{\square}_{\mathbf{S}}| \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni).$$

which we call the *functor of (unframed) derived deformations of type* $\mathbf{S}$. We make the analogous definitions for a characteristic 0 derived deformation problem.

If we have an equality $\mathcal{D}_v = \mathcal{D}^{\square}_{\bar\rho_v}$, we say $\mathbf{S}$ has trivial (or no) local condition at v.

Example 3.5.8. Although Definition 3.5.7 is quite general, our local conditions will arise from classical deformation rings in the following familiar way. Let $\mathbf{G} = \mathrm{GL}_n/F$ and $\bar\rho_v \colon \Gamma_{F_v} \to \mathrm{GL}_n(k)$. The local derived deformation functor $\mathcal{D}^{\square}_{\bar\rho_v}$ is in fact represented by the classical framed deformation ring $R^{\square}_{\bar\rho_v}$ by [BIP23, Thm. 1.1] and [GV18, Lem. 7.5]. Any conjugation-invariant quotient $R^{\square}_{\bar\rho_v} \twoheadrightarrow R^{D}_{\bar\rho_v}$ defines a local condition $\mathcal{D}_v$ via

$$\mathcal{D}_v \simeq \mathrm{Map}_{/k}(R^{D}_{\bar\rho_v}, -) \to \mathrm{Map}_{/k}(R^{\square}_{\bar\rho_v}, -) \simeq \mathcal{D}^{\square}_{\bar\rho_v}.$$

Definition 3.5.9. Let $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ be a deformation problem and suppose $T \subseteq S$ is a subset. We define the *functor of derived T-framed liftings of type* $\mathbf{S}$ as

$$\mathcal{D}^{\square,T}_{\mathbf{S}} = \mathcal{D}^{\square}_{\mathbf{S}} \times \prod_{v\in T} \widehat{\mathbf{G}}_k \colon \mathcal{A}rt_k \to \mathcal{A}ni.$$

There exists an action of $\widehat{\mathbf{G}}_k$ on $\mathcal{D}^{\square,T}_{\mathbf{S}}$ given by

$$\gamma \colon (\rho_A, \{\alpha_v\}_{v\in T}) \mapsto (\gamma\rho_A\gamma^{-1}, \{\gamma\alpha_v\}_{v\in T}).$$

We define the *functor of T-framed derived deformations of type* $\mathbf{S}$ as the geometric realisation

$$\mathcal{D}^{T}_{\mathbf{S}} \coloneqq |\mathcal{D}^{\square,T}_{\mathbf{S}} \times \widehat{\mathbf{G}}^{\bullet}_k| \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni).$$

Proposition 3.5.10. *Let $\bar\rho \colon \Gamma_{F,S} \to \mathrm{GL}_n(k)$ and suppose $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ is a deformation problem. Then, for any nonempty subset $T \subset S$, the natural morphism $\mathcal{D}^{T}_{\mathbf{S}} \to \mathcal{D}_{\mathbf{S}}$ is formally smooth of relative dimension $n^2|T| - 1$.*

Proof. See e.g. [A'C24, Lem. 5.1.5]. □

At $v \in S \setminus S_p$, our local conditions will be trivial, defined by the usual local deformation rings $R^{\square}_{\bar\rho_v}$.

Proposition 3.5.11. [All16, Prop. 1.2.2] *Let $v \nmid p$. Then $\dim R^{\square}_{\bar\rho_v}[1/\varpi] = n^2$, and if $\rho_v \colon \Gamma_{F_v} \to E$ is a representation such that the Weil–Deligne representation $\mathrm{WD}(\rho_v)$ is generic, the corresponding point of $\operatorname{Spec} R^{\square}_{\bar\rho_v}[1/\varpi]$ is smooth.*

At the places dividing p, we will consider local conditions defined by a variant of Kisin's [Kis08] potentially semistable deformation rings introduced in [Hev24, §5.1], defined by imposing conditions on Hodge–Tate weights and the action of inertia.

Definition 3.5.12. Let $v \mid p$. A *Weil–Deligne inertial type* of F_v is an isomorphism class of pairs $\tau = (\rho_\tau, N_\tau)$ where $\rho_\tau \colon I_{F_v} \to \mathrm{GL}_n(E)$ is a representation with open kernel and $N_\tau \in M_n(E)$ is a nilpotent matrix such that τ is isomorphic to the restriction $(r, N)|_{I_{F_v}}$ of a Weil–Deligne representation of Γ_{F_v}.

Hevesi defines a partial order $\preceq$ on Weil–Deligne inertial types, where $(\rho_\tau, N_\tau) \preceq (\rho_{\tau'}, N_{\tau'})$ if $\rho_\tau \cong \rho_{\tau'}$ and, roughly, the nilpotence of N_τ is bounded from below by that of $N_{\tau'}$; see [Hev24, Def. 2.11] for the precise definition.

Theorem 3.5.13. *Let $v \mid p$. Fix a dominant weight $\lambda \in (\mathbb{Z}^n)^{\mathrm{Hom}(F,E)}$ and a Weil–Deligne inertial type τ of F_v. Suppose $\bar\rho_v \colon \Gamma_{F_v} \to \mathrm{GL}_n(k)$. Then we have the following:*

1. [Hev24, Thm. 5.2] *There exists a local condition $\mathcal{D}^{\lambda_v, \preceq \tau_v}_{\bar\rho_v} \to \mathcal{D}^{\square}_{\bar\rho_v}$ represented by an $\mathcal{O}$-flat reduced quotient $R^{\square}_{\bar\rho_v} \twoheadrightarrow R^{\lambda_v, \preceq \tau_v}_{\bar\rho_v}$ such that a morphism $R^{\square}_{\bar\rho_v} \to E$ factors through $R^{\lambda_v, \preceq \tau_v}_{\bar\rho_v}$ if and only if the corresponding representation is potentially semistable with labelled Hodge–Tate weights $(\lambda_{\iota,1} + n - 1 > \cdots > \lambda_{\iota,n})_{\iota \in \mathrm{Hom}(F,E)}$ and $\mathrm{WD}(\rho_v)^{\mathrm{ss}}|_{I_{F_v}} \preceq \tau_v$.*
2. [Kis08, Thm. 3.3.8] $\dim R^{\lambda_v, \preceq \tau_v}_{\bar\rho_v}[1/\varpi] \le n^2 + \frac{1}{2} n(n-1)[F_v \colon \mathbb{Q}_p]$.
3. [All16, Prop. 1.3.12] *If ρ_v is a characteristic 0 lift of $\bar\rho_v$ such that the Weil–Deligne representation $\mathrm{WD}(\rho_v)$ is generic, the corresponding point of $\operatorname{Spec}(R^{\lambda_v, \preceq \tau_v}_{\bar\rho_v}[1/\varpi])$ is smooth with tangent space equal to the local Bloch–Kato Selmer group $H^1_g(\Gamma_{F_v}, \mathrm{ad}\,\rho)$.*

3.6. **Tangent complexes and representability.** In this section, we recall the definition of the tangent complex and Lurie's representability theorem for formally cohesive functors, [Lur04, §6.2].

Definition 3.6.1. [Lur04, Def. 6.2.1] A functor $\mathcal{D} \colon \mathcal{A}rt_k \to \mathcal{A}ni$ (resp. $\mathcal{D} \colon \mathcal{A}rt_E \to \mathcal{A}ni$) is *formally cohesive* if the following hold:

1. $\mathcal{D}(k)$ is contractible (resp. $\mathcal{D}(E)$ is contractible).
2. If $B' \to B$ and $B'' \to B$ are surjective (on π_0) morphisms in $\mathcal{A}rt_k$ (resp. $\mathcal{A}rt_E$), then
$$\mathcal{D}(B' \times_B B'') \simeq \mathcal{D}(B') \times_{\mathcal{D}(B)} \mathcal{D}(B'').$$

Lemma 3.6.2. *The derived deformation functors introduced in Section 3.5 are formally cohesive.*

Proof. Property (1) follows from Lemma 3.4.2. To prove (2), let $\mathcal{X}^{\square} \colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ be a functor satisfying Lemma 3.4.3, and let $\mathcal{D}^{\square} := \mathcal{X}^{\square}_{\bar\rho}$ be the deformation functor at a point $\bar\rho \in \mathcal{X}^{\square}(k)$. Then if $B' \to B \leftarrow B''$ is a diagram in $\mathcal{A}rt_k$, we have

$$\mathcal{D}(B' \times_B B'') = \mathcal{X}^{\square}(B' \times_B B'') \times_{\mathcal{X}^{\square}(k)} \{\bar\rho\} \simeq \mathcal{X}^{\square}_{\bar\rho}(B') \times_{\mathcal{X}^{\square}_{\bar\rho}(B)} \mathcal{X}^{\square}_{\bar\rho}(B'') \simeq \mathcal{D}(B') \times_{\mathcal{D}(B)} \mathcal{D}(B''),$$

as needed. This equivalence is equivariant for the action of

$$\widehat{\mathbf{PG}}(B' \times_B B'') = \widehat{\mathbf{PG}}(B') \times_{\widehat{\mathbf{PG}}(B)} \widehat{\mathbf{PG}}(B''),$$

which implies the claim for the unframed version $|\widehat{\mathbf{PG}}^\bullet \times \mathcal{D}^\square|$. The proof for a characteristic 0 deformation functor is identical. □

Given a formally cohesive functor $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$, $r \geq 1$ and $\xi_r \in \mathcal{D}(\mathcal{O}_r)$ where $\mathcal{O}_r = \mathcal{O}/\varpi^r$, one can define a functor on perfect connective $\mathcal{O}_r$-modules by

$$V \mapsto \mathcal{D}(\mathcal{O}_r \oplus V).$$

By [Lur04, Prop. 2.5.5, Lem. 6.2.2], this functor has a unique filtered colimit-preserving and finite limit-preserving extension to the category $\mathcal{M}od_{\mathcal{O}_r}$ of all $\mathcal{O}_r$-modules, which is representable in the sense that there exists an $\mathcal{O}_r$-module $\mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{D}) \in \mathcal{M}od_{\mathcal{O}_r}$ such that

$$V \mapsto \big(V \otimes_{\mathcal{O}_r} \mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{D})\big)_0,$$

the 0:th space of the spectrum $V \otimes_{\mathcal{O}_r} \mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{D})$. Concretely, the homotopy groups of $\mathrm{T}^\bullet_A(\mathcal{D})$ are given by

$$\mathrm{T}^i_A(\mathcal{D}) \coloneqq \pi_i\big(\mathrm{T}^\bullet_A(\mathcal{D})\big) \cong \pi_0\big(\mathcal{D}(A \oplus A[i]) \times_{\mathcal{D}(A)} \{x\}\big).$$

The $\mathcal{O}_r$-module $\mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{D}) \in \mathcal{M}od_{\mathcal{O}_r}$ is called the *tangent complex of $\mathcal{D}$ at $\mathcal{O}_r$*. Similarly, for a formally cohesive functor $\mathcal{D}\colon \mathcal{A}rt_E \to \mathcal{A}ni$, we define its tangent complex $\mathrm{T}^\bullet_E(\mathcal{D})$ at E in terms of the functor mapping a perfect connective $V \in \mathcal{M}od_E$ to $\mathcal{D}(E \oplus V)$. The construction of the tangent complex is functorial in $\mathcal{D}$ and $\mathcal{O}_r$, and we define

$$\mathrm{T}^\bullet_{\mathcal{O}}(\mathcal{D}) \coloneqq \lim_r \mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{D}).$$

If $\mathcal{D}$ is a formally cohesive functor represented by $\mathcal{R}$, we write $\mathrm{T}^\bullet_?(\mathcal{R}) \coloneqq \mathrm{T}^\bullet_?(\mathcal{D})$. The tangent complex of $\mathcal{R}$ at x is given by

$$\mathrm{T}^\bullet_A(\mathcal{R}) = \mathrm{Map}_{\mathcal{M}od_{\mathcal{R}}}(\mathrm{L}_{\mathcal{R}/\mathcal{O}}, A) \cong \mathrm{Map}_{\mathcal{M}od_A}(\mathrm{L}_{\mathcal{R}/\mathcal{O}} \otimes^{\mathbf{L}}_{\mathcal{R}} A, A),$$

where $\mathrm{L}_{\mathcal{R}/\mathcal{O}}$ denote the algebraic cotangent complex of $\mathcal{R}$. If $A \to B \to \mathcal{O}$ is a morphism in $\mathcal{A}lg_{\mathcal{O}/}$, we denote by $\mathrm{T}^\bullet_{\mathcal{O}}(B/A)$ the fibre of the induced map $\mathrm{T}^\bullet_{\mathcal{O}}(B) \to \mathrm{T}^\bullet_{\mathcal{O}}(A)$.

Theorem 3.6.3. [Lur04, Cor. 6.2.14] *Let $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ be a formally cohesive functor. The following are equivalent:*

(1) *There exists an $\mathcal{R} \in \mathcal{CNL}_k$ (Definition 2.2.6) and an equivalence of functors*
$$\mathcal{D} \simeq \mathrm{Hom}_{(\mathcal{A}lg_{\mathcal{O}/})_{/k}}(\mathcal{R}, -).$$

(2) *The k-vector spaces $\mathrm{T}^i_k(\mathcal{D})$ vanish for $i < 0$ and are finite dimensional for $i \geq 0$.*

The analogous statement holds for a formally cohesive functor $\mathcal{D}\colon \mathcal{A}rt_E \to \mathcal{A}ni$.

Given $\mathcal{R} \in \mathcal{CNL}_k$ with maximal ideal $\mathfrak{m} \subset \pi_0(\mathcal{R})$ and $B \in \mathcal{A}rt_k$, we have equivalences

$$\mathrm{Map}_{/k}(\mathcal{R}, B) \simeq \mathrm{Map}_{/k}(\lim_r \mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r, B) \simeq \operatorname*{colim}_r \mathrm{Map}_{/k}(\mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r, B).$$

The first equivalence holds because $\mathcal{R}$ is derived $\mathfrak{m}$-adically complete and the second holds by Proposition 2.2.4 and the artinian assumption on B. Since $\mathcal{R}$ is noetherian, [Lur04, Prop. 6.1.8] implies

$$\pi_0(\lim \mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r) \cong \varprojlim \pi_0(\mathcal{R}).$$

The quotient $\mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r$ need not be truncated, but one can rephrase Theorem 3.6.3(1) in terms of pro-representability using the pro-artinian completion $\{\tau_{\leq j}(\mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r)\}_{j,r} \in \mathcal{P}ro(\mathcal{A}rt_k)$, as in [GV18].

Proposition 3.6.4. *Let $\mathcal{R} \in \mathcal{CNL}_k$ and let $\mathcal{R} \to \mathcal{O}_r = \mathcal{O}/\varpi^r$ be a morphism over k. Then, for every $i \geq 0$, $\mathrm{T}^i_{\mathcal{O}_r}(\mathcal{R})$ is a finitely generated $\mathcal{O}_r$-module.*

Proof. The case $r = 1$ is Theorem 3.6.3. By considering the short exact sequence

$$0 \to \mathcal{O}_r \to \mathcal{O}_{r+1} \to k \to 0$$

we deduce the analogous statement for $\mathcal{O}_r$-coefficients using induction. □

Lemma 3.6.5. *Suppose we have morphisms $A \to B \to \mathcal{O}$ in $\mathcal{A}lg_{\mathcal{O}/}$ and that B is almost of finite presentation over A. Then, if $\mathrm{T}^i_{\mathcal{O}}(A)$ is a finitely generated $\mathcal{O}$-module for every $i \geq 0$, the same holds for $\mathrm{T}^i_{\mathcal{O}}(B)$.*

Proof. By [Lur04, Prop. 3.2.14], the relative cotangent complex $\mathrm{L}_{B/A}$ is an almost perfect animated B-module, i.e. $\pi_i(\mathrm{L}_{B/A})$ is a finitely generated $\pi_0(B)$-module for every $i \geq 0$. It follows that $\mathrm{L}_{B/A} \otimes^{\mathbf{L}}_B \mathcal{O}$ is an almost perfect animated $\mathcal{O}$-module. Dualising the fibre sequence

$$\mathrm{L}_{A/\mathcal{O}} \otimes^{\mathbf{L}}_A \mathcal{O} \to \mathrm{L}_{B/\mathcal{O}} \otimes^{\mathbf{L}}_B \mathcal{O} \to \mathrm{L}_{B/A} \otimes^{\mathbf{L}}_B \mathcal{O} \overset{+1}{\to}$$

we obtain a fibre sequence of tangent complexes with $\mathcal{O}$-coefficients

$$\mathrm{T}^\bullet_{\mathcal{O}}(B/A) \to \mathrm{T}^\bullet_{\mathcal{O}}(B) \to \mathrm{T}^\bullet_{\mathcal{O}}(A) \overset{+1}{\to}.$$

By assumption, $\mathrm{T}^i_{\mathcal{O}}(A)$ is finitely generated over $\mathcal{O}$ for every i. The same now follows for $\mathrm{T}^\bullet_{\mathcal{O}}(B)$. □

The tangent complexes of the deformation functors introduced in Section 3.5 are given by continuous cohomology in the usual sense.

Proposition 3.6.6. [Zhu25, Prop. 2.59] *Let Γ be a profinite group, $\mathbf{G} = \mathrm{GL}_n/\mathcal{O}$ and $\bar\rho \in \mathcal{X}^\square_{\Gamma,\mathbf{G}}(k)$. The tangent complex of $\mathcal{D}_{\bar\rho}$ at k is given by an animated k-module denoted $C^\bullet_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]$ such that*

$$\mathrm{T}^i_k(\mathcal{D}_{\bar\rho}) \cong \pi_{-i}\big(C^\bullet_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]\big) \cong \begin{cases} H^0(\Gamma, \mathrm{ad}^0\,\bar\rho) & \text{if } i = -1, \\ H^1(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 0, \\ H^2(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 1, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, if $\bar\rho$ is absolutely irreducible, then

$$\mathrm{T}^i_k(\mathcal{D}_{\bar\rho}) \cong H^{i+1}(\Gamma, \mathrm{ad}\,\bar\rho).$$

The tangent complex of $\mathcal{D}^\square_{\bar\rho}$ at k is given by an animated k-module $\overline{C}^\bullet_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]$ such that

$$\mathrm{T}^i_k(\mathcal{D}^\square_{\bar\rho}) \cong \pi_{-i}\big(\overline{C}^\bullet_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]\big) \cong \begin{cases} Z^1(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 0, \\ H^2(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The analogous statements are true with $\xi \in \mathcal{X}^\square_{\Gamma,\mathbf{G}}(\mathcal{O})$, $\xi_r \in \mathcal{X}^\square_{\Gamma,\mathbf{G}}(\mathcal{O}_r)$ or $\rho \in \mathcal{X}^\square_{\Gamma,\mathbf{G}}(E)$ in place of $\bar\rho$.

Corollary 3.6.7. *Let $\Gamma = \mathrm{Gal}(F_S/F)$ or $\mathrm{Gal}(\overline{F}_v/F_v)$ and fix $\bar\rho \in \mathcal{X}^\square_{\Gamma,\mathbf{G}}(k)$. The framed derived deformation functor $\mathcal{D}^\square_{\bar\rho} = (\mathcal{X}^\square_{\Gamma,\mathbf{G}})_{\bar\rho}$ is representable and if $\bar\rho$ is Schur, the derived deformation functor $\mathcal{D}_{\bar\rho}$ is representable. The same statement is true for $\rho \in \mathcal{X}^\square_{\Gamma,\mathbf{G}}(E)$.*

Proof. This follows from Theorem 3.6.3 and Proposition 3.6.6. □

We deduce the same statement for deformations of type $\mathbf{S}$.

Proposition 3.6.8. *Let $\mathbf{S} = (\bar{\rho}, S, \{\mathcal{D}_v\}_{v\in S})$ be a deformation problem and let $T \subseteq S$ be a subset. Suppose $\bar{\rho}$ is Schur. Then the functors $\mathcal{D}_{\mathbf{S}}$ and $\mathcal{D}^T_{\mathbf{S}}$ are representable.*

3.7. **Localisation at a characteristic 0 point.** In this section, we prove two results which generalise results of Kisin, relating the deformation ring of a characteristic 0 representation to that of its reduction mod p [Kis09, Lem. 2.3.3, Prop. 2.3.5].

In the following definition, which is a direct analogue of [Kis09, §2.3.1], we use the functor $\mathcal{I}ntB \to (\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$ of Lemma 3.2.7 to view $A \in \mathcal{I}ntB$ as an object of $(\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$.

Definition 3.7.1. Let $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ be a functor and $\xi \in \mathcal{D}(\mathcal{O}) = \lim \mathcal{D}(\mathcal{O}_r)$. We define

$$\begin{aligned}\mathcal{D}_{(\xi)}\colon \mathcal{A}rt_E &\to \mathcal{A}ni\\ (B \to E) &\mapsto \operatorname*{colim}_{A\in\mathcal{I}ntB} \mathcal{D}(A) \times_{\mathcal{D}(\mathcal{O})} \{\xi\}.\end{aligned}$$

As before, we extend $\mathcal{D}_{(\xi)}$ to $\mathrm{Pro}(\mathcal{A}rt_E)$ by defining $\mathcal{D}_{(\xi)}(\{A_i\}) \coloneqq \lim \mathcal{D}_{(\xi)}(A_i)$.

Proposition 3.7.2. *Let $\mathcal{X}^{\square}\colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ be a functor commuting with filtered colimits, finite limits and limits of the form $A = \lim A/^{\mathbf{L}}\varpi^r$ when $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module. Fix $\bar{\rho} \in \mathcal{X}^{\square}(k)$ and $\xi \in \mathcal{X}^{\square}_{\bar{\rho}}(\mathcal{O})$, and let ρ be the image of ξ in $\mathcal{X}^{\square}(E)$. Then there is an equivalence of functors*

$$(\mathcal{X}^{\square}_{\bar{\rho}})_{(\xi)} \simeq \mathcal{X}^{\square}_{\rho}.$$

Proof. Let $(B \to E) \in \mathcal{A}rt_E$. By definition,

$$(\mathcal{X}^{\square}_{\bar{\rho}})_{(\xi)}(B) = \operatorname*{colim}_{A\in\mathcal{I}ntB} (\mathcal{X}^{\square}_{\bar{\rho}}(A) \times_{\mathcal{X}_{\bar{\rho}}(\mathcal{O})} \{\xi\}).$$

Now, $A \simeq \lim A/^{\mathbf{L}}\varpi^r$ and $A/^{\mathbf{L}}\varpi^r \in \mathcal{A}rt_k$ by Lemma 3.2.6(3,4) Hence, if we let $\xi_r \in \mathcal{X}^{\square}(\mathcal{O}_r)$ denote the $\bmod \varpi^r$ reduction of ξ, we have by definition that

$$\mathcal{X}^{\square}_{\bar{\rho}}(A) \times_{\mathcal{X}^{\square}_{\bar{\rho}}(\mathcal{O})} \{\xi\} \simeq \lim_r \mathcal{X}^{\square}_{\bar{\rho}}(A/^{\mathbf{L}}\varpi^r) \times_{\mathcal{X}^{\square}_{\bar{\rho}}(\mathcal{O}_r)} \{\xi_r\} \simeq \mathcal{X}^{\square}(A) \times_{\mathcal{X}^{\square}(\mathcal{O})} \{\xi\}.$$

The category $\mathcal{I}ntB$ is filtered by Proposition 3.2.2 and filtered colimits commute with finite limits. Hence, by the assumption on $\mathcal{X}^{\square}$ and Lemma 3.2.6(1),

$$\operatorname*{colim}_{A\in\mathcal{I}ntB} (\mathcal{X}_{\bar{\rho}}(A) \times_{\mathcal{X}_{\bar{\rho}}(\mathcal{O})} \{\xi\}) \simeq \mathcal{X}^{\square}(B \times_E \mathcal{O}) \times_{\mathcal{X}^{\square}(\mathcal{O})} \{\xi\} \simeq \mathcal{X}^{\square}(B) \times_{\mathcal{X}^{\square}(E)} \{\rho\} = \mathcal{X}^{\square}_{\rho}(B),$$

as required. □

Theorem 3.7.3. *Let $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ be a formally cohesive functor represented by $\mathcal{R} \in \mathcal{CNL}_k$, and $\xi \in \mathcal{D}(\mathcal{O})$. Then $\mathcal{D}_{(\xi)}\colon \mathcal{A}rt_E \to \mathcal{A}ni$ is represented by*

$$(\mathcal{R} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi} \coloneqq \lim_s (\mathcal{R} \otimes_{\mathcal{O}} E)/^{\mathbf{L}} I^s_{\xi},$$

the derived completion of $\mathcal{R} \otimes_{\mathcal{O}} E$ at the maximal ideal $I_\xi = \ker(\pi_0(\mathcal{R}) \otimes_{\mathcal{O}} E \to E)$ defined by ξ.

Proof. Let $(B \to E) \in \mathcal{A}rt_E$ and let $\xi_r \in \mathcal{D}(\mathcal{O}_r)$ denote the $\bmod \varpi^r$ reduction of ξ. By definition and Lemma 3.2.6(3),

$$\mathcal{D}_{(\xi)}(B) = \operatorname*{colim}_{A\in\mathcal{I}ntB} (\mathcal{D}(A) \times_{\mathcal{D}(\mathcal{O})} \{\xi\}) \simeq \operatorname*{colim}_{A\in\mathcal{I}ntB} \big(\lim_r \mathcal{D}(A/^{\mathbf{L}}\varpi^r) \times_{\mathcal{D}(\mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\}\big).$$

Now, by representability of $\mathcal{D}$, we have

$$\lim_r \big(\mathcal{D}(A/^{\mathbf{L}}\varpi^r) \times_{\mathcal{D}(\mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\}\big) \simeq \lim_r \mathrm{Map}_{/k}(\mathcal{R}, A/^{\mathbf{L}}\varpi^r) \times_{\mathrm{Map}_{/k}(\mathcal{R},\mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\},$$

and from Lemma 3.2.6(4) and commutativity of limits, we obtain

$$\lim_r \mathrm{Map}_{/k}(\mathcal{R}, A/^{\mathbf{L}}\varpi^r) \times_{\mathrm{Map}_{/k}(\mathcal{R},\mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\} \simeq \mathrm{Map}_{/\mathcal{O}}(\mathcal{R}, A).$$

For any $A \in \mathcal{I}ntB$, the map $\pi_0(A) \to \pi_0(B)$ is injective and hence, for any $\mathcal{R} \to A$ in $(\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$, the induced map $\pi_0(\mathcal{R}) \to \pi_0(A)$ factors through a quotient $\pi_0(\mathcal{R})/(I_\xi^\circ)^s$ for some $s \geq 1$, where $I_\xi^\circ \subset \pi_0(\mathcal{R})$ is the inverse image of $I_\xi = \ker(\pi_0(\mathcal{R}) \otimes_{\mathcal{O}} E \to E)$. Proposition 2.2.4 implies

$$\mathrm{Map}_{/\mathcal{O}}(\mathcal{R}, A) \simeq \operatorname*{colim}_s \mathrm{Map}_{/\mathcal{O}}\left(\mathcal{R}/^{\mathbf{L}}(I_\xi^\circ)^s, A\right).$$

For any $A \in \mathcal{I}ntB$, $\pi_*(A)$ injects into $\pi_*(B)$, which is truncated. Hence, A is n-truncated for n sufficiently large depending only on B. Now, $\mathcal{R}/^{\mathbf{L}}(I_\xi^\circ)^s$ is almost of finite presentation over $\mathcal{O}$, since $\pi_0(\mathcal{R}/^{\mathbf{L}}(I_\xi^\circ)^s) \cong \pi_0(\mathcal{R})/(I_\xi^\circ)^s$ is a finitely generated $\mathcal{O}$-module and $\mathcal{R}/^{\mathbf{L}}(I_\xi^\circ)^s$ is a derived tensor product of noetherian animated algebras (Proposition 2.2.3). Moreover, $\mathcal{I}ntB$ is filtered (Proposition 3.2.2), and hence

$$\mathcal{D}_{(\xi)}(B) \simeq \operatorname*{colim}_{A\in\mathcal{I}ntB} \operatorname*{colim}_s \mathrm{Map}_{/\mathcal{O}}(\mathcal{R}/^{\mathbf{L}}(I_\xi^\circ)^s, A) \simeq \operatorname*{colim}_s \mathrm{Map}_{/\mathcal{O}}(\mathcal{R}/^{\mathbf{L}}(I_\xi^\circ)^s, \operatorname*{colim}_{A\in\mathcal{I}ntB} A).$$

By Lemma 3.2.6(1), $\operatorname{colim} A \simeq B \times_E \mathcal{O}$ and the space above is equivalent to

$$\operatorname*{colim}_s \mathrm{Map}_{/\mathcal{O}}\left((\mathcal{R}/^{\mathbf{L}}(I_\xi^\circ)^s, B \times_E \mathcal{O}\right) \simeq \operatorname*{colim}_s \mathrm{Map}_{/E}\left((\mathcal{R} \otimes_{\mathcal{O}} E)/^{\mathbf{L}} I_\xi^s, B\right).$$

Thus, $\mathcal{D}_{(\xi)}(B \to E) \simeq \mathrm{Map}_{/E}\left((\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi, B\right)$ as required. □

The construction behaves well with respect to finite limits, as the following simple observation shows.

Lemma 3.7.4. *Suppose given a pullback of formally cohesive functors*

$$\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'' \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni),$$

and let $\tilde{\xi} \in (\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')(\mathcal{O})$. Then

$$(\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')_{(\tilde{\xi})} \simeq \mathcal{D}'_{(\xi')} \times_{\mathcal{D}_{(\xi)}} \mathcal{D}''_{(\xi'')}$$

where ξ, ξ', ξ'' denote the images of $\tilde{\xi}$ in $\mathcal{D}(\mathcal{O}), \mathcal{D}'(\mathcal{O})$ and $\mathcal{D}''(\mathcal{O})$, respectively.

Proof. This follows from the fact that filtered colimits commute with finite limits, and that $\mathcal{I}ntB$ is filtered (Proposition 3.2.2). Indeed, for any $B \in \mathcal{A}rt_E$,

$$(\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')_{(\tilde{\xi})}(B) \simeq \operatorname*{colim}_{A\in\mathcal{I}ntB} (\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')(A) \simeq \mathcal{D}'_{(\xi')}(B) \times_{\mathcal{D}_{(\xi)}(B)} \mathcal{D}''_{(\xi'')}(B),$$

as needed. □

Recall that we defined derived deformation functors $\mathcal{D}^\square_{\bar{\rho}}, \mathcal{D}_{\bar{\rho}}, \mathcal{D}^\square_\rho, \mathcal{D}_\rho$ in Definition 3.5.2. The following is a typical application of Proposition 3.7.2 and Theorem 3.7.3.

Corollary 3.7.5. *Let $\mathcal{X}^\square_{\Gamma,\mathbf{G}}$ be the derived moduli stack of framed $\mathbf{G}$-valued Γ-representations (see Definition 3.4.1). Suppose $\bar{\rho} \in \mathcal{X}^\square_{\Gamma,\mathbf{G}}(k)$, let $\mathcal{D}^\square_{\bar{\rho}} = (\mathcal{X}^\square_{\Gamma,\mathbf{G}})_{\bar{\rho}}$ and fix $\xi \in \mathcal{D}^\square_{\bar{\rho}}(\mathcal{O})$ with image $\rho = \xi \otimes E \in \mathcal{D}^\square_{\bar{\rho}}(E)$. Then there is an equivalence of functors*

$$(\mathcal{D}^\square_{\bar{\rho}})_{(\xi)} \simeq \mathcal{D}^\square_\rho.$$

The same statement holds for the functor $\mathcal{D}_{\bar{\rho}}$ of unframed deformations (see Definition 3.5.2).

Proof. The statement for the framed functors follows from Lemma 3.4.3 and Proposition 3.7.2. To prove the analogous statement for the unframed functor, we let $(B \to E) \in \mathcal{A}rt_E$. The proof of Proposition 3.7.2 comprises an equivalence

$$(\mathcal{D}^{\square}_{\bar\rho})_{(\xi)}(B) = \operatorname*{colim}_{A\in\mathcal{I}ntB} \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(A) \times_{\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(\mathcal{O})} \{\xi\} \simeq \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(B) \times_{\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(E)} \{\rho\},$$

which is equivariant for the action of

$$\operatorname*{colim}_{A\in\mathcal{I}ntB} \widehat{\mathbf{PG}}_k(A) \times_{\widehat{\mathbf{PG}}_k(\mathcal{O})} \{\mathrm{id}\} \simeq \operatorname*{colim}_{A\in\mathcal{I}ntB} \mathbf{PG}(A) \times_{\mathbf{PG}(\mathcal{O})} \{\mathrm{id}\} \simeq \mathbf{PG}(B) \times_{\mathbf{PG}(E)} \{\mathrm{id}\} \simeq \widehat{\mathbf{PG}}_E(B),$$

where we have subscripted $\widehat{\mathbf{PG}}_k$ and $\widehat{\mathbf{PG}}_E$ to distinguish the functors defined on $\mathcal{A}rt_k$ resp. $\mathcal{A}rt_E$. Hence, taking geometric realisations now prove the claim. □

Proposition 3.7.6. *Let $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ be a formally cohesive functor such that $\mathrm{T}^i_k(\mathcal{D})$ is finite for every i, and fix $\xi \in \mathcal{D}(\mathcal{O})$. Then $\mathcal{D}_{(\xi)}\colon \mathcal{A}rt_E \to \mathcal{A}ni$ has tangent complex*

$$\mathrm{T}^{\bullet}_E(\mathcal{D}_{(\xi)}) = \big(\lim_r \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D})\big) \otimes_{\mathcal{O}} E.$$

Proof. It suffices to prove that, for every perfect connective $V \in \mathcal{M}od_{\mathcal{O}}$, we have an equivalence

$$\mathcal{D}(\mathcal{O}\oplus V) \times_{\mathcal{D}(\mathcal{O})} \{\xi\} \simeq \tau_{\geq 0}\big(\lim_r \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}} V\big) \simeq \lim_r \tau_{\geq 0}\big(\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}} V\big).$$

Arguing as in Proposition 3.6.4, one sees that $\mathrm{T}^i_{\mathcal{O}_r}(\mathcal{D})$ is finitely generated for every i. Thus, if $V_r := V/^{\mathbf{L}}\varpi^r$, and ξ_r is the image of ξ in $\mathcal{D}(\mathcal{O}_r)$, then

$$\begin{aligned}\tau_{\geq 0}\big(\lim_r \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}} V\big) \simeq \lim_r \tau_{\geq 0}\big(\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}_r} V_r\big) &\simeq \lim_r \mathcal{D}(\mathcal{O}_r \oplus V_r) \times_{\mathcal{D}(\mathcal{O}_r)} \{\xi_r\} \\ &\simeq \mathcal{D}(\mathcal{O}\oplus V) \times_{\mathcal{D}(\mathcal{O})} \{\xi\},\end{aligned}$$

where we have used that $\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}_r} V_r$ is almost perfect, so that the inverse limit has no higher derived functors. □

Proposition 3.7.7. *Let $R = \pi_0(R) \in \mathcal{CNL}_k$ be static and suppose $\xi\colon R \to \mathcal{O}$ is a morphism such that $\mathfrak{p} = \ker(R\otimes_{\mathcal{O}} E \to E)$ is a smooth point. Then, for any $r \geq 1$ and $i \geq 1$, the $\mathcal{O}$-module $\mathrm{T}^i_{\mathcal{O}_r}(R)$ has finite length bounded independently of r.*

Proof. Let $i, r \geq 1$. Consider the short exact sequence $0 \to \mathcal{O} \to \mathcal{O} \to \mathcal{O}_r \to 0$ given by multiplication by ϖ^r, and the induced exact sequence

$$\mathrm{T}^i_{\mathcal{O}}(R) \to \mathrm{T}^i_{\mathcal{O}_r}(R) \to \mathrm{T}^{i+1}_{\mathcal{O}}(R).$$

By Proposition 3.6.4 and Lemma 3.6.5, these are finitely generated $\mathcal{O}$-modules. Since ξ defines a smooth point, $\mathrm{T}^i_{\mathcal{O}}(R)$ is torsion for $i \geq 1$. It follows that

$$\mathrm{length}_{\mathcal{O}}\big(\mathrm{T}^i_{\mathcal{O}_r}(R)\big) \leq \mathrm{length}_{\mathcal{O}}\big(\mathrm{T}^i_{\mathcal{O}}(R)\big) + \mathrm{length}_{\mathcal{O}}\big(\mathrm{T}^{i+1}_{\mathcal{O}}(R)\big)$$

which is finite and independent of r. □

Proposition 3.7.8. *Let $\mathcal{D}_{\mathbf{S}}$ be the derived deformation functor of the mod p deformation problem*

$$\mathbf{S} = (\bar\rho, S, \{\mathrm{Map}_{/k}(R^D_{\bar\rho_v}, -)\}_{v\in S}).$$

Suppose $\xi \in \mathcal{D}_{\mathbf{S}}(\mathcal{O})$ and let $\rho = \xi[1/p]\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$. Then $(\mathcal{D}_{\mathbf{S}})_{(\xi)} \simeq \mathcal{D}_{\mathbf{S}(\rho)}$, where $\mathbf{S}(\rho)$ is the characteristic 0 deformation problem

$$\mathbf{S}(\rho) := (\rho, S, \{\mathrm{Map}_{/E}((R^D_{\bar\rho_v} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi_v}, -)\}_{v\in S})$$

and $(R^D_{\bar\rho_v} \otimes_{\mathcal{O}} E)^\wedge_{\xi_v}$ *denotes the completion of* $R^D_{\bar\rho_v} \otimes_{\mathcal{O}} E$ *at the ideal* $I_{\xi_v} \subset \pi_0(R^D_{\bar\rho_v} \otimes E \to E)$ *defined by* $\xi_v = \xi|_{\Gamma_{F_v}}$*, as in Theorem 3.7.3.*

Proof. Let $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ and $(B \to E) \in \mathcal{A}rt_E$. By definition, the functor of framed liftings of type $\mathbf{S}$ is given by the pullback $\mathcal{D}^\square_{\mathbf{S}} = \mathcal{D}^\square_{\bar\rho} \times_{\prod_{v\in S}\mathcal{D}^\square_{\bar\rho_v}} \prod_{v\in S}\mathcal{D}_v$. By Lemma 3.7.4, if we let $\xi_v \coloneqq \xi|_{\Gamma_{F_v}} \in \mathcal{D}^\square_{\bar\rho_v}(\mathcal{O})$, we have

$$(\mathcal{D}^\square_{\mathbf{S}})_{(\xi)} \simeq \left(\mathcal{D}^\square_{\bar\rho} \times_{\prod_{v\in S}\mathcal{D}^\square_{\bar\rho_v}} \prod_{v\in S}\mathcal{D}_v\right)_{(\xi)} \simeq (\mathcal{D}^\square_{\bar\rho})_{(\xi)} \times_{\prod_{v\in S}(\mathcal{D}^\square_{\bar\rho_v})_{(\xi_v)}} \prod_{v\in S}(\mathcal{D}_v)_{(\xi_v)}.$$

Applying Proposition 3.7.2 and Theorem 3.7.3 yields an equivalence $(\mathcal{D}^\square_{\mathbf{S}})_{(\xi)} \simeq \mathcal{D}^\square_{\mathbf{S}(\rho)}$ which is conjugation-equivariant. Taking geometric realisations completes the proof. □

Corollary 3.7.9. *Let* $\mathcal{D}_{\mathbf{S}}$ *be the deformation functor of the deformation problem*

$$\mathbf{S} = (\bar\rho, S, \{\mathcal{D}^{\lambda_v,\preceq\tau_v}_{\bar\rho_v}\}_{v\in S_p} \cup \{\mathcal{D}^\square_{\bar\rho_v}\}_{v\in S\setminus S_p})$$

where $\bar\rho$ *is absolutely irreducible, and suppose* $\xi \in \mathcal{D}_{\mathbf{S}}(\mathcal{O})$ *is a point corresponding to a representation* $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ *such that the Weil–Deligne representations* $\mathrm{WD}(\rho|_{\Gamma_{F_v}})$ *are generic for every* $v \in S$*. Then we have a weak equivalence* $(\mathcal{D}_{\mathbf{S}})_{(\xi)} \simeq \mathcal{D}^{\mathrm{ss}}_\rho$*, where* $\mathcal{D}^{\mathrm{ss}}_\rho$ *is the deformation functor of the characteristic 0 deformation problem*

$$(\rho, S, \{\mathcal{D}^{\lambda_v}_{\rho_v}\}_{v\in S_p} \cup \{\mathcal{D}^\square_{\rho_v}\}_{v\in S^p}).$$

Moreover, the tangent complex of $(\mathcal{D}_{\mathbf{S}})_{(\xi)}$ *is concentrated in degrees* $[0,1]$ *where we have natural identifications*

$$\mathrm{T}^0_E\left((\mathcal{D}_{\mathbf{S}})_{(\xi)}\right) \cong H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho), \quad \mathrm{T}^1_E\left((\mathcal{D}_{\mathbf{S}})_{(\xi)}\right) \cong H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho(1))^\vee.$$

Proof. By Proposition 3.7.8, we have an equivalence $(\mathcal{D}_{\mathbf{S}})_{(\xi)} \simeq \mathcal{D}_{\mathbf{S}(\rho)}$, where $\mathbf{S}(\rho)$ has local conditions at $v \in S \setminus S_p$ given by the unrestricted local framed deformation rings $R^\square_{\rho_v}$. At $v \in S_p$, the local conditions are defined by

$$(R^{\lambda_v,\preceq\tau_v}_{\bar\rho_v} \otimes_{\mathcal{O}} E)^\wedge_\xi \simeq \varprojlim_s (R^{\lambda_v,\preceq\tau_v}_{\bar\rho_v} \otimes_{\mathcal{O}} E)/I^s_\xi \simeq R^{\lambda_v}_{\rho_v},$$

where we have used Theorem 3.5.13(3) and the assumption on $\mathrm{WD}(\rho_v)$, and $\varprojlim$ denotes the 1-categorical inverse limit in the category of rings. Note that the local conditions are defined by smooth algebras by Proposition 3.5.11 and Theorem 3.5.13(3). To describe the tangent complex, we use the argument in [GV18, Thm. 9.2, App. B] which proves the corresponding statement for mod p deformations. The same argument, applied to $\mathcal{D}^{\mathrm{ss}}_\rho$, identifies the tangent complex with the complex with local conditions of Corollary 2.4.2; the result follows. □

Having determined the tangent complex $\mathrm{T}^\bullet_E(\mathcal{R}^{\mathrm{ss}}_\rho)$, we are able to identify the graded ring $\pi_*(\mathcal{R}^{\mathrm{ss}}_\rho)$ under the assumption that the adjoint Bloch–Kato Selmer group vanishes.

Corollary 3.7.10. *Suppose* $H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho) = 0$*. Then there is a natural isomorphism*

$$\pi_*(\mathcal{R}^{\mathrm{ss}}_\rho) \cong \bigwedge H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)).$$

Proof. By Corollary 3.7.9 and the assumed vanishing of $H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho)$, the tangent complex $\mathrm{T}^\bullet_E(\mathcal{R}^{\mathrm{ss}}_\rho)$ is concentrated in degree 1. It follows from [AG15, Cor. 2.1.6] that $\pi_*(\mathcal{R}^{\mathrm{ss}}_\rho)$ is an exterior algebra on

$\pi_1(\mathcal{R}_\rho^{\mathrm{ss}})$ which is of dimension $l_0 = \dim_E \mathrm{T}_E^1(\mathcal{R}_\rho^{\mathrm{ss}})$ over E (Proposition 2.3.1). Thus, it suffices to define a natural surjection

$$\pi_1(\mathcal{R}_\rho^{\mathrm{ss}}) \twoheadrightarrow H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)).$$

By Corollary 3.7.9, we have an isomorphism

$$H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1))^\vee \cong \mathrm{T}_E^1(\mathcal{R}_\rho^{\mathrm{ss}}) = \pi_0\Big(\mathrm{Map}_{/E}\left(\mathcal{R}_\rho^{\mathrm{ss}}, E \oplus E[1]\right)\Big).$$

Any class $[f]$ in this π_0 induces a morphism $\pi_1(\mathcal{R}_\rho^{\mathrm{ss}}) \to E$. This defines a map

$$\varphi\colon\ \mathrm{T}_E^1(\mathcal{R}_\rho^{\mathrm{ss}}) \to \pi_1(\mathcal{R}_\rho^{\mathrm{ss}})^\vee.$$

We claim that φ is injective. Indeed, suppose $\varphi([f]) = 0$, i.e. f is a morphism $\mathcal{R}_\rho^{\mathrm{ss}} \to E \oplus E[1]$ such that its induced map on π_1 is trivial. Repeated application of [Lur04, Prop. 3.3.3] shows that the space of such maps is equivalent to

$$\mathrm{Map}_{/E}\left(\pi_0(\mathcal{R}_\rho^{\mathrm{ss}}), E \oplus E[1]\right) \hookrightarrow \mathrm{Map}_{/E}\left(\mathcal{R}_\rho^{\mathrm{ss}}, E \oplus E[1]\right).$$

Now, $\pi_0(\mathcal{R}_\rho^{\mathrm{ss}}) \cong E$ and thus, this space is contractible and $[f]$ is trivial. Passing to duals, we obtain a surjection $\varphi^\vee\colon \pi_1(\mathcal{R}_\rho^{\mathrm{ss}}) \cong H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1))$, as required. This completes the proof. □

4. Patching and Venkatesh's conjecture

In this section, we prove the following result.

Theorem 4.0.1. *Let Π be a regular algebraic cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_F)$, cohomological with respect to an algebraic representation $\mathcal{V}_\lambda$ where $\lambda \in (\mathbb{Z}^n)^{\mathrm{Hom}(F,E)}$, and let $\rho = \rho_{\Pi,\iota}\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ denote the Galois representation attached to (Π, ι) where $\iota\colon \overline{\mathbb{Q}}_p \xrightarrow{\sim} \mathbb{C}$ and $E/\mathbb{Q}_p$ is a finite extension containing all embeddings $F \to \overline{\mathbb{Q}}_p$. Suppose the following:*

1. *$p \nmid 2n$.*
2. *Π is tempered at almost all finite places.*
3. *$\zeta_p \notin F$ and $\bar{\rho}|_{\Gamma_{F(\zeta_p)}}$ has enormous image* [ACC+23, Def. 6.2.28].
4. *$\bar{\rho}$ is absolutely irreducible and decomposed generic* [ACC+23, Def. 4.3.1].
5. *F contains an imaginary quadratic subfield in which p splits.*
6. *For every pair $\bar{v} \neq \bar{v}'$ of places of F^+ above p, we have (with S_p^+ the places of F^+ above p)*

$$\sum_{\bar{w} \in S_p^+ \setminus \{\bar{v}, \bar{v}'\}} [F_{\bar{w}}^+ : \mathbb{Q}_p] > \frac{1}{2}[F^+ : \mathbb{Q}].$$

7. *For every $v \in S$, the Weil-Deligne representation $\mathrm{WD}(\rho|_{\Gamma_{F_v}})$ is generic.*

Let $\mathcal{R}_\rho^{\mathrm{ss}}$ denote the derived deformation ring representing the characteristic 0 deformation problem $(\rho, S, \{\mathcal{D}_{\rho_v}^{\lambda_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\rho_v}^{\square}\}_{v \in S \setminus S_p})$. Then

$$\pi_*(\mathcal{R}_\rho^{\mathrm{ss}}) \cong \bigwedge H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)),$$

and there is an action of $\mathcal{R}_\rho^{\mathrm{ss}}$ on $R\Gamma(K, \mathcal{V}_\lambda)_{\mathfrak{p}}$ inducing a free action

$$\pi_*(\mathcal{R}_\rho^{\mathrm{ss}}) \circlearrowright H_*(X_K, \mathcal{V}_\lambda)_\Pi.$$

Remark 4.0.2. We briefly describe the roles played by the assumptions in Theorem 4.0.1 and what is currently known.

- *Taylor–Wiles method:* We require (1)-(4) to prove the existence of Taylor–Wiles primes in Proposition 4.1.2 below. We do not expect (1) to be essential. The Ramanujan conjecture predicts that (2) always holds; the case $n = 2$ is known by [BCG+25] and the polarisable case holds by [Car12, Car14]. One might expect to be able to replace the condition in (3) on $\bar{\rho}|_{\Gamma_{F(\zeta_p)}}$ with a 'big image' assumption on the characteristic 0 representation, and to remove the 'decomposed generic' part of (4).
- *Local-global compatibility:* Assumptions (5) and (6) are necessary to apply Hevesi's local-global compatibility result [Hev24, Thm. 1.4] which implies that ρ_v has the correct Hodge–Tate weights, in particular that $\mathbf{S}$ is a well-defined deformation problem; this compatibility is expected to hold in general – when stronger results are available in the literature, we expect to be able to relax assumptions (5) and (6).
- *Vanishing of the Bloch–Kato Selmer group:* All of the assumptions (1)-(7) are required to apply the abstract patching lemma of A'Campo [A'C24, Lem. 2.6.4], though our assumptions are somewhat weaker than those in *loc.cit.*, thanks to Hevesi's local-global compatibility result [Hev24]. Assumption (7) would follow from a strong form of local-global compatibility at the ramified places. In the conjugate self-dual case, it follows from the main results of [Car14, Car12].

Remark 4.0.3. A priori, the action of $\mathcal{R}^{\mathrm{ss}}_{\rho}$ depends on non-canonical choices made in the Taylor–Wiles argument below. With stronger local assumptions on $\bar{\rho}$ at $v \mid p$, the naturality is established in [GV18, §15] by making a comparison between the characteristic p derived deformation ring $\mathcal{R}_{\mathbf{S}}$ and the derived Hecke algebra introduced in [Ven19].

From now on, we fix Π and ρ as in Theorem 4.0.1, and an $\mathcal{O}$-lattice $\xi\colon \Gamma_{F,S} \to \mathrm{GL}_n(\mathcal{O})$ of ρ with reductions $\xi_r\colon \Gamma_{F,S} \to \mathrm{GL}_n(\mathcal{O}_r)$. The main result of [Hev24] then implies the following.

Theorem 4.0.4. [Hev24] *With assumptions as in Theorem 4.0.1,*

$$\mathbf{S} = (\bar{\rho}, S, \{\mathcal{D}_{\bar{\rho}_v}^{\lambda_v, \preceq \tau_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S \setminus S_p})$$

is a well-defined deformation problem.

4.1. **Taylor–Wiles primes.**

The basis of our patching argument later on is a delicate choice of Taylor–Wiles places, detailed in this subsection.

Definition 4.1.1. Let $\mathbf{S} = (\bar{\rho}, S, \{\mathcal{D}_v\}_{v \in S})$ be a deformation problem. A Taylor–Wiles datum of level N for $\mathbf{S}$ is a tuple $(Q_N, \{(\alpha_{v,1}, \dots, \alpha_{v,n})\}_{v \in Q_N})$ where:

(1) Q_N is a finite set of places of F, disjoint from S, such that, for every $v \in Q_N$, $v \equiv 1 \bmod p^N$.
(2) For every $v \in Q_N$, $(\alpha_{v,1}, \dots, \alpha_{v,n}) \in k^n$ are the eigenvalues of $\bar{\rho}(\mathrm{Frob}_v)$, assumed to be pairwise distinct and k-rational.

We generally omit $(\alpha_{v,i})$ and write simply Q_N for Taylor–Wiles data. Our main result Theorem 4.0.1 contains the assumption (3) that $\bar{\rho}|_{\Gamma_{F(\zeta_{p^\infty})}}$ has enormous image. In [NT23, Def. 2.23], a notion of 'enormous image' for characteristic 0 representations is defined, which is implied by the 'enormous image' property of its reduction $\bar{\rho}$.

Proposition 4.1.2. *Let $\mathbf{S}' = (\bar{\rho}, S, \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S})$ be the deformation problem with no conditions at $v \in S$, and suppose that*

- $p \nmid 2n$
- $\zeta_p \notin F$ *and* $\bar{\rho}|_{\Gamma_{F(\zeta_p)}}$ *has enormous image.*

- Π *is tempered at almost all finite places.*

Then there exist integers $q, l \geq 0$ such that, for any $N \geq 1$, there exists a Taylor–Wiles datum Q_N for $\mathbf{S}'$ such that

(1) $|Q_N| = q$
(2) *For every $v \in Q_N$, $v \equiv 1 \bmod p^N$ and the rational prime below v splits in F_0.*
(3) *There is a local $\mathcal{O}$-algebra surjection $R^{S,\mathrm{loc}}_{\mathbf{S}}[[X_1, \dots, X_g]] \twoheadrightarrow R^S_{\mathbf{S}_{Q_N}}$.*
(4) *For every $r \leq N$, with $SQ_N := S \cup Q_N$ we have*

$$\operatorname{length}_{\mathcal{O}} H^1(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)) \leq l,$$

hence if $\mathcal{L}_S = \{\mathcal{L}_{v,r}\}_{v\in S}$ is a Selmer system for $\operatorname{ad}\xi_r$ and

$$\mathcal{L}_{SQ_N} = \mathcal{L}_S \cup \{H^1(\Gamma_{F_v}, \operatorname{ad}\xi_r)\}_{v\in Q_N},$$

then

$$\operatorname{length}_{\mathcal{O}} H^1_{\mathcal{L}^\perp_{SQ_N}}(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)) \leq l.$$

Proof. By [A'C24, Prop. 6.2.32], for q large enough we can find, for every $N \geq 1$, a Taylor–Wiles datum $(Q_N, \{(\alpha_{v,1}, \dots, \alpha_{v,n})\}_{v\in Q_N})$ such that (1)-(3) hold. To achieve (4), we will enlarge the sets Q_N by adding suitable Taylor–Wiles primes, following the proof of [NT23, Lem. 2.30]. Thus, in (1) we replace if necessary q by a larger number. The splitting condition in (2) defines a set of Dirichlet density 1, so we may restrict our attention to places satisfying the condition. Enlarging Q_N does not affect (3), as the statement is equivalent to the vanishing of

$$H^1(\Gamma_{F,SQ_N}, \operatorname{ad}\bar\rho(1)) = \ker\left(H^1(\Gamma_{F,S}, \operatorname{ad}\bar\rho(1)) \to \prod_{v\in Q_N} H^1(\Gamma_{F_v}, \operatorname{ad}\bar\rho(1))\right)$$

and adding more factors to the product over v preserves the vanishing of this kernel. The second statement in (4) follows from the first, since we have an inclusion

$$H^1_{\mathcal{L}^\perp_{SQ_N}}(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)) \hookrightarrow H^1(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)).$$

Let $\operatorname{ad}_{E/\mathcal{O}} = \operatorname{ad}\xi \otimes_{\mathcal{O}} E/\mathcal{O}$ and similarly for $\operatorname{ad}_{\mathcal{O}}, \operatorname{ad}_E, \operatorname{ad}_{\mathcal{O}_r}$ so that $\operatorname{ad}_{E/\mathcal{O}} = \operatorname{colim}\operatorname{ad}_{\mathcal{O}_r}$. Fix an integer

$$s \geq \operatorname{corank}_{\mathcal{O}} H^1(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1)) := \operatorname{rank}_{\mathcal{O}} \operatorname{Hom}_{\mathcal{O}}\left(H^1(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1)), E/\mathcal{O}\right).$$

Our goal is to find Frobenius elements $\sigma_1, \dots, \sigma_s \in G_{F(\zeta_{p^\infty})}$ such that:

(i) $\rho(\sigma_i)$ has n distinct eigenvalues in E.
(ii) The kernel of the map

$$H^1(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1)) \to \bigoplus_{i=1}^s H^1(\langle\sigma_i\rangle, \operatorname{ad}_{E/\mathcal{O}}(1)) \cong \bigoplus_{i=1}^s \operatorname{ad}_{E/\mathcal{O}}/(\sigma_i - 1)\operatorname{ad}_{E/\mathcal{O}}$$

is of finite length as an $\mathcal{O}$-module.

Given such $\sigma_i \in G_{F(\zeta_{p^\infty})}$, for every $r \leq N$, consider the commutative diagram with exact columns

$$\begin{array}{ccc}
0 & & 0 \\
\downarrow & & \downarrow \\
H^0(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1))/\varpi^r & \longrightarrow & \oplus_{i=1}^s H^0(\langle\sigma_i\rangle, \mathrm{ad}_{E/\mathcal{O}}(1))/\varpi^r \\
\downarrow & & \downarrow \\
H^1(\Gamma_{F,S}, \mathrm{ad}_{\mathcal{O}_r}(1)) & \longrightarrow & \oplus_{i=1}^s H^1(\langle\sigma_i\rangle, \mathrm{ad}_{\mathcal{O}_r}(1)) \\
\downarrow & & \downarrow \\
H^1(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1))[\varpi^r] & \longrightarrow & \oplus_{i=1}^s H^0(\langle\sigma_i\rangle, \mathrm{ad}_{E/\mathcal{O}}(1))[\varpi^r] \\
\downarrow & & \downarrow \\
0 & & 0.
\end{array}$$

Note that $H^0(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1))$ has finite length by irreducibility of ρ. It follows from the snake lemma that the kernel of

$$H^1(\Gamma_{F,S}, \mathrm{ad}_{\mathcal{O}_r}(1)) \to \bigoplus_{i=1}^s H^1(\langle\sigma_i\rangle, \mathrm{ad}_{\mathcal{O}_r}(1))$$

has length bounded independently of r and N. By the Chebotarev density theorem, we can then find places $v_1, \dots v_s$ of F with Frobenius elements $\mathrm{Frob}_{v_1}, \dots, \mathrm{Frob}_{v_s}$ approximating $\sigma_1, \dots, \sigma_s$, and thus we obtain statement (4) of the proposition. Now, we have an isomorphism

$$H^1(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)) \cong (E/\mathcal{O})^{s_0} \oplus M$$

where M is a finite length $\mathcal{O}$-module and $s_0 \leq s$ is the corank defined above. Thus, the problem reduces to finding, for every non-zero

$$f\colon E/\mathcal{O} \to H^1(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)),$$

an element $\sigma \in G_{F(\zeta_{p^\infty})}$ such that $\rho(\sigma)$ has n distinct eigenvalues and the composition

$$E/\mathcal{O} \to H^1(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)) \to H^1(\langle\sigma\rangle, \mathrm{ad}_{E/\mathcal{O}}(1)) \cong \frac{\mathrm{ad}_{E/\mathcal{O}}(1)}{(\sigma - 1)\,\mathrm{ad}_{E/\mathcal{O}}(1)}$$

is non-zero. Let $F_\infty = F(\zeta_{p^\infty})$ and $L_\infty = \overline{F}^{\ker \mathrm{ad}_E(1)}$. Then $F_\infty \subseteq L_\infty$ since $E(1) \subset \mathrm{ad}_E(1)$. The short exact sequence

$$0 \to \mathrm{ad}_{\mathcal{O}} \to \mathrm{ad}_E \to \mathrm{ad}_{E/\mathcal{O}} \to 0$$

induces an exact sequence (with $\Gamma(L_\infty/F) = \mathrm{Gal}(L_\infty/F)$)

$$H^1\big(\Gamma(L_\infty/F), \mathrm{ad}_E(1)\big) \to H^1\big(\Gamma(L_\infty/F), \mathrm{ad}_{E/\mathcal{O}}(1)\big) \to H^2\big(\Gamma(L_\infty/F), \mathrm{ad}_{\mathcal{O}}(1)\big).$$

The first term vanishes by [Kis04, Lem. 6.2] and the assumption that Π is tempered at all finite places. The third term is a finitely generated $\mathcal{O}$-module. It follows that the middle term is annihilated by a power of p. Thus, the exactness of the inflation-restriction sequence

$$0 \to H^1\big(\Gamma(L_\infty/F), \mathrm{ad}_{E/\mathcal{O}}(1)\big) \to H^1\big(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)\big) \to H^1\big(\Gamma(F_S/L_\infty), \mathrm{ad}_{E/\mathcal{O}}(1)\big)^{\Gamma(L_\infty/F)}$$

implies that the composition

$$E/\mathcal{O} \xrightarrow{f} H^1\big(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)\big) \to H^1\big(\Gamma(F_S/L_\infty), \mathrm{ad}_{E/\mathcal{O}}(1)\big)^{\Gamma(L_\infty/F)} \cong \mathrm{Hom}_{\Gamma(F_S/F)}\big(\Gamma(L_{\infty,S_{L_\infty}}/L_\infty), \mathrm{ad}_{E/\mathcal{O}}(1)\big)$$

is non-zero. Let $W \subset \mathrm{ad}_{E/\mathcal{O}}(1)$ be the $\mathcal{O}$-submodule spanned by $\{f(x)(\sigma) \mid x \in E/\mathcal{O}, \sigma \in G_{L_\infty}\}$. Then W is non-zero and divisible. By [NT23, Lem. 2.22] there exists $\sigma \in G_{F_\infty}$ such that $\rho(\sigma)$ has n distinct eigenvalues in E and $W \not\subset (\sigma - 1)\,\mathrm{ad}_{E/\mathcal{O}}(1)$. Thus, for some $m \geq 0$ and $\tau \in G_{L_\infty}$ we have

$$f(1/\varpi^m)(\tau) \notin (\sigma - 1)(\mathrm{ad}_{E/\mathcal{O}}(1)).$$

If $f(1/\varpi^m)(\sigma) \notin (\sigma - 1)(\mathrm{ad}_{E/\mathcal{O}}(1))$ then we are done. If not, then $\tau\sigma$ has the sought property, i.e.

$$E/\mathcal{O} \to H^1(F_S, \mathrm{ad}_{E/\mathcal{O}}(1)) \to \frac{\mathrm{ad}_{E/\mathcal{O}}(1)}{(\tau\sigma - 1)\,\mathrm{ad}_{E/\mathcal{O}}(1)}$$

is non-zero, since τ acts trivially on $\mathrm{ad}_E(1)$. □

4.2. **Patching.** In this section, we set up the patching argument, basically following [A'C24] and [GV18] but with some adjustments. Define deformation problems

$$\mathbf{S} := \big(\bar\rho, S, \{\mathcal{D}_{\bar\rho_v}^{\lambda_v, \preceq \tau_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\bar\rho_v}^{\square}\}_{v \in S \setminus S_p}\big)$$
$$\mathbf{S}' := \big(\bar\rho, S, \{\mathcal{D}_{\bar\rho_v}^{\square}\}_{v \in S}\big)$$

We let $\mathcal{R}_{\mathbf{S}}, \mathcal{R}_{\mathbf{S}'} \in \mathcal{CNL}_k$ denote the corresponding derived deformation rings.

For a Taylor–Wiles-datum Q, we define an augmented deformation problem

$$\mathbf{S}_Q = \big(\bar\rho, S \cup Q, \{\mathcal{D}_{\bar\rho_v}^{\lambda_v, \preceq \tau_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\bar\rho_v}^{\square}\}_{v \in S \setminus S_p} \cup \{\mathcal{D}_{\bar\rho_v}^{\square}\}_{v \in Q}\big).$$

and similarly for $\mathbf{S}'_Q$. Applying Proposition 4.1.2, we obtain $q, l \geq 0$ and for every $N \geq 1$ a Taylor–Wiles datum Q_N satisfying the conclusions of the proposition. We let

$$\mathcal{R}_N := \mathcal{R}^S_{\mathbf{S}_{Q_N}} \text{ and } \mathcal{R}'_N := \mathcal{R}^S_{\mathbf{S}'_{Q_N}}$$

be the S-framed type-$\mathbf{S}_{Q_N}$ and type-$\mathbf{S}'_{Q_N}$ derived deformation rings. We also let

$$R^S_{\mathbf{S}_{Q_N}} := \pi_0(\mathcal{R}^S_{\mathbf{S}_{Q_N}}) \text{ and } R^S_{\mathbf{S}'_{Q_N}} := \pi_0(\mathcal{R}^S_{\mathbf{S}'_{Q_N}})$$

denote the underlying classical deformation rings. These objects and the ones introduced below depend on the specific choice of Q_N and not just N, but in our applications, we will fix a choice of Q_N for every $N \geq 1$, motivating our simplified notation.

In [GV18], the authors consider deformations of $\bar\rho_v$ for $v \in Q_N$ valued in the diagonal torus $\mathbf{T} \subset \mathbf{G}$. The local $\mathbf{T}$-valued deformation functors at Taylor–Wiles places are proved in [GV18, Lem. 8.3] to be equivalent to their $\mathbf{G}$-valued counterparts, and for typographical convenience we will not make a distinction.

We let $\mathcal{T} := \mathcal{O}[[T_1, \ldots, T_{n^2|S|-1}]]$ be the power series ring in the 'S-frame variables' and define, for any Taylor–Wiles datum Q_N,

$$\mathcal{S}_v := \text{representing object of } \mathcal{D}_v^{\square},\ v \in Q_N,$$
$$\mathcal{S}_N := \big(\widehat{\bigotimes}_{v \in Q_N} \mathcal{S}_v\big)\widehat{\otimes}_{\mathcal{O}}\mathcal{T}.$$

We also define an unramified counterpart; let $\mathcal{S}_v^{\mathrm{ur}}$ represent the subfunctor of $\mathcal{D}_v^{\square}$ consisting of deformations factoring through the quotient by inertia and set (note the lack of $\mathcal{T}$)

$$\mathcal{S}_N^{\mathrm{ur}} \coloneqq \widehat{\bigotimes}_{v\in Q_N} \mathcal{S}_v^{\mathrm{ur}}.$$

Lemma 4.2.1. [GV18, Eq. 11.4] *For every $N \geq 1$, we have natural maps $\mathcal{S}_N \to \mathcal{R}'_N \to \mathcal{R}_N$ and R'_N and R_N are local S_N-algebras. Moreover, there are equivalences*

$$\mathcal{R}_{\mathbf{S}} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{\mathcal{S}_N} \mathcal{S}_N^{\mathrm{ur}} \text{ and } \mathcal{R}_{\mathbf{S}'} \simeq \mathcal{R}'_N \otimes^{\mathbf{L}}_{\mathcal{S}_N} \mathcal{S}_N^{\mathrm{ur}}.$$

The rings $\mathcal{S}_N$ and $\mathcal{S}_N^{\mathrm{ur}}$ are in fact static, and the latter is equivalent to a power series ring over $\mathcal{O}$ (see [GV18, Lem. 8.6]). It follows that the descent formula afforded by Lemma 4.2.1 can be simplified by, informally speaking, cancelling these power series variables.

Lemma 4.2.2. *Let Q_N be a Taylor–Wiles datum and define*

$$S_N^\circ \coloneqq \widehat{\bigotimes}_{v\in Q_N} \mathcal{O}[[\mathbf{T}(k(v))_p)]],$$

where $\mathbf{T} \subset \mathbf{G}$ is the diagonal torus. Then we have the following:

1. *There is a natural morphism $S_N^\circ \to \mathcal{S}_N$ which is formally smooth. That is, $\mathcal{S}_N \simeq \pi_0(\mathcal{S}_N)$ is equivalent to a power series ring over S_N°.*
2. *There is an equivalence $\mathcal{S}_N^{\mathrm{ur}} \simeq \mathcal{S}_N \otimes^{\mathbf{L}}_{S_N^\circ} \mathcal{O}$. Consequently, there are equivalences*

$$\mathcal{R}_{\mathbf{S}} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{S_N^\circ} \mathcal{O} \text{ and } \mathcal{R}_{\mathbf{S}'} \simeq \mathcal{R}'_N \otimes^{\mathbf{L}}_{S_N^\circ} \mathcal{O}.$$

3. *Let $r \leq N$. Then, for the $\mathcal{O}_r$-point defined by the mod ϖ^r-reduction of the augmentation $S_N^\circ \to \mathcal{O}$, we have*

$$\mathrm{T}^1_{\mathcal{O}_r}(S_N^\circ) \cong \prod_{v\in Q_N} H^2(\Gamma_{F_v}, \operatorname{ad}\xi_r).$$

Proof. (1) and (2) are contained in (the proof of) [GV18, Lem. 8.6]. We obtain (3) from Proposition 3.6.6 and (1). □

It is worth pointing out that

$$S_N^\circ \coloneqq \mathcal{T}[[Y_1, \dots, Y_{qn}]]/\langle (1+Y_i)^{p^{N_i}} - 1\rangle,$$

for exponents N_i all satisfying $N_i \geq N$. This equivalence holds after reduction mod ϖ^N, so that

$$S_N^{\mathrm{ur}}/\varpi^N \simeq S_N/\varpi^N \otimes^{\mathbf{L}}_{S_N^\circ/\varpi^N} \mathcal{O}_N.$$

As is typical in the Taylor–Wiles method, we will pass to a certain limit over N and thus obtain objects of a simple form. We define

$$\begin{aligned} S_\infty^\circ &\coloneqq \mathcal{T}[[Y_1, \dots, Y_{qn}]], \\ \mathfrak{a}_N &= \left\langle (1+Y_i)^{p^N} - 1, (1+T_i)^{p^N} - 1 \right\rangle \subset S_\infty^\circ, \\ \mathfrak{c}_N &= \left\langle \varpi^N, \mathfrak{a}_N \right\rangle \subset S_\infty^\circ, \\ \overline{S}_N^\circ &\coloneqq S_\infty^\circ/\mathfrak{c}_N, \end{aligned}$$

so that $S_\infty^\circ \cong \varprojlim_N \overline{S}_N^\circ$ as profinite topological rings. For every N, we fix an isomorphism $\overline{S}_N^\circ \cong S_N^\circ/\mathfrak{c}_N$. Proposition 4.1.2 implies that there is a surjection

$$R_\infty \coloneqq R^{S,\mathrm{loc}}_{\mathbf{S}}[[X_1, \dots, X_{nq-n^2[F^+:\mathbb{Q}]}]] \twoheadrightarrow R_N = R^S_{\mathbf{S}_{Q_N}}.$$

We fix arbitrary an arbitrary such surjection for every $N \geq 1$. Note that $\mathfrak{m}_{R_N} = \mathfrak{m}_{R_\infty} R_N$; we write simply $\mathfrak{m}$ for the maximal ideal of R_N.

Theorem 4.2.3. [A'C24] *With the notation above and under the assumptions of Theorem 4.0.1, we have the following:*

(i) *There exists an increasing sequence $\mathbb{N}^* = \{N_s\}_{s\geq 2} \subseteq \mathbb{N}$ such that, with*

$$\overline{R}_N \coloneqq R_N/(\mathfrak{a}_N R_N + \mathfrak{m}^{s(N)}),$$
$$s(N) \coloneqq \#\{n \in \mathbb{N}^* \colon n \leq N\} + 1,$$

there are isomorphisms of R_∞-algebras $\overline{R}_N/\mathfrak{m}^{s(M)} \cong \overline{R}_M$ for $M \leq N$ in $\mathbb{N}^*$. Moreover, there is an isomorphism

$$\widehat{R}_{\infty,\mathfrak{p}} \xrightarrow{\sim} \big(\varprojlim_{N\in\mathbb{N}^*} \overline{R}_N\big)^\wedge_{\mathfrak{p}},$$

where $\mathfrak{p} = \ker(R_\infty \to \mathcal{O})$ corresponds to ξ and the transition maps are given by

$$\overline{R}_N \twoheadrightarrow \overline{R}_N/\mathfrak{m}^{s(M)} \cong \overline{R}_M.$$

(ii) *There exists an R_∞-module M_∞ such that $\widehat{M}_{\infty,\mathfrak{p}}$ is free over $\widehat{R}_{\infty,\mathfrak{p}}$ and*

$$\widehat{M}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E \simeq \mathrm{RHom}\left(R\Gamma(K,\mathcal{V}_\lambda)_{\mathfrak{p}}, E\right).$$

Consequently, the graded E-algebra

$$\pi_*\big(\widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E\big) \cong \mathrm{Tor}_*^{\widehat{S}^\circ_{\infty,\mathfrak{a}}}\left(\widehat{R}_{\infty,\mathfrak{p}}, E\right)$$

acts freely on the graded module

$$\pi_*\big(\widehat{M}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E\big) \cong H_*(K,\mathcal{V}_\lambda)_\Pi.$$

(iii) *The geometric adjoint Bloch–Kato Selmer group of ρ vanishes, i.e.*

$$H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho) = 0.$$

Proof. All three essentially follow from the proof of [A'C24, Thm. 5.3.5]. Since our local conditions at $v \mid p$ differ slightly, we spell out the proof which is a straightforward application of the abstract patching lemma [A'C24, Lem. 2.6.4]. For $N \geq 1$, one defines (see [A'C24, Prop. 5.3.3] for details):

- An open compact subgroup $K_0(Q_N) = \prod_v K_{0,v}(Q_N) \subset K$ where $K_{0,v}(Q_N) = K_v$ for $v \notin Q_N$ and $K_{0,v}(Q_N) = \mathrm{Iw}_v$ is the matrices which are upper triangular modulo ϖ_v for $v \in Q_N$.
- An open compact subgroup $K_1(Q_N) = \prod_v K_{1,v}(Q_N) \subset K$ where $K_{1,v}(Q_N) = K_v$ for $v \notin Q_N$ and $K_{1,v}(Q_N) = \mathrm{Iw}_{1,v} \subset \mathrm{Iw}_v$ is the matrices which are upper triangular and unipotent modulo ϖ_v for $v \in Q_N$.
- For any object M with a $\mathbb{T}^S$-action, an enlarged (faithful) Hecke algebra $\mathbb{T}^{S\cup Q_N}_{Q_N}(M)$ obtained by adjoining Hecke operators at the places $v \in Q_N$.
- Maximal ideals $\mathfrak{m}_1, \mathfrak{m}_0$ such that, if $\mathfrak{m} \subset \mathbb{T}^S(R\Gamma(X_K, \mathcal{V}_{\lambda,\tau}))$ is the ideal corresponding to Π, we have morphisms

$$\mathbb{T}^{S\cup Q_N}_{Q_N}\big(R\Gamma(X_{K_1(Q_N)}, \mathcal{V}_{\lambda,\tau})\big)_{\mathfrak{m}_1} \to \mathbb{T}^{S\cup Q_N}\big(R\Gamma(X_{K_0(Q_N)}, \mathcal{V}_{\lambda,\tau})\big)_{\mathfrak{m}_0} \xrightarrow{\sim} \mathbb{T}^S\big(R\Gamma(X_K, \mathcal{V}_{\lambda,\tau})\big)_{\mathfrak{m}}.$$

where the first is given by taking $K_0(Q_N)/K_1(Q_N)$-coinvariants, and corresponding quasi-isomorphisms of complexes

$$R\Gamma\big(K_0(Q_N)/K_1(Q_N), R\Gamma(X_{K_1(Q_N)}, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}_1}\big) \xrightarrow{\sim} R\Gamma(X_{K_0(Q_N)}, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}_0} \xrightarrow{\sim} R\Gamma(X_K, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}}.$$

In the numbering of [A'C24, Lem. 2.6.4], we have 7 statements to verify.

(1) R_∞ is a complete noetherian local $\mathcal{O}$-algebra.
(2) $R_N/\mathfrak{a}_N$ is a local S_N°-algebra and a quotient of R_∞.
(3) Define complexes of $\mathcal{O}$-modules

$$C_0 \coloneqq \mathrm{RHom}_{\mathcal{O}}\left(R\Gamma(X_K, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}}, \mathcal{O}\right)$$
$$C_N \coloneqq \mathrm{RHom}_{\mathcal{O}}\left(R\Gamma(X_{K_1(Q_N)}, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}_1}, \mathcal{T}\right)/\mathfrak{a}_N$$

where the right-hand sides denote minimal representatives in the derived category of $\mathcal{O}$ and S_N, respectively. Then $C_N^\bullet/\mathfrak{a} \simeq C_0^\bullet$ by the fourth bullet point above.
(4) By Proposition 2.1.1, $C_0[1/\varpi]$ has homology concentrated in degrees $[q_0, q_0 + l_0]$ and is not exact. Moreover, $\dim(S_\infty^\circ)_{\mathfrak{a}} = n^2|S| - 1 + qn$, and from Proposition 3.5.11 and Theorem 3.5.13(2) we deduce

$$\dim R_\infty[1/\varpi] \leq n^2|S| + qn - n[F^+ \colon \mathbb{Q}] = \dim(S_\infty^\circ)_{\mathfrak{a}} - l_0.$$

(5) For $N \geq 1$, we let

$$T_N \coloneqq \mathbb{T}_{Q_N}^{S\cup Q_N}(C_N)_{\mathfrak{m}_1} \cdot \mathcal{T} \subset \mathrm{End}_{\mathbf{D}(S_N)}(C_N),$$

where $\mathfrak{m}_1, \mathfrak{m}_0$ are maximal ideals as in [A'C24, Prop. 5.3.3], so that the image of T_N in $\mathrm{End}_{\mathbf{D}(S_0)}(C_0)$ equals

$$\mathbb{T}^{S\cup Q_N}(C_0) \subset \mathbb{T}^S(C_0) =: T_0.$$

(6) Using Proposition 3.5.10 and [Hev24, Thm. 2.24], we obtain for every N an S_N-algebra surjection (after taking a tensor product with $\mathcal{T}$)

$$R_N \to T_N/I_N$$

where $I_N^\delta = 0$ for some δ independent of N, obtaining a commutative square

$$\begin{array}{ccc} R_N & \twoheadrightarrow & T_N/I_N \\ \downarrow & & \downarrow \\ R_0 & \twoheadrightarrow & T_0/I_0 \end{array}$$

(7) Since the Weil–Deligne representations $\mathrm{WD}(\rho_v)$ are generic for every $v \in S$, the representation ρ defines a smooth point of $R_\infty[1/\varpi] \cong R_{\mathbf{S}}^{S,\mathrm{loc}}[[X_1, \ldots, X_{nq-n^2[F^+ \colon \mathbb{Q}]}]]$.

Now, [A'C24, Lem. 2.3.3, Lem. 2.6.4] provide a subsequence $\mathbb{N}^* \subseteq \mathbb{N}$ indexed by $s \colon \mathbb{N}^* \to \mathbb{N}_{\geq 2}$ and morphisms $\overline{R}_N \to \overline{R}_M$ for every $M \leq N$ in $\mathbb{N}^*$ as in the statement, and an isomorphism as in (i): in the notation of loc. cit.,

$$\widehat{R}_{\infty,\mathfrak{p}} \cong \mathbf{P}_{R_\infty}\big((R_N)_{N\geq 1}\big)_{\mathfrak{p}}^\wedge \cong (\varprojlim_N \overline{R}_N)_{\mathfrak{p}}^\wedge,$$

where $\mathbf{P}_{R_\infty}$ is the patching functor. Statements (ii) and (iii) follow from [A'C24, Thm. 5.3.5] and [A'C24, Lem. 2.6.4], respectively. □

In Theorem 4.2.3, the subset $\mathbb{N}^* \subseteq \mathbb{N}$ and the maps $\overline{R}_N \to \overline{R}_M$ depend on a choice of non-principal ultrafilter on $\mathbb{N}$. As such, the isomorphism provided by (1) is non-canonical. From now on, we implicitly fix a choice of ultrafilter and consider the objects afforded by Theorem 4.2.3 to be fixed. In the sequel, limits and colimits over N are taken over $N \in \mathbb{N}^*$.

In light of part (ii) of Theorem 4.2.3, we see that to define an action as in Theorem 4.0.1, it suffices to prove a weak equivalence

$$(\mathcal{R}_{\mathbf{S}} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi} \simeq \widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^{\circ}_{\infty,\mathfrak{a}}} E,$$

and to identify the right-hand side with the exterior algebra on the Bloch–Kato Selmer group.

4.3. **The approximating deformation functor.** Our proof of Theorem 4.0.1 will rely on constructing rings which approximate $\mathcal{R}_{\mathbf{S}}$, at least locally around the point corresponding to ρ. In this section, we construct these approximating rings and prove a few statements about their tangent complexes.

For every $M \leq N$, Theorem 4.2.3(1) provides a commutative diagram

$$\begin{array}{ccccc} \overline{R}_N & \longleftarrow & \overline{S}^{\circ}_N & \longrightarrow & \mathcal{O}_N \\ \downarrow & & \downarrow & & \downarrow \\ \overline{R}_M & \longleftarrow & \overline{S}^{\circ}_M & \longrightarrow & \mathcal{O}_M. \end{array}$$

We let $\mathcal{C}_N \coloneqq \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}_N} \overline{S}^{\mathrm{ur}}_N$ and denote by $e_{N,M} \colon \mathcal{C}_N \to \mathcal{C}_M$ the induced map on derived tensor products. Note that there is a morphism

$$\mathcal{R}_{\mathbf{S}} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{S^{\circ}_N} \mathcal{O} \to \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}^{\circ}_N} \mathcal{O}_N.$$

We will show that, as $N \to \infty$, the right-hand converges to a ring which approximates $\mathcal{R}_{\mathbf{S}}$ near ρ. In order for this to be a reasonable strategy, the artinian quotients $\overline{R}_N$ and $\overline{S}^{\circ}_N$ ought to be, informally speaking, reasonably large. The following proposition establishes this in a precise sense.

Proposition 4.3.1. *Let $r < s(N)$. Then the surjections $R_N \twoheadrightarrow \overline{R}_N$ and $S^{\circ}_N \twoheadrightarrow \overline{S}^{\circ}_N = S^{\circ}_N/\mathfrak{c}_N$ induce isomorphisms on tangent spaces at the point corresponding to ξ_r; that is,*

$$\mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_N) \cong \mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_N) \text{ and } \mathrm{T}^0_{\mathcal{O}_r}(\overline{S}^{\circ}_N) \cong \mathrm{T}^0_{\mathcal{O}_r}(S^{\circ}_N).$$

Proof. The natural maps on tangent spaces is given by precomposition of derivations:

$$\mathrm{Der}_{\mathcal{O}}(\overline{R}_N, \mathcal{O}_r) \hookrightarrow \mathrm{Der}_{\mathcal{O}}(R_N, \mathcal{O}_r).$$

Thus, the statement for $S^{\circ}_N \twoheadrightarrow \overline{S}^{\circ}_N$ is clear. For the other statement, it suffices to show that any derivation $d \colon R_N \to \mathcal{O}_r$ annihilates $\mathfrak{a}_N R_N$ and $\mathfrak{m}^{s(M)}$. We have homomorphisms

$$S^{\circ}_{\infty} \to R_N \to \overline{R}_N \to \mathcal{O}_r$$

where the composite is the mod ϖ^r reduction of the augmentation map $S^{\circ}_{\infty} \to \mathcal{O}$. Reduced mod ϖ^r, the generators $(1+Y_i)^{p^N} - 1$ of $\mathfrak{a}_N \subset S^{\circ}_{\infty}$ have only quadratic and higher terms. Hence, we have the containment $\mathfrak{a}_N \subseteq \ker(S^{\circ}_{\infty} \to \mathcal{O}_r)^2$ and consequently

$$\mathfrak{a}_N R_N \subseteq \ker(R_N \to \mathcal{O}_r)^2.$$

The Leibniz rule now forces any derivation $d \colon R_N \to \mathcal{O}_r$ annihilates $\mathfrak{a}_N R_N$. Moreover, the fixed map $\xi_r \colon R_N \to \mathcal{O}_r$ factors through $R_N/\mathfrak{m}^r$, so that any derivation $d \colon R_N \to \mathcal{O}_r$ factors through $R_N/\mathfrak{m}^{r+1}$ by the Leibniz rule. Since $r+1 \leq N$, this completes the proof for $R_N \twoheadrightarrow \overline{R}_N$. □

Lemma 4.3.2. *With notation as above, let*

$$\mathcal{F}_N = \mathrm{Map}_{/k}(\mathcal{C}_N, -)\colon \mathcal{A}rt_k \to \mathcal{A}ni$$

and $\mathcal{F}_\infty = \mathrm{colim}_N \mathcal{F}_N$. *For our fixed representation* $\xi \in \mathcal{F}_\infty(\mathcal{O})$, *we have the following:*

(1) *For every* i *and every* $r \geq 1$, $\mathrm{T}^i_{\mathcal{O}_r}(\mathcal{F}_\infty)$ *is a finitely generated* $\mathcal{O}_r$*-module.*

(2) *The functor* $(\mathcal{F}_\infty)_{(\xi)}$ *(see Definition 3.7.1) is represented by* $\widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E \in \mathcal{CNL}_E$.

(3) *The tangent complex of* $(\mathcal{F}_\infty)_{(\xi)}$ *has cohomology concentrated in degrees* $[0,1]$ *and satisfies*

$$\dim_E \mathrm{T}^0_E\left((\mathcal{F}_\infty)_{(\xi)}\right) - \dim_E \mathrm{T}^1_E\left((\mathcal{F}_\infty)_{(\xi)}\right) = -l_0.$$

Remark 4.3.3. Due to the lack of information about the rings $\overline{R}_N$, it is unclear to the author whether $\mathcal{F}_\infty$ is necessarily represented by $\mathcal{C}_\infty := \lim_N \mathcal{C}_N$, i.e. whether $\{\mathcal{C}_N\}_{N\geq 1} \in \mathcal{P}ro(\mathcal{A}rt_k)$ is equivalent to the pro-artinian completion $\{\tau_{\leq j}(\mathcal{C}_\infty/^{\mathbf{L}}\mathfrak{m}^r_{\pi_0(\mathcal{C}_\infty)})\}_{j,r} \in \mathcal{P}ro(\mathcal{A}rt_k)$.

Proof. The fact that filtered colimits commute with finite limits implies that there is an equivalence of functors

$$\mathcal{F}_\infty \simeq \left(\operatorname*{colim}_N \mathrm{Map}_{/k}(\overline{R}_N, -)\right) \times_{\mathrm{colim}_N \mathrm{Map}_{/k}(\overline{S}^\circ_N, -)} \left(\mathrm{colim}_N \mathrm{Map}_{/k}(\mathcal{O}_N, -)\right).$$

Now, by [Sta23, Tag 06SF] and the fact that R_∞ surjects onto $\overline{R}_N$ for every $N \geq 1$, the $\mathfrak{m}$-adic topology on the inverse limit $\varprojlim \overline{R}_N$ can be defined using the system of ideals $\ker(\varprojlim \overline{R}_N \to \overline{R}_N)$. Similarly, $S^\circ_\infty \cong \varprojlim \overline{S}^\circ_N$ is an isomorphism of topological rings. It then follows from [HLP23, Prop. 2.1.4] that there are equivalences of functors

$$\operatorname*{colim}_N \mathrm{Map}_{/k}(\overline{R}_N, -) \simeq \mathrm{Map}_{/k}(\varprojlim \overline{R}_N, -),$$
$$\operatorname*{colim}_N \mathrm{Map}_{/k}(\overline{S}^\circ_N, -) \simeq \mathrm{Map}_{/k}(S^\circ_\infty, -),$$

and similarly for $\mathcal{O}$. Thus, we have proved

$$\mathcal{F}_\infty \simeq \mathrm{Map}_{/k}(\varprojlim \overline{R}_N, -) \times_{\mathrm{Map}_{/k}(S^\circ_\infty, -)} \mathrm{Map}_{/k}(\mathcal{O}, -).$$

It follows that there is a fibre sequence of tangent complexes

$$\mathrm{T}^0_{\mathcal{O}_r}(\mathcal{F}_\infty) \to \mathrm{T}^\bullet_{\mathcal{O}_r}(\varprojlim \overline{R}_N) \to \mathrm{T}^\bullet_{\mathcal{O}_r}(S^\circ_\infty) \xrightarrow{+1}.$$

Since the rings $\varprojlim \overline{R}_N$ and S°_∞ are noetherian, Proposition 3.6.4 and the fibre sequence above implies that $\mathrm{T}^i_{\mathcal{O}_r}(\mathcal{F}_\infty)$ is a finitely generated $\mathcal{O}_r$-module.

To prove (2), we note that Theorem 4.2.3 implies an isomorphism $\widehat{R}_{\infty,\mathfrak{p}} \cong (\varprojlim \overline{R}_N)^\wedge_{\mathfrak{p}}$, and thus applying Theorem 3.7.3, Lemma 3.7.4, and Proposition 3.7.6, we obtain an equivalence

$$(\mathcal{F}_\infty)_{(\xi)} \simeq \mathrm{Map}_{/E}(\widehat{R}_{\infty,\mathfrak{p}}, -) \times_{\mathrm{Map}_{/E}(\widehat{S}^\circ_{\infty,\mathfrak{a}}, -)} \mathrm{Map}_{/E}(E, -) \simeq \mathrm{Map}_{/E}(\widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E, -).$$

Thus, we obtain a fibre sequence of tangent complexes over E:

$$\mathrm{T}^\bullet_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \to \mathrm{T}^\bullet_E(\widehat{R}_{\infty,\mathfrak{p}}) \to \mathrm{T}^\bullet_E(\widehat{S}^\circ_{\infty,\mathfrak{a}}) \xrightarrow{+1},$$

which induces an exact sequence

$$0 \to \mathrm{T}^0_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \to \mathrm{T}^0_E(\widehat{R}_{\infty,\mathfrak{p}}) \to \mathrm{T}^0_E(\widehat{S}^\circ_{\infty,\mathfrak{a}}) \to \mathrm{T}^1_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \to 0,$$

where we have used that $\widehat{R}_{\infty,\mathfrak{p}}$ is formally smooth by Theorem 3.5.13(3) and Conjecture 1, under the genericity assumption on $\mathrm{WD}(\rho_v)$ for every $v \in S$. The higher terms of the same exact sequence yield isomorphisms $\mathrm{T}^i_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \cong 0$ for $i \neq 0, 1$, which completes the proof. □

Proposition 4.3.4. *Let $r \leq N$. With assumptions and notation as above, the kernel*

$$\ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\overline{S}^\circ_N)\right)$$

is a finite length $\mathcal{O}$-module with length bounded independently of r and N.

Proof. To begin, we prove the same statement for $\mathcal{R}'_N$, the derived deformation ring with trivial local conditions at all $v \in S$. That is, we let $r \leq N$ and prove that the kernel

$$\ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N)\right)$$

is a finite length $\mathcal{O}$-module with length bounded independently of r and N. By Proposition 3.6.6 and Lemma 4.2.2 and the discussion in Section 2.4, the map in question is the product of restrictions

$$\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \cong H^2(\Gamma_{F,SQ_N}, \mathrm{ad}\,\xi_r) \to \prod_{v \in Q_N} H^2(\Gamma_{F_v}, \mathrm{ad}\,\xi_r) \cong \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N)$$

We factor $\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S_N)$ as $\beta \circ \alpha$, where

$$H^2(\Gamma_{F,SQ_N}, \mathrm{ad}\,\xi_r) \xrightarrow{\alpha} \prod_{v \in SQ_N} H^2(\Gamma_{F_v}, \mathrm{ad}\,\xi_r) \xrightarrow{\beta} \prod_{v \in Q_N} H^2(\Gamma_{F_v}, \mathrm{ad}\,\xi_r)$$

to obtain an exact sequence

$$0 \to \ker(\alpha) \to \ker(\beta \circ \alpha) \to \ker(\beta).$$

To prove the proposition, it suffices to show that $\ker(\alpha)$ and $\ker(\beta)$ have finite length bounded independently of r and N. To see this, consider the Selmer system

$$\{\mathcal{L}_{v,r}\}_{v \in SQ_N} = \{H^1(\Gamma_{F_v}, \mathrm{ad}\,\xi_r)\}_{v \in SQ_N} \cup \{H^1_{\mathrm{ur}}(\Gamma_{F_v}, \mathrm{ad}\,\xi_r)\}_{v \notin SQ_N}.$$

It follows from the Poitou–Tate sequence that we have an exact sequence

$$H^1(\Gamma_{F,SQ_N}, \mathrm{ad}\,\xi_r(1)) \to H^2(\Gamma_{F,SQ_N}, \mathrm{ad}\,\xi_r) \xrightarrow{\alpha} \prod_{v \in SQ_N} H^2(\Gamma_{F_v}, \mathrm{ad}\,\xi_r).$$

In particular, we have

$$\mathrm{length}_{\mathcal{O}} \ker(\alpha) \leq \mathrm{length}_{\mathcal{O}} H^1(\Gamma_{F,SQ_N}, \mathrm{ad}\,\xi_r(1)),$$

which is bounded from above independently of r, N by Proposition 4.1.2(4). Now,

$$\ker(\beta) = \prod_{v \in S} H^2(\Gamma_{F_v}, \mathrm{ad}\,\xi_r) = \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}'})$$

and since the characteristic 0 point of $\mathrm{Spec}\, R^{S,\mathrm{loc}}_{\mathbf{S}'}$ corresponding to ξ is smooth by assumption, the $\mathcal{O}$-module $\ker(\beta)$ has finite length bounded independently of r (and N) by Proposition 3.7.7. This completes the proof of the initial claim. Consider now the kernel

$$\ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N)\right)$$

in the statement of the proposition. The equivalence

$$\mathcal{R}_N \simeq \mathcal{R}'_N \otimes^{\mathbf{L}}_{R^{S,\mathrm{loc}}_{\mathbf{S}'}} R^{S,\mathrm{loc}}_{\mathbf{S}}$$

induces a commutative diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & Q_{r,N} & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \longrightarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \oplus \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \\ & & & & \downarrow{\scriptstyle \gamma} \qquad\qquad \downarrow \\ & & & & \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N) \longleftarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \end{array}$$

where

$$Q_{r,N} \coloneqq \mathrm{coker}\left(\mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R}'_N) \oplus \mathrm{T}^0_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \to \mathrm{T}^0_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}'})\right).$$

The top row is exact and the right vertical map is the projection onto the first coordinate. Thus, we see that $Q_{r,N} \hookrightarrow \ker(\gamma)$, so that γ factors as

$$\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \twoheadrightarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N)/Q_{r,N} \xrightarrow{\tilde{\gamma}} \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N),$$

and there is a short exact sequence

$$0 \to Q_{r,N} \to \ker(\gamma) \to \ker(\tilde{\gamma}) \to 0.$$

By Proposition 4.1.2(4), $Q_{r,N}$ is annihilated by a power of p which is independent of r and N. Thus, it suffices to establish the same property for $\ker(\tilde{\gamma})$. By the first diagram, we have

$$\begin{aligned} \ker(\tilde{\gamma}) &\subseteq \ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \oplus \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \twoheadrightarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N)\right) \\ &\cong \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \oplus \ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N)\right), \end{aligned}$$

whereby it suffices to show that $\mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}})$ and $\ker(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N))$ are annihilated by powers of p that are independent of r and N. The former statement follows from Proposition 3.7.7 together with the genericity assumption on $\mathrm{WD}(\rho_v)$, Proposition 3.5.11 and Theorem 3.5.13; the latter statement is the claim we proved in the beginning of this proof. □

4.4. **Proof of the main theorem.** Recall that we have defined, for every $N \geq 1$,

$$\mathcal{C}_N = \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}^\circ_N} \mathcal{O}_N \in (\mathcal{A}lg_{\mathcal{O}/})/k.$$

Our fixed choice of $\xi\colon \Gamma_{F,S} \to \mathrm{GL}_n(\mathcal{O})$ lifting $\bar{\rho}$ determines morphisms

$$\xi_r\colon \mathcal{C}_N \to \mathcal{O}_r,\ r \leq N,$$

and Theorem 4.2.3 provides compatible $e_{N,M}\colon \mathcal{C}_N \to \mathcal{C}_M$ lifting ξ_r for $M, N \in \mathbb{N}^*$.

Theorem 4.4.1. *With $\mathcal{C}_N$ as above, let*

$$\mathcal{F}_\infty = \operatorname*{colim}_N \mathrm{Map}_{/k}(\mathcal{C}_N, -)\colon \mathcal{A}rt_k \to \mathcal{A}ni$$

be the functor considered in Section 4.3. There exists a morphism $\mathcal{F}_\infty \to \mathcal{D}_{\mathbf{S}}$ inducing an equivalence

$$(\mathcal{F}_\infty)_{(\xi)} \simeq (\mathcal{D}_{\mathbf{S}})_{(\xi)} \in \mathcal{F}un(\mathcal{A}rt_E, \mathcal{A}ni).$$

Remark 4.4.2. Recall that the inverse system $(\mathcal{C}_N)_{N\in\mathbb{N}^*}$ is is not canonically defined. The proof of Theorem 4.4.1 is also non-constructive and, a priori, the provided equivalence depends on our fixed choice of non-principal ultrafilter on $\mathbb{N}$. In [GV18, §15], the authors prove, with stronger hypotheses, that the equivalence is canonical. This is done by establishing a certain compatibility with the derived Hecke algebra of [Ven19]. It would be interesting to prove a similar result in characteristic 0 without introducing extra assumptions.

The proof of Theorem 4.4.1 is subdivided into separate results. The morphism in the theorem is induced by a morphism of rings

$$\mathcal{R}_{\mathbf{S}} \to \mathcal{C}_\infty = \lim_N \mathcal{C}_N.$$

We begin by showing that a homotopy class of maps $\mathcal{R} \to \mathcal{C}_\infty$ is simply a compatible sequence of homotopy classes of maps $\mathcal{R} \to \mathcal{C}_N$. We use the notation

$$[\mathcal{R}, -] \coloneqq \pi_0\big(\operatorname{Map}_{/k}(\mathcal{R}, -)\big).$$

Lemma 4.4.3. *Let $\mathcal{R} \in \mathcal{CNL}_k$ and suppose $(\mathcal{C}_N)_{N\geq 1}$ is a projective system in $\mathcal{CNL}_k$ such that $\tau_{\leq j}\mathcal{C}_N \in \mathcal{A}rt_k$ for every $j \geq 0$, with (∞-categorical) limit $\mathcal{C}_\infty = \lim \mathcal{C}_N$. Then the natural maps*

$$[\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_N [\mathcal{R}, \mathcal{C}_N] \to \varprojlim_{j,N} [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N]$$

are bijections (of sets), where $\varprojlim$ denotes the 1-categorical inverse limit in $\mathbf{Set}$.

Proof. By [GV18, Lem. A.9], there exists a short exact sequence of pointed sets (for any basepoint)

$$0 \to \varprojlim{}^1_{j,N} \pi_1\big(\operatorname{Map}_{/k}(\mathcal{R}, \tau_{\leq j}\mathcal{C}_N)\big) \to [\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_{j,N} [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N] \to 0.$$

Since $\tau_{\leq j}\mathcal{C}_N \in \mathcal{A}rt_k$, Lemma 3.1.5 implies that $\operatorname{Map}_{/k}(\mathcal{R}, \tau_{\leq j}\mathcal{C}_N)$ has finite homotopy groups (for any choice of basepoint). Hence, by the Mittag–Leffler criterion, the first term in the sequence vanishes, and we obtain

$$[\mathcal{R}, \mathcal{C}_\infty] \cong \varprojlim_{j,N} [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N].$$

It follows that $[\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_N [\mathcal{R}, \mathcal{C}_N]$ is injective. It is automatically surjective by the existence of the short exact sequence

$$0 \to \varprojlim{}^1_N \pi_1\big(\operatorname{Map}_{/k}(\mathcal{R}, \mathcal{C}_N)\big) \to [\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_N [\mathcal{R}, \mathcal{C}_N] \to 0,$$

also deduced from [GV18, Lem. A.9]. □

Proposition 4.4.4. *For an integer $a \geq 0$, let $X_N(a) \subseteq [\mathcal{R}, \mathcal{C}_N]$ denote the subset consisting of classes $[f]$ such that, for every $r \leq M \leq N$,*

(i) *$f \circ e_{N,M} \colon \mathcal{R} \to \mathcal{C}_M$ lifts ξ_r,*
(ii) *$\mathrm{T}^0_{\mathcal{O}_r}(e_{N,M} \circ f)$ is an isomorphism,*
(iii) *$p^a \operatorname{coker}(\mathrm{T}^1_{\mathcal{O}_r}(e_{N,M} \circ f)) = 0$,*

where $e_{N,M} \colon \mathcal{C}_N \to \mathcal{C}_M$ is the morphism defined at the beginning of Section 4.3, and

$$\mathrm{T}^i_{\mathcal{O}_r}(e_{N,M} \circ f) \colon \mathrm{T}^i_{\mathcal{O}_r}(\mathcal{C}_M) \to \mathrm{T}^i_{\mathcal{O}_r}(\mathcal{R})$$

is the induced homomorphism of $\mathcal{O}_r$-modules. Then we have the following:

$(X_N(a))_{N\in\mathbb{N}^*}$ is an inverse system of non-empty profinite sets with continuous transition maps induced by $e_{N,M}$, and there exists an $a \in \mathbb{Z}$ such that $X_N(a)$ is nonempty; in particular,

$$\varprojlim_N X_N(a) \subset [\mathcal{R}, \mathcal{C}_\infty]$$

is nonempty.

Proof. We subdivide the proof into four steps. In Steps 1-3, we show that $(X_N(a))$ form an inverse system of profinite sets with continuous transition maps. This will hold independently of choice of $a \in \mathbb{Z}$. Finally, in Step 4, we show that there exists an a such that $X_N(a) \neq \emptyset$ for all N; the conclusion then follows from an application of Tychonoff's theorem.

Step 1: $(X_N(a))$ **is an inverse system.** Let $[f] \in X_N(a)$. Then, for any $L \leq M \leq N$, we have

$$[e_{M,L} \circ e_{N,M} \circ f] = [e_{N,L} \circ f]$$

and hence $[e_{N,M} \circ f] \in X_M(a)$ and the transition maps are compatible. Hence, $(X_N(a))_{N \in \mathbb{N}^*}$ is an inverse system.

Step 2: $X_N(a)$ **is profinite.** By Lemma 3.1.5,

$$[\mathcal{R}, \mathcal{C}_N] = \varprojlim_j [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N]$$

is profinite. To prove that $X_N(a)$ is profinite, we show that it is the limit of an inverse system of subsets (by functoriality, $e_{N,M}$ defines a morphism between truncations, also denoted $e_{N,M}$ below)

$$X_N^j(a) \subseteq [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N],\ j \geq 0.$$

In other words, we may represent any $[f] \in [\mathcal{R}, \mathcal{C}_N]$ by a compatible sequence of classes

$$[f_j \colon \mathcal{R} \to \tau_{\leq j}\mathcal{C}_N] \in [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N],\ j \geq 0,$$

and the claim is that $[f] \in X_N(a)$ if and only if, for every $j \leq 0$ and $r \leq M \leq N$:

(i') The composition $e_{N,M} \circ f_j \colon \mathcal{R} \to \tau_{\leq j}\mathcal{C}_N \to \tau_{\leq j}\mathcal{C}_M$ lifts $\xi_r \colon \mathcal{R} \to \mathcal{O}_r$.
(ii') $\mathrm{T}^0_{\mathcal{O}_r}(e_{N,M} \circ f_j)$ is an isomorphism.
(iii') $p^a \operatorname{coker}\left(\mathrm{T}^1_{\mathcal{O}_r}(e_{N,M} \circ f_j)\right) = 0$.

We compare these conditions to (i), (ii) and (iii) in the statement of the theorem. Condition (i') is equivalent to (i) since ξ_r factors as

$$\mathcal{R} \to \pi_0(\mathcal{R}) \to \pi_0(\mathcal{C}_M) \to \mathcal{O}_r,$$

and condition (ii') is equivalent to (ii) since $\mathrm{T}^0_k(\tau_{\leq j}\mathcal{C}_M) \cong \mathrm{T}^0_k(\pi_0(\mathcal{C}_M))$ for every $j \geq 0$. To prove that (iii') is equivalent to (iii), it suffices to show that

$$\mathrm{T}^1_{\mathcal{O}_r}(\tau_j\mathcal{C}_M) \to \mathrm{T}^1_{\mathcal{O}_r}(\tau_{j+1}\mathcal{C}_M)$$

is an isomorphism for j large enough, independent of M. Let $\mathcal{C}_M^i = \tau_{\leq i}\mathcal{C}_M$. We have a cofibre sequence of cotangent complexes

$$\mathcal{C}_M^j \otimes^{\mathbf{L}}_{\mathcal{C}_M^{j+1}} \mathrm{L}_{\mathcal{C}_M^{j+1}/\mathcal{O}} \to \mathrm{L}_{\mathcal{C}_M^j/\mathcal{O}} \to \mathrm{L}_{\mathcal{C}_M^j/\mathcal{C}_M^{j+1}} \xrightarrow{-1}.$$

By [Lur04, Prop. 3.2.16] (see also [Lur17, Cor. 7.4.3.6]), $\mathrm{L}_{\mathcal{C}_M^j/\mathcal{C}_M^{j+1}}$ has trivial homology in degrees below $j+2$. Applying $\mathrm{RHom}(- \otimes^{\mathbf{L}}_{\mathcal{C}_M^j} \mathcal{O}_r, \mathcal{O}_r)$ and passing to cohomology, we find that

$$\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_M^j) \cong \mathrm{T}^1_{\mathcal{O}_R}(\mathcal{C}_M^{j+1}) \text{ for } j \geq 2,$$

as needed. Thus, we define finite subsets $X_N^j(a) \subseteq [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N]$ using (i'), (ii') and (iii') and conclude that $X_N(a) = \varprojlim_j X_N^j(a)$ is profinite.

Step 3: $X_N(a) \to X_M(a)$ **is continuous.** Let $M \leq N$ and consider $e_{N,M} \colon \mathcal{C}_N \to \mathcal{C}_M$. For any j, we have commutative diagrams (the right one a consequence of the left one)

$$\begin{array}{ccc} \mathcal{C}_N & \longrightarrow & \mathcal{C}_M \\ \downarrow & & \downarrow \\ \tau_{\leq j}\mathcal{C}_N & \longrightarrow & \tau_{\leq j}\mathcal{C}_M \end{array} \qquad \begin{array}{ccc} [\mathcal{R}, \mathcal{C}_N] & \longrightarrow & [\mathcal{R}, \mathcal{C}_M] \\ \downarrow & & \downarrow \\ [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N] & \longrightarrow & [\mathcal{R}, \tau_{\leq j}\mathcal{C}_M] \end{array}$$

It follows that $[\mathcal{R}, \mathcal{C}_N] \to [\mathcal{R}, \mathcal{C}_M]$ is continuous for the profinite topologies, and hence the restriction $X_N(a) \to X_M(a)$ is also continuous.

Step 4: $X_N(a)$ is nonempty. Let $N \geq 1$ be arbitrary. We prove that there exists an $a \in \mathbb{Z}$, independent of N, such that $[f] \in X_N(a)$, where $f = f_N$ is the composite morphism

$$f \colon \mathcal{R} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{S^\circ_N} \mathcal{O} \to \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}^\circ_N} \mathcal{O}_N = \mathcal{C}_N,$$

where the equivalence follows from Lemma 4.2.2. Note that we do not prove any compatibility between f_N's for varying N.

For any $r \leq M \leq N$, the map $e_{N,M} \circ f$ induces a morphism of fibre sequences; passing to cohomology we obtain a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{C}_M) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_M) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\overline{S}_M) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R}) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R}_N) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{S}_N). \end{array}$$

The two rightmost vertical maps are isomorphisms by Proposition 4.3.1 and the fact that $\mathrm{T}^0_{\mathcal{O}_r}(A) \cong \mathrm{T}^0_{\mathcal{O}_r}(\pi_0(A))$ for any $A \in (\mathcal{CNL}_k)_{/\mathcal{O}_r}$. It follows from the five lemma that the leftmost vertical map is also an isomorphism. This establishes property (ii) in the definition of $X_N(a)$, irrespective of a. To prove property (iii), we consider the continuation of the diagram above:

$$\begin{array}{ccccccc} \mathrm{T}^0_{\mathcal{O}_r}(\overline{S}^\circ_M) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_M) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\overline{R}_M) \oplus \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{O}_M) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\overline{S}^\circ_M) \\ \downarrow \cong & & \downarrow \alpha & & \downarrow & & \downarrow \\ \mathrm{T}^0_{\mathcal{O}_r}(S^\circ_N) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N). \end{array}$$

The claim to be proven is that $p^a \operatorname{coker}(\alpha) = 0$ for an $a \geq 0$ which is independent of r, M and N. By the diagram, we have a surjection

$$\ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N)\right) \cong \operatorname{coker}\left(\mathrm{T}^0_{\mathcal{O}_r}(S^\circ_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R})\right) \twoheadrightarrow \operatorname{coker}(\alpha).$$

We recall that $\ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S^\circ_N)\right)$ has finite length bounded independently of r and N by Proposition 4.3.4. Thus, $\operatorname{coker}(\alpha)$ is annihilated by a power of p which is independent of N, and there exists an absolute constant a such that $[f] \in X_N(a)$. In particular, $X_N(a) \neq \emptyset$ for every N. □

Proof of Theorem 4.4.1. By Lemma 4.4.3 and Theorem 4.4.4, for $a \in \mathbb{Z}$ sufficiently large, there exists $[g] \in \varprojlim_N X_N(a)$ represented by a compatible sequence (g_N) of morphisms

$$g_N \colon \mathcal{R} \to \mathcal{C}_N$$

such that $\mathrm{T}^0_{\mathcal{O}_r}(g_N) \colon \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{C}_N) \to \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R})$ is an isomorphism and the cokernel

$$Q_{r,N}(g) = \operatorname{coker}\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R})\right)$$

is p^a-torsion. We will prove that the morphism

$$(\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi \to (\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi$$

induced by g is a weak equivalence, by showing that it is an isomorphism on T^0_E and an epimorphism on T^1_E, then counting dimensions. The first statement follows from the fact that g_N induces isomorphisms on T^0. To prove the second statement, we note that p^a-torsion modules are preserved under

filtered colimits and inverse limits, so that

$$p^a Q_{r,N}(g) = 0 \implies \left(\varprojlim_r \operatorname*{colim}_N Q_{r,N}(g)\right) \otimes_{\mathcal{O}} E = 0.$$

Now, $\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_\infty)$ is finite by Lemma 4.3.2(2). It follows from Proposition 3.7.6 that we have an isomorphism

$$\begin{aligned}\left(\varprojlim_r \operatorname*{colim}_N Q_{r,N}(g)\right) \otimes_{\mathcal{O}} E &\cong \operatorname{coker}\left(\left(\varprojlim_r \operatorname*{colim}_N \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_N)\right) \otimes_{\mathcal{O}} E \to \left(\varprojlim_r \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R})\right) \otimes_{\mathcal{O}} E\right) \\ &\cong \operatorname{coker}\left(\mathrm{T}^1_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \to \mathrm{T}^1_E\left((\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi\right)\right)\end{aligned}$$

since colimits commute and $\varprojlim$ is exact on inverse systems of finite abelian groups. Thus, this cokernel vanishes and we have proved that the map

$$\mathrm{T}^*_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \to \mathrm{T}^*_E\left((\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi\right)$$

induced by g is an isomorphism in degree 0 and an epimorphism in degree 1. Both are graded vector spaces concentrated in degrees $[0,1]$ and have the same dimensions by Lemma 4.3.2 and Proposition 3.7.8. It follows that $(\mathcal{F}_\infty)_{(\xi)} \simeq (\mathcal{D}_{\mathbf{S}})_{(\xi)}$ in $\mathcal{F}un(\mathcal{A}rt_E, \mathcal{A}ni)$. □

We are now in a position to piece together our results to deduce our main theorem.

Proof of Theorem 4.0.1. We have already seen in Lemma 4.3.2(2) that

$$\mathcal{F}_\infty \simeq \mathrm{Map}_{/E}(\widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E, -),$$

and thus Theorem 4.4.1 implies that

$$\mathcal{R}^{\mathrm{ss}}_\rho \simeq \widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E.$$

By Theorem 4.2.3, there exists a free $\widehat{R}_{\infty,\mathfrak{p}}$-module $\widehat{M}_{\infty,\mathfrak{p}}$ such that

$$\widehat{M}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E \simeq R\Gamma(K, \mathcal{V}_{\lambda,\tau})^\vee_{\mathfrak{p}}.$$

We obtain an action of the graded ring $\pi_*(\mathcal{R}^{\mathrm{ss}}_\rho)$ on the homology of this complex. Thus, by Corollary 3.7.10, we obtain a free action

$$\bigwedge H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \cong \pi_*\left(\mathcal{R}^{\mathrm{ss}}_\rho\right) \circlearrowright H_*(K, \mathcal{V}_{\lambda,\tau})_\Pi.$$

□

References

[A'C24] Lambert A'Campo. Rigidity of automorphic Galois representations over CM fields. *Int. Math. Res. Not. IMRN*, (6):4541–4623, 2024.

[ACC+23] Patrick B. Allen, Frank Calegari, Ana Caraiani, Toby Gee, David Helm, Bao V. Le Hung, James Newton, Peter Scholze, Richard Taylor, and Jack A. Thorne. Potential automorphy over CM fields. *Ann. of Math. (2)*, 197(3):897–1113, 2023.

[AG15] D. Arinkin and D. Gaitsgory. Singular support of coherent sheaves and the geometric Langlands conjecture. *Selecta Math. (N.S.)*, 21(1):1–199, 2015.

[All16] Patrick B. Allen. Deformations of polarized automorphic Galois representations and adjoint Selmer groups. *Duke Math. J.*, 165(13):2407–2460, 2016.

[BCG+25] George Boxer, Frank Calegari, Toby Gee, James Newton, and Jack A. Thorne. The Ramanujan and Sato-Tate conjectures for Bianchi modular forms. *Forum Math. Pi*, 13:Paper No. e10, 65, 2025.

[BIP23] Gebhard Böckle, Ashwin Iyengar, and Vytautas Paškūnas. On local Galois deformation rings. *Forum Math. Pi*, 11:Paper No. e30, 54, 2023.
[BK90] Spencer Bloch and Kazuya Kato. L-functions and Tamagawa numbers of motives. In *The Grothendieck Festschrift, Vol. I*, volume 86 of *Progr. Math.*, pages 333–400. Birkhäuser Boston, Boston, MA, 1990.
[BS73] A. Borel and J.-P. Serre. Corners and arithmetic groups. *Comment. Math. Helv.*, 48:436–491, 1973.
[BS15] Bhargav Bhatt and Peter Scholze. The pro-étale topology for schemes. *Astérisque*, (369):99–201, 2015.
[Cai21] Yichang Cai. Derived deformation rings allowing congruences, 2021.
[Car12] Ana Caraiani. Local-global compatibility and the action of monodromy on nearby cycles. *Duke Math. J.*, 161(12):2311–2413, 2012.
[Car14] Ana Caraiani. Monodromy and local-global compatibility for $l = p$. *Algebra Number Theory*, 8(7):1597–1646, 2014.
[CG18] Frank Calegari and David Geraghty. Modularity lifting beyond the Taylor–Wiles method. *Invent. Math.*, 211(1):297–433, 2018.
[Cis19] Denis-Charles Cisinski. *Higher categories and homotopical algebra*, volume 180 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2019.
[ČS24] Kestutis Česnavičius and Peter Scholze. Purity for flat cohomology. *Ann. of Math. (2)*, 199(1):51–180, 2024.
[CT22] Yichang Cai and Jacques Tilouine. Simplicial Galois deformation functors. In *Perfectoid spaces*, Infosys Sci. Found. Ser., pages 349–389. Springer, Singapore, 2022.
[GV18] S. Galatius and A. Venkatesh. Derived Galois deformation rings. *Adv. Math.*, 327:470–623, 2018.
[Hev24] Bence Hevesi. Ordinary parts and local-global compatibility at $\ell = p$, 2024.
[HLP23] Daniel Halpern-Leistner and Anatoly Preygel. Mapping stacks and categorical notions of properness. *Compos. Math.*, 159(3):530–589, 2023.
[HLTT16] Michael Harris, Kai-Wen Lan, Richard Taylor, and Jack Thorne. On the rigid cohomology of certain Shimura varieties. *Res. Math. Sci.*, 3:Paper No. 37, 308, 2016.
[HT17] David Hansen and Jack A. Thorne. On the GL_n-eigenvariety and a conjecture of Venkatesh. *Selecta Math. (N.S.)*, 23(2):1205–1234, 2017.
[HV19] Michael Harris and Akshay Venkatesh. Derived Hecke algebra for weight one forms. *Exp. Math.*, 28(3):342–361, 2019.
[Kis04] Mark Kisin. Geometric deformations of modular Galois representations. *Invent. Math.*, 157(2):275–328, 2004.
[Kis08] Mark Kisin. Potentially semi-stable deformation rings. *J. Amer. Math. Soc.*, 21(2):513–546, 2008.
[Kis09] Mark Kisin. Moduli of finite flat group schemes, and modularity. *Ann. of Math. (2)*, 170(3):1085–1180, 2009.
[Lur04] Jacob Lurie. *Derived algebraic geometry*. ProQuest LLC, Ann Arbor, MI, 2004. Thesis (Ph.D.)–Massachusetts Institute of Technology.
[Lur09] Jacob Lurie. *Higher topos theory*, volume 170 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 2009.
[Lur17] Jacob Lurie. Higher algebra, 2017. Available at https://www.math.ias.edu/~lurie/papers/HA.pdf.
[Lur18] Jacob Lurie. Kerodon. https://kerodon.net, 2018.
[Mil86] J. S. Milne. *Arithmetic duality theorems*, volume 1 of *Perspectives in Mathematics*. Academic Press, Inc., Boston, MA, 1986.
[Mol24] Douglas Molin. On completed homology and a conjecture of venkatesh, 2024. https://arxiv.org/abs/2407.04228.
[Nek06] Jan Nekovář. Selmer complexes. *Astérisque*, (310):viii+559, 2006.
[NSW00] Jürgen Neukirch, Alexander Schmidt, and Kay Wingberg. *Cohomology of number fields*, volume 323 of *Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer-Verlag, Berlin, 2000.
[NT23] James Newton and Jack A. Thorne. Adjoint Selmer groups of automorphic Galois representations of unitary type. *J. Eur. Math. Soc. (JEMS)*, 25(5):1919–1967, 2023.
[PV21] Kartik Prasanna and Akshay Venkatesh. Automorphic cohomology, motivic cohomology, and the adjoint L-function. *Astérisque*, (428):viii+132, 2021.
[Qui67] Daniel G. Quillen. *Homotopical algebra*, volume No. 43 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin-New York, 1967.
[Sch15] Peter Scholze. On torsion in the cohomology of locally symmetric varieties. *Ann. of Math. (2)*, 182(3):945–1066, 2015.
[Sta23] The Stacks project authors. The Stacks project. https://stacks.math.columbia.edu, 2023.

[SZ99] P. Schneider and E.-W. Zink. K-types for the tempered components of a p-adic general linear group. *J. Reine Angew. Math.*, 517:161–208, 1999. With an appendix by Schneider and U. Stuhler.

[Ven19] Akshay Venkatesh. Derived Hecke algebra and cohomology of arithmetic groups. *Forum Math. Pi*, 7:e7, 119, 2019.

[Zhu25] Xinwen Zhu. Coherent sheaves on the stack of Langlands parameters. In *The Langlands program. II. Geometrization of the Langlands correspondence*, volume 112.2 of *Proc. Sympos. Pure Math.*, pages 39–123. Amer. Math. Soc., Providence, RI, 2025.

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